

Development of an Analytical Bond Model for High Strength Reinforcing Bars

Farhan Anjum

This thesis is presented as part of the requirements for the conferral of the degree:

Bachelor of Civil Engineering (Honours)

Supervisor:
Professor Hamid Valipour

The University of New South Wales
School of Civil and Environmental Engineering

2026

Declaration

I, *Farhan Anjum*, declare that this thesis is submitted in partial fulfilment of the requirements for the conferral of the degree *Bachelor of Civil Engineering (Honours)*, from the University of New South Wales, is wholly my own work unless otherwise referenced or acknowledged. This document has not been submitted for qualifications at any other academic institution.

Farhan Anjum

November 9, 2026

Abstract

Bond between a deformed bar and the surrounding concrete governs anchorage design, yet the development length provisions of AS 3600:2018, Eurocode 2 and *fib* Model Code 2010 are empirical expressions fitted to test databases. They return a length but give no access to the stress field that produces it, and they require recalibration whenever a reinforcement grade falls outside the range from which they were derived, most recently for the SENSE600® bars introduced in Australia by InfraBuild. An analytical treatment carries no such dependence on bar grade.

This thesis develops a closed-form analytical model of a deformed bar bonded in concrete. The specimen is modelled as three concentric elastic cylinders, bar, rib layer and concrete; an assumed stress field satisfying equilibrium and interface continuity reduces the problem to one unknown function, the axial stress in the bar. Making the strain energy stationary yields a fourth-order differential equation solved in closed form, the radial-stress parameter following from a second stationarity condition. No numerical integration, discretisation or iteration is used. Failure is assessed by two criteria: yielding of the bar, and cracking of the concrete under a quadratic interaction of hoop and shear stress averaged over a semicircular region after Taylor's Theory of Critical Distances. Bond stiffness and slip follow from the same energy.

The formulation is linear elastic and represents no bond degradation, so it applies to short embedment lengths, where the stress distribution remains close to elastic up to failure. It is assessed against pull-out tests rather than anchorage at full development length. The implementation is verified against two published analytical solutions, reproducing both without parameter adjustment, and predictions are compared with pull-out tests from the literature.

Extending the model to the long embedments that govern development length, and hence to the anchorage of SENSE600® reinforcement, requires a softening bond-slip law and an understanding of material non-linearity, and is the principal direction for further work.

Acknowledgements

Contents

Abstract iii

Acknowledgements iv

List of Figures ix

List of Tables xi

1 Introduction 1

2 Literature Review 2

2.1 Bond Theory and the Bond Model . . . 2
2.1.1 Background on Bond . . . 2
2.1.2 Failure Modes . . . 4
2.1.3 Factors Influencing Bond Strength . . . 4
2.1.4 Prediction Models . . . 9
2.2 History of 500 MPa Reinforcement . . . 12
2.2.1 Study on Australian 500N Bars . . . 12
2.2.2 Splice Strength and Lap Length for 500 MPa Bars . . . 13
2.2.3 International Standards for 500 MPa and Beyond . . . 13
2.3 Empirical Work on Development Lengths . . . 13
2.3.1 What is development length? . . . 14
2.3.2 fib Model Code 2010 . . . 15
2.3.3 Eurocode 2 (EN 1992-1-1:2004, Clause 8.4.4) . . . 16
2.3.4 AS3600:2018 - Anchorage Chapter . . . 17
2.3.5 AS5100.5:2017 - Anchorage Chapter . . . 17
2.3.6 ACI 318-19 — Chapter 25.4 and Table 25.4.2.3 . . . 18
2.3.7 Foster et al. (2024) - SENSE600 Development Length Equation 19
2.4 Analytical Work on Development Length . . . 20
2.4.1 The Volkersen Shear-Lag Model - Analytical Baseline . . . 20
2.4.2 Elasticity-Based Cylindrical Stress Models . . . 21
2.4.3 Fracture Mechanics Failure Criteria . . . 24
2.5 Background on SENSE600 Bars and the Australian Context . . . 24
2.5.1 Introduction of 600 MPa Reinforcement in Australia . . . 24
2.5.2 AS/NZS 4671:2019 and Bar Geometry Requirements . . . 25
2.5.3 Ng and McGregor (2024) - Stage I Experimental Program . . . 25

2.5.4 Foster et al. (2024) - Stage II Experimental Program and Development Length Equation . . . 25
2.5.5 Ng and McGregor (2024) - Changes to Australian Standards . 26
2.6 Summary and Research Gap . . . 26
2.6.1 Summary . . . 26
2.6.2 Research Gap . . . 26

3 Methodology **27**
3.1 The bond pullout test . . . 27
3.1.1 Idealisation of the specimen . . . 28
3.2 Three-layer model and assumptions . . . 29
3.3 Assumed stress field . . . 29
3.4 Strain energy . . . 31
3.5 Governing differential equation and its solution . . . 32
3.6 Boundary conditions and the radial-stress parameter . . . 32
3.6.1 Boundary conditions . . . 32
3.6.2 Determination of α_1 . . . 33
3.7 Bond Stiffness . . . 34
3.8 Failure criteria . . . 35
3.8.1 Mode I: yielding of the bar . . . 35
3.8.2 Mode II: cracking of the concrete . . . 35
3.8.3 The theory of critical distances . . . 36
3.8.4 Predicted capacity . . . 38
3.9 Calibration curves . . . 39

4 Validation of the Proposed Method **42**
4.1 Approach . . . 42
4.2 Verification against Nemes and Lachaud (2009) . . . 43
4.2.1 Benchmark configuration . . . 43
4.2.2 Shear stress distribution . . . 44
4.2.3 Circumferential stress distribution . . . 46
4.2.4 Discussion . . . 47
4.3 Verification against Hassanieh et al. (2018) . . . 48
4.3.1 Rationale for a second benchmark . . . 48
4.3.2 Benchmark configuration . . . 49
4.3.3 Results . . . 49
4.3.4 Discussion . . . 55
4.4 What the two benchmarks establish . . . 56

5 Taylor's Theories of Critical Distances **58**
5.1 Three methods of Critical Distances . . . 59
5.1.1 Area Method . . . 59
5.1.2 Line Method . . . 61
5.1.3 Point Method . . . 63
5.1.4 Comparison of the three methods . . . 64
5.2 Sensitivity Analysis . . . 65
5.2.1 Reference specimen . . . 65

5.2.2 Mesh Size vs Pullout Load . . . 66
5.2.3 Mesh Size vs Computation Time . . . 68

6 Parametric Study of Factors Influencing Bond 70
6.1 Approach . . . 71
6.1.1 Baseline specimen . . . 71
6.1.2 Quantities reported . . . 71
6.1.3 Factors considered and the threshold for influence . . . 72
6.1.4 Isolating a mechanism by removing it . . . 72
6.1.5 Factors outside the scope of the model . . . 72
6.2 Geometric factors . . . 73
6.2.1 Embedment length . . . 73
6.2.2 Bar diameter . . . 74
6.2.3 Cover ratio . . . 76
6.2.4 Rib layer thickness . . . 77
6.3 Material factors . . . 78
6.3.1 Concrete strength . . . 78
6.3.2 Concrete stiffness . . . 80
6.3.3 Tensile-to-shear strength ratio . . . 81
6.3.4 Factors of minor influence . . . 81
6.4 Synthesis . . . 83
6.4.1 Ranking the factors . . . 83
6.4.2 Capacity and stiffness are not governed alike . . . 83
6.4.3 Where the uncertainty sits . . . 84
6.5 Conclusions . . . 84

7 Calibration Curves and an Analytical Development Length Equation 86
7.1 The calibration curve family . . . 86
7.1.1 Construction . . . 86
7.1.2 The effective length . . . 87
7.2 Normalisation and the master curve . . . 87
7.2.1 Single-variable normalisations fail . . . 87
7.2.2 The shear-lag form . . . 88
7.2.3 The master curve . . . 88
7.2.4 Why the form is mechanics and not curve fitting . . . 89
7.3 What sets the transfer length . . . 89
7.4 The development length equation . . . 90
7.4.1 Grade independence of the calibration curve . . . 90
7.4.2 Derivation . . . 90
7.4.3 Relation to the code form . . . 90
7.5 The limit imposed by the saturation capacity . . . 91
7.5.1 The condition . . . 91
7.5.2 The size of the discrepancy . . . 91
7.5.3 Calibration cannot close the gap . . . 92
7.5.4 Why the gap exists . . . 92
7.5.5 What the equation is therefore good for . . . 93

7.6 Comparison with code and research expressions 93
7.7 Conclusions 94

8 Comparison to Other Analytical Models **95**
8.1 Tepfers 95

9 Limitations and Future Work **96**
9.1 Material Non-Linearity 96

10 Conclusion **97**

A Your first appendix **98**
A.1 The title of the first section 98
A.2 Fitting of the power-law exponents 98

Bibliography **105**

List of Figures

2.1.1.1 Schematic of bond transfer mechanisms in reinforced concrete (ACI 408R-03, 2003). . . . 3
2.1.2.1 Schematic of splitting failure in reinforced concrete (Cairns, 2013). . 4
2.1.3.1 Relationship between concrete tensile strength and bond strength (Li et al. 2012). . . . 5
2.1.3.2 Relationship between concrete tensile strength and bar diameter (Li et al. 2012). . . . 6
2.1.3.3 Relationship between concrete tensile strength and concrete cover (Li et al. 2012). . . . 7
2.1.3.4 Relationship between concrete tensile strength and embedment length (Li et al. 2012). . . . 8
2.1.3.5 Relationship between concrete tensile strength and transverse confinement (Li et al. 2012). . . . 9
2.1.4.1 Definition of cover dimensions for fib MC2010 bond model (Cairns 2015). . . . 12
2.2.1.1 Rib pattern of 500N Class reinforcing bars (Li et al. 2012). . . . 12

3.1.0.1 Bond pullout test specimen: bar, bonded length L, reaction plate and clearance hole. . . . 28
3.8.3.1 The three Theory of Critical Distances averaging geometries . . . 36
3.8.3.2 Area Method averaging procedure . . . 37
3.9.0.1 Structure of the calibration sweep . . . 40

4.2.1.1 Diagram of the adhesively bonded joint from Nemes and Lachaud (2009). . . . 43
4.2.2.1 Interfacial shear stress: benchmark and present model . . . 44
4.2.3.1 Circumferential stress: benchmark and present model . . . 46
4.3.3.1 Rod axial stress: benchmark and present model . . . 51
4.3.3.2 Glue-line shear stress: benchmark and present model . . . 52
4.3.3.3 Tangential stress: benchmark and present model . . . 53
4.3.3.4 Shear distribution against embedment length . . . 54

5.1.1.1 Clipping of the Area Method averaging region near the loaded end . 59
5.1.2.1 Line Method sampling at the governing section . . . 63

6.2.1.1 Effect of embedment length on the baseline specimen ($d_b = 12\,\text{mm}$, $f'_c = 32\,\text{MPa}$, $r_e = 75\,\text{mm}$). (a) The predicted cracking load levels off by about 100 mm, gaining only 1.1 % over the remaining 200 mm. (b) The mean bond stress continues to fall as $1/L$, because a load that has stopped growing is being divided by a length that has not. . 74

6.2.2.1 Effect of bar diameter at fixed embedment ($L = 150\,\text{mm}$, $f'_c = 32\,\text{MPa}$, $r_e = 75\,\text{mm}$). (a) The cracking load rises almost in proportion to the diameter, following $d_b^{0.89}$. (b) The mean bond stress is nearly constant, falling only 12 % over a threefold increase in bar size, which is the same result seen another way. 75

6.2.3.1 Effect of concrete cover ($d_b = 12\,\text{mm}$, $L = 150\,\text{mm}$, $f'_c = 32\,\text{MPa}$) on (a) the cracking load and (b) the mean bond stress. The two open markers are the cases in which the averaging disc does not fit inside the concrete and is cut short at the outer face; they are excluded from the fit of (6.5). The remaining six points rise steadily. 77

6.3.1.1 Effect of concrete strength, with E_c following Standards Australia (2018) Table 3.1.2 ($d_b = 12\,\text{mm}$, $L = 150\,\text{mm}$, $r_e = 75\,\text{mm}$). (a) The predicted cracking load and (b) the mean bond stress, both rising as $(f'_c)^{0.390}$ and flattening gently as the strength increases. Because d_b and L are fixed throughout this sweep the two differ only by the constant bonded area. 79

6.3.2.1 Effect of concrete modulus at constant strength ($f'_c = 32\,\text{MPa}$, $d_b = 12\,\text{mm}$, $L = 150\,\text{mm}$, $r_e = 75\,\text{mm}$): (a) cracking load, (b) mean bond stress. Both *fall* as the concrete stiffens, because f_t and f_v are held constant and only the stress distribution changes. 80

7.1.1.1 Calibration curves: predicted cracking load against normalised embedment length for four bar diameters ($f'_c = 32\,\text{MPa}$, $d_{cyl} = 150\,\text{mm}$, $l_{char} = 20\,\text{mm}$, pull–push). 86

7.2.3.1 The calibration curves of Figure 7.1.1.1 replotted as F_u/F_∞ against L/L_0. The four families collapse onto $\tanh(L/L_0)$, reducing the spread from 88 to 12 %. 88

List of Tables

2.3.6.1 Simplified development length for deformed bars in tension (adapted from ACI 318-19 Table 25.4.2.3) 19

3.2.0.1 Zones of the three-layer idealisation. 29
3.9.0.1 Parameter matrix for the calibration sweep 41

4.2.1.1 Benchmark parameters, after Table 2 of Nemes and Lachaud (2009). 43
4.2.2.1 Shear stress τ_{rz} at the benchmark load, present model against Figure 3(b) of Nemes and Lachaud (2009). Tube values are taken at the mid-thickness of the layer, the adhesive value at the interface. 45
4.2.3.1 Circumferential stress σ_θ at the benchmark load, present model against Figure 4(b) of Nemes and Lachaud (2009). Tube values are taken at the mid-thickness of the layer, the adhesive value at the interface. . . 47
4.3.2.1 Benchmark parameters, after Hassanieh et al. (2018). Quantities marked † are not stated in the source and are assumed. 49
4.3.3.1 Present code against Figure 4 of Hassanieh et al. (2018), evaluated on the glue line. 50

5.1.2.1 Line Method sampling at the governing station $z = 0$, reference specimen of Table 5.2.1.1. The shear vanishes identically because $\tau_{rz} = 0$ is imposed at both ends of the bonded length. 62
5.1.4.1 The three critical-distance methods applied to the reference specimen of Table 5.2.1.1, $l_{\mathrm{ch}} = 20\,\mathrm{mm}$. Stresses and $\varphi_{\max}$ are at the production discretisation; the last column is the converged value from Section 5.2. 64
5.2.1.1 Reference specimen used throughout this chapter. 65
5.2.2.1 Line Method: convergence of the predicted Mode II capacity with the number of steps along the averaging line. Line length 40 mm; each run averages the stress at the n_{item} step centres, as in (5.3). . . 66
5.2.2.2 Line Method: convergence of the predicted Mode II capacity with the number of steps along the averaging line. Line length 40 mm; each run averages the stress at the n_{item} step centres, as in (5.3). . . 66
5.2.2.3 Converged Mode II capacities and the coarsest discretisation within 1% of each. 67
5.2.3.1 Computation time per evaluation of F_{II} against mesh refinement, with the discretisation error of Section 5.2.2 alongside. Times measured on an Intel Core i9-12900H (20 logical cores, 2.5 GHz, 32 GB RAM) running MATLAB R2026b under Windows 11. 68

6.1.1.1 Baseline specimen for the parametric study. . . . 71
6.2.1.1 Effect of embedment length ($d_b = 12\,\text{mm}$). . . . 73
6.2.2.1 Effect of bar diameter at fixed embedment length ($L = 150\,\text{mm}$). . . . 74
6.2.2.2 Effect of bar diameter at constant slenderness ($L/d_b = 10$, so the embedment varies with the bar). . . . 74
6.2.3.1 Effect of concrete cover ($d_b = 12\,\text{mm}$, $L = 150\,\text{mm}$). The first two rows are not comparable with the rest; see the text. . . . 76
6.2.4.1 Sensitivity to the assumed rib layer thickness. . . . 78
6.3.1.1 Effect of concrete strength, with E_c following Standards Australia (2018) Table 3.1.2. . . . 78
6.3.2.1 Effect of concrete modulus at constant strength ($f'_c = 32\,\text{MPa}$, so $f_t = f_v$ unchanged). . . . 80
6.3.3.1 Effect of the shear-to-tensile strength ratio ($f_t = 2.83\,\text{MPa}$ held). . . . 81
6.3.4.1 The three remaining factors, each swept over five values about the baseline. Changes are quoted across the range shown. . . . 82
6.4.1.1 Sensitivity of capacity and stiffness to every factor, ranked by $|n(F_{II})|$. Here n is the exponent of a power law $F_{II} \propto x^n$ fitted across each swept range, as defined in Section 6.1; the cover entries exclude the two clipped cases of Table 6.2.3.1. . . . 83

7.1.2.1 Effective embedment length, read from the calibration curves. . . . 87
7.2.2.1 Parameters of (7.1) fitted to each calibration curve. . . . 88
7.3.0.1 Fitted transfer length against the elastic decay length of the governing equation. . . . 89
7.5.1.1 Saturated capacity against the force required to yield the bar ($f_{sy} = 550\,\text{MPa}$, $f'_c = 32\,\text{MPa}$, $d_{cyl} = 150\,\text{mm}$, $l_{char} = 20\,\text{mm}$). . . . 91
7.6.0.1 Structure of the present expression against three code and research formulations. . . . 93

A.2.0.1 Least-squares working for $F_{II} \propto d_b^n$, from the bar-diameter sweep at $L = 150\,\text{mm}$. . . . 99
A.2.0.2 Residuals of the fit (A.6). . . . 99
A.2.0.3 Whether the averaging disc fits inside the concrete. Fourteen radial items are required. . . . 100
A.2.0.4 Least-squares working for $F_{II} \propto (c/d_b)^n$, six admissible points. . . . 100
A.2.0.5 The three fits behind the concrete-strength exponent. . . . 101
A.2.0.6 Local exponent on concrete strength, measured against the value predicted by (A.11) applied interval by interval. . . . 102
A.2.0.7 Least-squares working for $F_{II} \propto (f_v/f_t)^n$. . . . 102
A.2.0.8 The closed form (A.17) against the computed values and against the power-law fit. . . . 103
A.2.0.9 Power-law exponents for every factor, fitted by (A.2). A positive exponent means the output rises with the input. . . . 104

Chapter 1

Introduction

Chapter 2

Literature Review

2.1 Bond Theory and the Bond Model

2.1.1 Background on Bond

The structural performance of reinforced concrete fundamentally depends on the ability of embedded steel reinforcing bars to transfer forces to and from the surrounding concrete matrix. This force transfer mechanism, broadly referred to as 'bond', governs how tensile and compressive stresses develop along a bar's length, dictates anchorage and development length requirements, and influences crack spacing, crack width, and deformation behaviour at the member level. A thorough understanding of bond theory is therefore a prerequisite for the development of any bond model or analytically derived development length equation. This chapter reviews the fundamental definition and theoretical background of bond, the mechanisms of bond resistance, the modes of bond failure, and the principal predictive models that have been proposed in the literature.

Bond in reinforced concrete is defined as the interfacial interaction between the surface of an embedded reinforcing bar and the surrounding concrete that enables the transfer of longitudinal forces between the two materials. It is quantified as the bond stress (τ), defined as the shear stress per unit of nominal bar surface area acting parallel to the bar axis at the steel-concrete interface. For a bar of diameter d_b embedded over a length L and subject to a tensile force P, the average bond stress is given by:

$$\tau = \frac{P}{\pi d_b L} \tag{2.1}$$

In reality, bond stress is non-uniformly distributed along the anchorage length. It peaks near the loaded end of the bar and diminishes toward the free end, with the exact distribution depending on the local bond-slip constitutive relationship, the strain distribution in the bar, and the degree of confinement provided by the concrete cover and transverse reinforcement (Tepfers, 1973; fib Model Code 2010, 2013).

This non-uniformity has important implications for the definition of development length rather than being a simple product of average bond stress and perimeter, development length must be derived from integration of the local bond-slip law along the bar, a process that underpins the analytical development length approach pursued in the present thesis.

For deformed reinforcing bars, the total bond resistance between bar and concrete arises from three distinct physical mechanisms, each active at different stages of loading and slip (Zuo, Leon, et al. n.d.) (ACI 408R-03, 2003; Tepfers, 1973):

- **Chemical adhesion** – molecular attraction between the concrete and the steel surface; significant only at the very first stage of loading when slip is negligible.
- **Mechanical interlock** – bearing of the ribs (or deformations) on the concrete "keys" that form between adjacent ribs. For deformed bars, mechanical interlock dominates the bond capacity, providing far higher resistance than adhesion or friction alone.
- **Friction** – resistance generated by the normal pressure acting on the bar surface and by the roughness of the concrete.

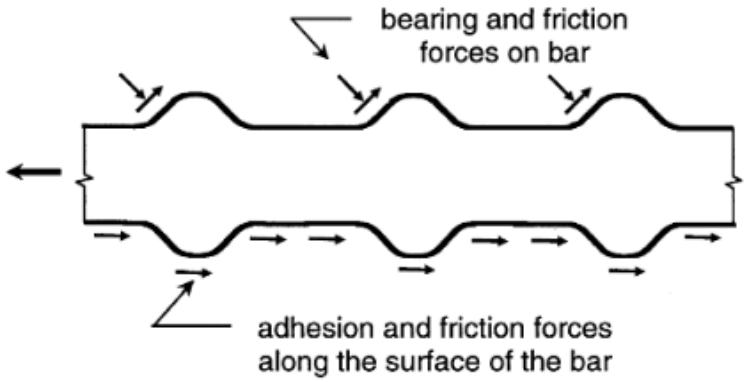


Figure 2.1.1.1: Schematic of bond transfer mechanisms in reinforced concrete (ACI 408R-03, 2003).

The relative magnitudes of these three components, and the slip levels at which they are mobilised, vary with bar surface geometry, concrete strength, cover, and confinement. For practical reinforced concrete design with deformed bars, mechanical interlock accounts for the large majority of total bond resistance and is the mechanism captured by the constitutive parameters in bond-slip models such as the fib Model Code 2010.

2.1.2 Failure Modes

2.1.2.1 Pull-Out Failure

Pull-out failure occurs when the concrete keys between ribs are sheared off, allowing the bar to slide axially without splitting the cover. It governs under well-confined conditions, large cover ($c/d_b \gtrsim$ 2.5-3.0) or dense transverse reinforcement, where hoop stresses are suppressed below the splitting threshold. The response is ductile: bond stress peaks at a slip of approximately 1–3 mm then declines gradually to a friction residual. The fib MC 2010 expresses peak bond stress as $\tau_{max} = 2.5\sqrt{f_c}$ (MPa) for well-confined bars. Foster et al. (2024) observed pull-out as the governing mode in all 54 SENSE600 Stage II specimens at $20d_b$ with helical confinement.

2.1.2.2 Splitting Failure

Splitting failure occurs when radial rib bearing forces generate hoop tensile stresses that exceed the concrete tensile capacity, driving longitudinal cracks through the cover. Once the crack reaches the external surface, bond resistance drops suddenly and the failure is brittle. This mode dominates in practice where cover is limited and transverse reinforcement is sparse, the typical situation in slabs and footings. Two geometries are recognised depending on bar arrangement: face-and-side splitting, governed by the minimum of side cover and half the clear bar spacing; and bottom splitting, governed by bottom cover. AS3600 and Eurocode 2 select the governing dimension accordingly. The transition between modes is gradual: increasing cover or reducing bar diameter favours pull-out, while very high-strength concrete can promote splitting at lower cover ratios due to the brittle fracture of the matrix (Harajli, 2004).

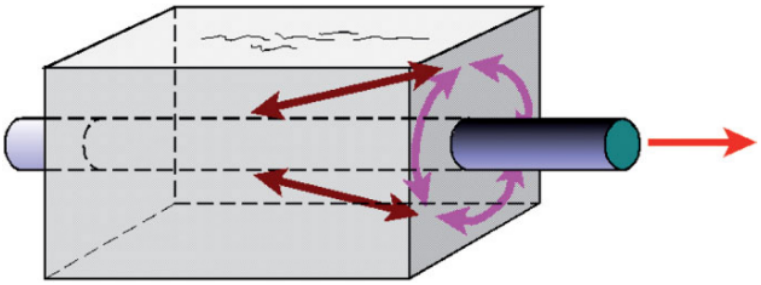


Figure 2.1.2.1: Schematic of splitting failure in reinforced concrete (Cairns, 2013).

2.1.3 Factors Influencing Bond Strength

Bond strength is governed by the interaction of several material and geometric factors. This section summarises the key factors identified consistently across the experimental literature, with reference to Li et al. (2012), whose systematic program on Australian 500N bars provides the most comprehensive single-source dataset for the Australian context and the direct experimental baseline for SENSE600 comparison.

2.1.3.1 Concrete Compressive Strength

Bond strength increases with concrete compressive strength, as higher-strength concrete provides greater resistance to both rib bearing (pull-out capacity) and hoop tension (splitting resistance). However, this relationship is non-linear: as concrete grade increases beyond approximately 40 MPa, the incremental benefit on bond diminishes because the fracture mode transitions from aggregate debonding toward aggregate splitting, where the matrix toughness grows more slowly than compressive strength. Darwin (2005) and ACI 408R-03 (2003) demonstrated that ${f'_c}^{0.25}$ better represents the concrete contribution to bond than the traditionally used ${f'_c}^{0.5}$ for unconfined bars across the full practical range of concrete strengths. Li et al. (2012) confirmed this in their regression for Australian 500N bars:

$$\tau_u = 23.315 {f'_c}^{0.25} - 37.579 \tag{2.2}$$

using the fourth-root term throughout their analysis of all five influencing factors.

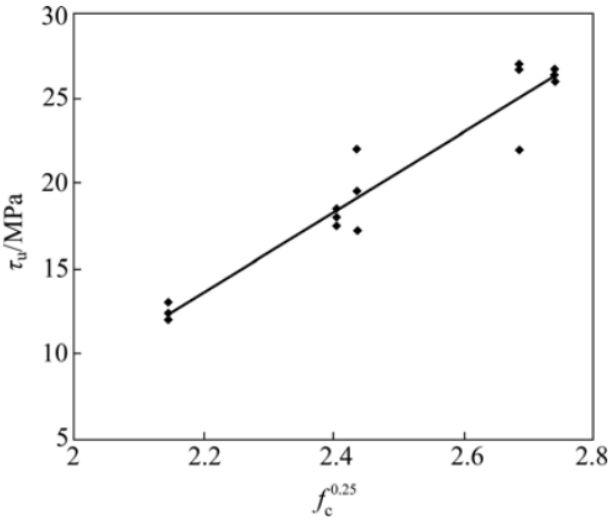


Figure 2.1.3.1: Relationship between concrete tensile strength and bond strength (Li et al. 2012).

2.1.3.2 Bar Diameter

Bond strength per unit area decreases with increasing bar diameter. Larger bars carry proportionally greater tensile forces relative to their bonded surface area, generate more non-uniform bond stress distributions over longer required embedment lengths, and produce greater radial bearing forces that increase the risk of splitting. Li et al. (2012) quantified this for Australian 500N bars through their Group BL tests across diameters from 8 to 36 mm, expressing the normalised bond strength as:

$$\frac{\tau_u}{f_c'^{0.25}} = \frac{83.043}{d^{0.5592}} \tag{2.3}$$

This size effect is codified through the k_2 factor in AS3600 and the η_3 factor in fib MC2010.

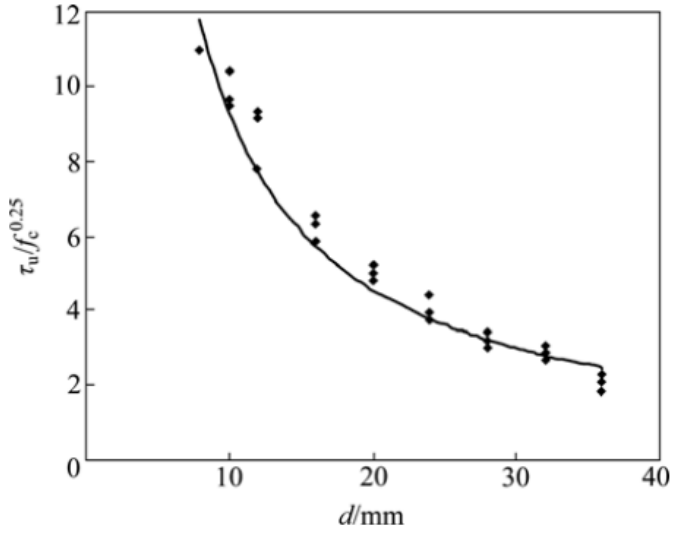


Figure 2.1.3.2: Relationship between concrete tensile strength and bar diameter (Li et al. 2012).

2.1.3.3 Concrete Cover

Increasing concrete cover raises bond strength by providing greater confinement against the radial bearing forces of the ribs, delaying the initiation and propagation of splitting cracks and allowing more of the concrete's tensile capacity to be mobilised before failure. Li et al. (2012) demonstrated a clear and consistent increase in normalised bond strength with cover in their Group CL tests:

$$\frac{\tau_u}{f_c'^{0.25}} = 0.191c + 1.60 \tag{2.4}$$

Critically, when bar diameter and cover were varied simultaneously in their Group DL tests, cover was found to have a stronger influence on bond strength than bar diameter meaning that generous cover can partially offset the adverse size effect of larger bars. This finding directly supports the expectation that SENSE600 bars, which are smaller in diameter than the equivalent-force 500 MPa bars they replace, will perform favourably under unchanged cover conditions.

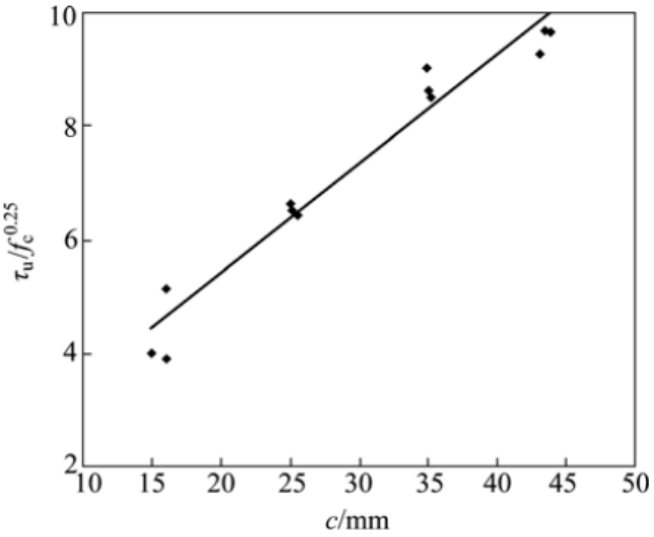


Figure 2.1.3.3: Relationship between concrete tensile strength and concrete cover (Li et al. 2012).

2.1.3.4 Embedment Length

Average bond stress decreases as embedment length increases, because bond stress is non-uniformly distributed and the more remote portions of the bar contribute less to resistance. Li et al. (2012) confirmed this inverse relationship through their Group EL tests, expressing it as:

$$\frac{\tau_u}{f_c'^{0.25}} = \frac{16.908d}{l_a} + 4.8974 \tag{2.5}$$

They also noted the important practical upper limit: if embedment is too long, the bar yields or fractures before bond failure, the fully developed condition that represents the design objective of development length provisions.

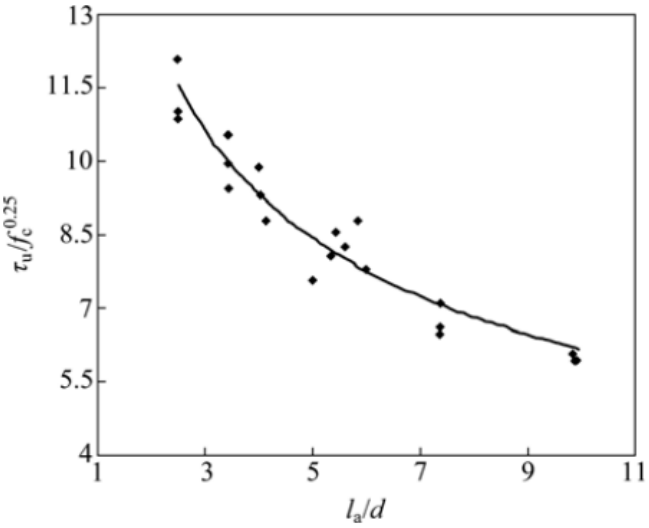


Figure 2.1.3.4: Relationship between concrete tensile strength and embedment length (Li et al. 2012).

2.1.3.5 Transverse Confinement

Transverse reinforcement (stirrups or helices) confines the propagation of longitudinal splitting cracks, resisting both vertical and horizontal splitting and allowing higher rib bearing forces to develop before failure. Li et al. (2012) demonstrated that stirrups can significantly enhance bond strength, expressing the relationship as:

$$\frac{\tau_u}{f_c'^{0.75}} = 33.057\frac{A_{sv1}}{cs_{sv}} + 0.8852 \tag{2.6}$$

where A_{sv1} is the stirrup bar area, c is cover, and s_{sv} is stirrup spacing. The use of $f_c'^{0.75}$ rather than $f_c'^{0.25}$ for unconfined bars, reflects the ACI 408R-03 finding that transverse reinforcement engages the concrete in a stress state where strength scales with a higher power of compressive strength. An upper limit exists beyond which additional confinement produces no further bond strength increase, and bar pull-out or fracture governs instead.

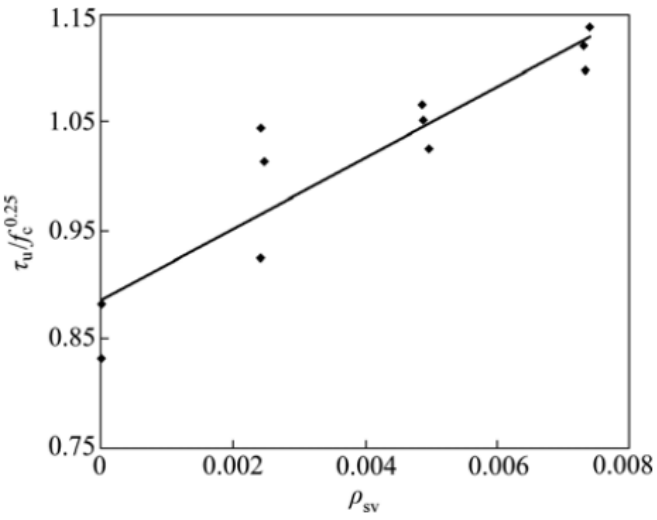


Figure 2.1.3.5: Relationship between concrete tensile strength and transverse confinement (Li et al. 2012).

2.1.3.6 Bar Surface Geometry and Relative Rib Area

The relative rib area f_R, the ratio of projected rib area perpendicular to the bar axis to the nominal bar surface area, quantifies the mechanical interlock efficiency of a deformed bar. Higher f_R produces greater mechanical interlock per unit slip, increasing peak bond stress and reducing required development lengths. AS/NZS 4671:2019 specifies minimum f_R values for Class N bars as a function of diameter. The SENSE600 bar's rib pattern was designed to exceed this minimum, providing a positive bond performance margin. Pauletta et al. (2020) demonstrated that bars with different rib profiles, even within the broad category of deformed steel, cannot be assumed to follow the same bond-slip parameters without experimental recalibration, reinforcing the need for SENSE600-specific characterisation.

2.1.4 Prediction Models

The following models are reviewed in chronological order. They are selected because they form a direct theoretical and methodological lineage: Orangun, Jirsa and Breen (1977) provide the empirical regression framework that underpins all code development length expressions; Darwin et al./ACI 408R-03 (2003) provide the reliability-calibrated database analysis that identifies the limitations of the Orangun model and motivates key refinements; fib Model Code 2010 (Cairns, 2015) provides the current international benchmark that unifies the preceding work into a comprehensive semi-mechanical model; and Andreasen and Nielsen (1989) provide the plasticity-based anchorage framework directly used by Foster et al. (2024) to derive the SENSE600 development length equation, and which will be extended in the present thesis. Together, these five models encompass the full progression from empirical observation to mechanistically grounded prediction, and each build on the limitations of its predecessor.

2.1.4.1 Orangun, Jirsa and Breen (1977) - The Foundational Empirical Regression

Orangun, Jirsa and Breen (1977) represented the pivotal transition from working-stress allowable bond values to strength-based development length design. Using a large database of beam splice and pull-out tests, they derived a multivariate regression expressing normalised bond strength as:

$$\frac{u}{\sqrt{f'_c}} = 0.1 + 0.25\frac{c}{d_b} + 4.15\frac{d_b}{l_d} + \frac{A_{tr}f_{tr}}{500sd_b} \tag{2.7}$$

where u is average bond stress at failure (MPa), c is the governing cover dimension (mm), l_d is development or splice length (mm), A_{tr} is the total area of transverse reinforcement within spacing s (mm), and f_{tr} is the transverse reinforcement yield strength (MPa). The four terms capture: a baseline bond capacity; the benefit of cover (consistent with Tepfers' prediction); the inverse relationship between embedment length and average bond stress; and the confinement benefit of transverse reinforcement. Rearranging to solve for l_d directly yielded the structural template for ACI 318 and AS3600 development length provisions that remains in use today.

Orangun et al. also established that development length should be designed with a capacity-reduction philosophy, treated as a strength limit state analogous to flexure or shear, rather than simply by inflating bar stress. The key limitation is the use of $\sqrt{f'_c}$ as the concrete term, which overestimates the concrete contribution to bond for f'_c above approximately 40 MPa. This is addressed by Darwin et al. in Section 1.4.2.

2.1.4.2 Darwin et al. / ACI Committee 408 (2003) - Database Reliability

The work of Darwin and colleagues, culminating in ACI 408R-03 (2003), represents the most thorough analytical audit of the empirical development length literature. Compiling over 1,000 beam splice and pull-out specimens, the Committee applied formal reliability analysis to the ACI 318 provisions and identified three important findings.

First, and most significant for the present thesis: the concrete contribution to bond is more accurately represented by $f_c'^{0.25}$ than by $\sqrt{f'_c}$. Darwin (2005) showed through regression analysis of the full dataset that $f_c'^{0.25}$ produces a substantially tighter and less biased fit across the range of 17–110 MPa. The physical explanation is that at higher concrete grades, fracture transitions from aggregate debonding (where strength scales with $\sqrt{f'_c}$.) toward aggregate splitting (where matrix toughness grows more slowly), saturating the bond benefit of higher-strength concrete. Li et al. (2012) adopted this recommendation directly, using $f_c'^{0.25}$ throughout all their regression expressions for Australian 500N bars.

Second, for bars confined by transverse reinforcement, the concrete contribution scales with a higher power, between $f_c'^{0.75}$ and $f_c'^{1.0}$, because confinement engages the

matrix in a stress state where aggregate friction and compression are more intensively mobilised. Li et al. (2012) reflected this in their stirrup regression by using $f_c'^{0.75}$ for confined specimens, consistent with the Committee's recommendation.

Third, the ACI 318 provisions were shown to provide inconsistent reliability across bar sizes and cover conditions, unconservative in some cases (e.g., the 0.8 bar size factor for smaller bars, which was subsequently removed) and excessively conservative in others. The report recommended introducing an explicit ϕ-factor for bond, analogous to those used for flexure and shear.

2.1.4.3 fib Model Code 2010 (Cairns, 2015) — Current International Benchmark

The fib Model Code 2010, reviewed by Cairns (2015), represents the most comprehensive bond model currently codified in an international standard. It synthesises the empirical foundations of Orangun et al., the analytical splitting framework of Tepfers, and the reliability insights of ACI 408R-03 into a unified model that explicitly accounts for concrete strength, bar surface geometry, cover, transverse reinforcement, steel grade, and transverse confining pressure.

The basic design bond strength $f_{bd,0}$ is defined for benchmark conditions and modified by factors $\eta_1 - \eta_4$:

$$f_{bd,0} \;=\; \eta_1\eta_2\eta_3\eta_4 \;\times\; (1/\gamma_c) \;\times\; \sqrt{f_{ck}} \tag{2.8}$$

where η_1 accounts for bond quality and casting position, η_2 for concrete strength relative to benchmark, η_3 for bar diameter (size effect), and most importantly η_4 for steel grade, ranging from 1.2 for 400 MPa steel down to 0.68 for 800 MPa steel. This explicit η_4 factor is the most significant analytical innovation of fib MC2010 relative to Orangun et al. and ACI 408R-03: it recognises that higher-grade bars have lower bond efficiency because the average bond stress mobilised over the longer required development length is reduced. For the present thesis, η_4 provides analytical justification for why a purpose-built bond model is needed for SENSE600 bars, the bond efficiency of a 600 MPa bar cannot be inferred from 500 MPa data by simple yield stress scaling.

The above equation was then calibrated using the ACI408 bond test database and validated using additional data gathers from fib TG4.5 to get a mean strength expression:

$$\begin{aligned} f_{\text{stm}} &= 54\left(\frac{f_{\text{cm}}}{25}\right)^{0.25}\left(\frac{25}{\phi}\right)^{0.2}\left(\frac{l_b}{\phi}\right)^{0.55}\left[\left(\frac{c_{\min}}{\phi}\right)^{0.25}\left(\frac{c_{\max}}{c_{\min}}\right)^{0.1} + k_m K_{\text{tr}}\right] \\ &\leq f_y \\ &\leq 10\sqrt{f_{\text{cm}}}\,\frac{l_b}{\phi} \end{aligned} \tag{2.9}$$

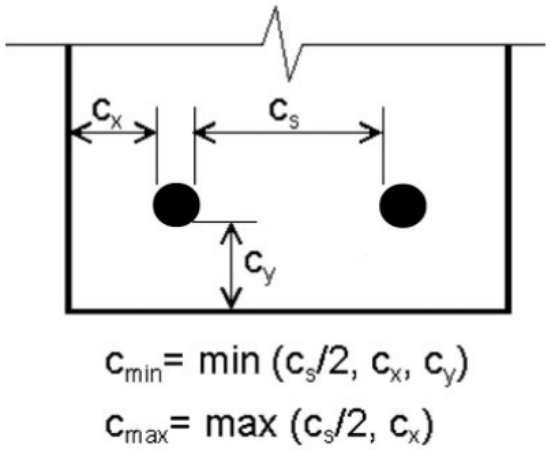


Figure 2.1.4.1: Definition of cover dimensions for fib MC2010 bond model (Cairns 2015).

2.2 History of 500 MPa Reinforcement

2.2.1 Study on Australian 500N Bars

Li et al. (2012) conducted a systematic experimental program characterising the tensile bond and anchorage properties of Australian-manufactured 500N Class reinforcing bars in concrete with compressive strengths from 25 to 65 MPa, using both pull-out specimens and beam anchorage tests. Their results demonstrated that AS 3600:2009 provisions were generally conservative for 500N bars across the tested range of concrete strengths and cover conditions, but that the degree of conservatism was variable, in some cases less than 10%, which is insufficient for robust design given the inherent scatter in bond test data. The study also confirmed that the square root of compressive strength $\left(\sqrt{f'_c}\right)$ adequately described the concrete contribution to bond for the 500N bars tested, though the authors noted that the fourth-root representation might be more accurate for higher concrete strengths.

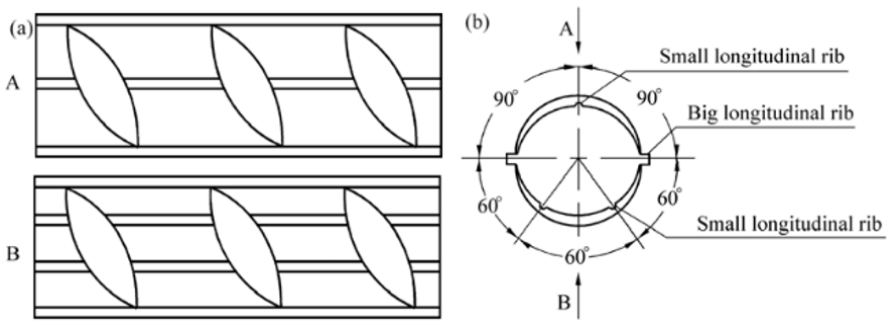


Figure 2.2.1.1: Rib pattern of 500N Class reinforcing bars (Li et al. 2012).

This work remains the primary Australian experimental reference for 500 MPa bar bond behaviour and establishes the direct benchmark against which SENSE600

bond performance should be compared. The experimental methodology employed by Li et al., pull-out cylinders with helical confinement at multiple embedment lengths, across a range of concrete strengths, closely resembles the approach adopted by Foster et al. (2024) for SENSE600, enabling a direct comparison of bond performance between the two grades.

2.2.2 Splice Strength and Lap Length for 500 MPa Bars

Canbay and Frosch investigated the bond strength of lap-spliced bars, analysing existing test data to assess the reliability of ACI 318 lap splice provisions. They concluded that the modification factors used in ACI 318 for lap splices were based on limited and heterogeneous data and recommended a more unified approach in which splice strength is expressed directly as a function of development length without separate empirical multipliers. While their work was focused on 500 MPa class steel, the methodological framework is directly applicable to the derivation of lap splice provisions for SENSE600 bars. Similarly, Cairns (2015) reviewed the lap splice provisions in fib MC2010, noting that the model provides a more mechanistically transparent framework for splice design than empirical code multipliers, and that its calibration for higher steel grades required further experimental data.

Ahlborn, French and Shield (2000) demonstrated that bars with high relative rib area consistently achieved higher splice strengths than conventional bars of the same diameter, and that the advantage was most pronounced at shorter splice lengths — a finding with practical significance for the design of heavily congested connection zones where the shorter absolute lengths achievable with SENSE600 bars would be most beneficial.

2.2.3 International Standards for 500 MPa and Beyond

Internationally, the treatment of 500 MPa reinforcement in design codes has followed a consistent pattern: development length expressions from lower-grade calibration databases are scaled to 500 MPa using yield stress ratios, with the scaling validated through targeted experimental programs. Eurocode 2 (EN 1992-1-1:2004) accomodates yield strengths up to 600 MPa within its general development length framework, using the design bond stress f_{bd} as the concrete capacity term and applying modification factors for cover and confinement. The fib Model Code 2010 similarly provides a general framework applicable to a range of yield strengths, though its bond-slip model parameters were primarily calibrated for 500 MPa class steel. ACI 318-19 provides development length expressions up to 550 MPa, beyond which code provisions are not directly applicable and independent experimental validation is required.

2.3 Empirical Work on Development Lengths

Empirical work on development length has progressed from simple average bond-stress concepts to more refined code-based formulations that attempt to represent

confinement, bar geometry, steel grade, and stress state. Across the literature, the common mechanical basis is that development length is the embedment required to transfer steel force into the surrounding concrete without bond failure. Cairns (2015) notes that this can be expressed through the average bond relation:

$$f_b = \frac{\Delta f_s A_s}{\pi \phi l_b} \tag{2.10}$$

but stresses that this is only a convenient simplification because actual bond transfer is governed by rib bearing, splitting resistance, cover, spacing, and confinement rather than by a uniform interface shear stress alone. This chapter reviews what development length is, traces the progression of empirical and analytical models from the classic Orangun et al. regression to the current major code provisions, and concludes with the most recent Australian contribution, the Foster et al. (2024) development length equation for SENSE600 bars.

2.3.1 What is development length?

Development length is the minimum embedment length required for a reinforcing bar to develop its design yield strength at a critical section without bond failure occurring between the bar and the surrounding concrete. It is the macroscopic expression of the bond-slip constitutive relationship, translated into a single design length through integration of the local bond resistance over the bar's perimeter and embedment length. The conceptual basis is straightforward: at a critical section, the reinforcing bar must carry its full design stress (f_{sy}). For this stress to be developed, the force in the bar ($A_s \times f_{sy}$) must be transferred into the concrete through bond over the available embedment length. If the embedment is shorter than the development length, bond failure will occur before yield is reached, and the bar is unable to fulfil its structural role. Mathematically, the development length is derived from the equilibrium of the bar force and the total bond resistance:

$$A_s \times f_{sy} = \tau_{avg} \times \pi \times d_b \times L_{sy.t} \tag{2.11}$$

Rearranging, the development length is:

$$L_{sy.t} = \frac{A_s \times f_{sy}}{\tau_{avg} \times \pi \times d_b} = \frac{d_b \times f_{sy}}{4\tau_{avg}} \tag{2.12}$$

The challenge in code development has always been the accurate determination of τ_{avg} the average bond stress mobilised over the development length at the point of bar yielding, which depends on concrete strength, cover, bar geometry, and confinement in ways that are not captured by a single constant. The entire history of development length research can be read as successive attempts to improve the prediction of τ_{avg} across the full range of design conditions.

A useful way to frame the analytical literature is to distinguish between older empirical development-length expressions and newer code formats derived from semi-mechanical bond models. The classic Orangun, Jirsa and Breen (1977) work remains important because it explicitly linked development length to bar stress, concrete strength, bar diameter, cover, and transverse reinforcement. They argued that development length should be treated with a capacity-reduction philosophy — not simply by inflating bar stress — because bond is a strength problem like flexure or shear. Their proposed regression directly informed ACI 318 and remains the methodological ancestor of all current code provisions.

2.3.2 fib Model Code 2010

Within the fib framework, the 2010 Model Code represents a significant analytical shift toward a mechanistically grounded bond model. Cairns (2015) explains that fib MC2010 first defines a basic design bond strength $f_{bd,0}$ for benchmark conditions and then modifies it for cover, transverse reinforcement, and transverse pressure. One of the most important changes relative to earlier codes is the explicit inclusion of steel grade through factor η_4, with values reducing from 1.2 for 400 MPa steel to 0.68 for 800 MPa steel. This reflects the physical reality that as yield strength increases, the development length required to reach bar yield also increases, so the average bond stress mobilised over that longer length is lower. In other words, fib MC2010 recognises analytically that higher steel grades cannot be treated using the same bond benchmark as lower-grade bars, a principle directly relevant to the SENSE600 context.

For straight bars in tension, Cairns (2015) reproduces the fib Model Code 2010 design bond strength as Equation 9:

$$f_{bd} \;=\; (\alpha_2 \;+\; \alpha_3)\, f_{bd,0} \;-\; 2p_{tr} < \; 2.0 f_{bd,0} \;-\; 0.4 p_{tr} \;\leq\; 1.5/\gamma_c \sqrt{f_{ck}} \qquad (2.13)$$

where α_2 accounts for the effect of concrete cover and bar spacing on the splitting resistance, α_3 accounts for the contribution of transverse reinforcement, $f_{bd,0}$ is the basic design bond strength for benchmark conditions (proportional to $\sqrt{f_{ck}}/\gamma_c$, with the steel grade factor η_4 incorporated), and p_{tr} is the average compressive stress perpendicular to the bar axis (taken as positive in compression). Because transverse compression increases confining pressure and thereby enhances bond strength, a compressive p_{tr} reduces the required anchorage length, one of the key conceptual advances in fib MC2010 relative to its predecessors.

Cairns (2015) argues that this is one of the most important conceptual changes in fib MC2010: shorter anchorages are no longer justified by a blanket distinction between anchorage and lap splice conditions, but by whether a beneficial transverse compression field is present. This distinction has direct practical relevance for the design of SENSE600 bar anchorages in regions of transverse compression (e.g., within beam-column joints or at well-confined supports) versus regions without beneficial compression (e.g., simply supported beam ends with minimum cover). The

basic design anchorage length in fib MC2010 is then derived from the equilibrium condition:

$$l_{bd} = \frac{A_s \ f_{sd}}{\pi \ d_b \ f_{bd}} \tag{2.14}$$

where f_{sd} is the design bar stress. For bars stressed to their design yield strength, this gives

$$l_{bd} = \frac{d_b f_{yd}}{4 f_{bd}} \tag{2.15}$$

which is the direct application of the average bond stress concept introduced in Section 3.1, but with f_{bd} now rigorously derived from the bond-slip model and the confinement conditions of the specific design situation.

2.3.3 Eurocode 2 (EN 1992-1-1:2004, Clause 8.4.4)

Eurocode 2 (EN 1992-1-1:2004) treats the design anchorage length for straight bars in tension through a clearly structured sequence in Clause 8.4, culminating in the design anchorage length expression in Clause 8.4.4. The approach retains the classical 'basic required length plus modification factors' framework but embeds the same underlying bond mechanics as fib MC2010.

The starting point is the ultimate design bond stress (Clause 8.4.2):

$$f_{bd} = 2.25 \ \eta_1 \eta_2 f_{ctd} \tag{2.16}$$

where η_1 accounts for bond conditions and casting position (1.0 for good bond; 0.7 for poor bond, including top-cast bars with more than 250 mm of concrete below), η_2 accounts for bar diameter (1.0 for $d_b \leq 32$ mm; $(132 - d_b)/100$ for $d_b > 32$ mm), and f_{ctd} is the design tensile strength of concrete. The use of concrete tensile strength, rather than compressive strength, reflects the mechanistic reality that splitting failure, which governs bond in most practical situations, is fundamentally a tensile fracture problem.

The basic required anchorage length (Clause 8.4.3) for a straight bar stressed to its design value σ_{sd} is then:

$$l_{b,rqd} = \frac{d_b}{4} \times \frac{\sigma_{sd}}{f_{bd}} \tag{2.17}$$

The design anchorage length (Clause 8.4.4) applies a set of modification factors to $l_{b,rqd}$:

$$l_{bd} = \ \alpha_1 \alpha_2 \alpha_3 \alpha_4 \alpha_5 \times \ l_{b,rqd} \ \geq l_{b,min} \tag{2.18}$$

where: α_1 accounts for bar shape (1.0 for straight bars); α_2 accounts for the effect of minimum concrete cover (c_d/d_b, limited to a minimum product $\alpha_2\alpha_3\alpha_5 \geq 0.7$); α_3 accounts for confinement by transverse reinforcement not welded to the main bar; α_4 accounts for welded transverse bars along the design anchorage length; and α_5 accounts for the effect of transverse compressive pressure along the design anchorage length. The minimum anchorage length is:

$$l_{b,min} = max\{\ 0.3 \times l_{b,rqd}\ ;\ 10d_b\ ; 100\ mm\} \tag{2.19}$$

This minimum ensures a practical lower bound on anchorage length regardless of the degree of confinement. The Eurocode framework is applicable for yield strengths up to 600 MPa and is therefore directly relevant to SENSE600 bar design in Australian bridge projects governed by European-derived specifications. Its explicit use of concrete tensile strength in the bond stress expression aligns it more closely with the mechanics of splitting failure than the $\sqrt{f_c'}$ formulations used in ACI 318 and AS 3600.

2.3.4 AS3600:2018 - Anchorage Chapter

AS 3600:2018 Clause 13.1 governs development length in Australian building practice. The tension development length is:

$$L_{sy.tb} \ = \ \frac{0.5k_1k_3f_{sy}\ d_b}{k_2\sqrt{f_c'}} \ \geq \ 0.058f_{sy}k_1d_b \tag{2.20}$$

Where:

$k_1 = 1.3$ for top-cast bars (> 300 mm of concrete below), 1.0 otherwise $k_2 = (132$ - $d_b)/100$ $k_3 = 1.0$ - $0.15(c_d$ - $d_b)/\ d_b$ (cover factor, $0.7 \leq k_3 \leq 1.0$).

Compression development is shorter because end bearing contributes to force transfer; no bar position factor applies. Both expressions use $\sqrt{f_c'}$ despite ACI 408R-03 and Li et al. (2012) showing $f_c'^{0.25}$ is more accurate above 40 MPa. Ng et al. (2024) showed that substituting equivalent-capacity SENSE600 bars into the tension expression increases predicted development lengths by 2.5 – 9.5%, because the higher yield strength (×1.2) outweighs the reduced diameter (×0.913), the analytical finding that motivated Foster et al. (2024) to develop an improved-rib bar and purpose-built equation.

2.3.5 AS5100.5:2017 - Anchorage Chapter

AS5100.5 governs concrete bridge design in Australia and adopts the same core tension development length expression as AS3600:

$$L_{sy.tb} \ = \ \frac{0.5k_1k_3f_{sy}\ d_b}{k_2\sqrt{f_c'}} \ \geq \ 0.058f_{sy}k_1d_b \tag{2.21}$$

Section 13 was substantially revised in 2017 to treat development length as a discrete analytical design problem, with separate clauses for tension, compression, headed bars, mesh, bundled bars, and lap splices, mirroring the AS 3600 structure. The bridge code supplements these provisions with requirements for fatigue, durability, and stress-reversal detailing relevant to live load cycling. For SENSE600 applications, the same equivalent-capacity substitution approach permitted under the 2024 amendments to AS 3600 and AS/NZS 4671 would apply.

2.3.6 ACI 318-19 — Chapter 25.4 and Table 25.4.2.3

ACI 318-19 places development length provisions for deformed bars in tension in Chapter 25.4.2, with the detailed tabulated expressions in Table 25.4.2.3. The ACI approach uses a simplified two-formula format depending on bar size and confinement conditions, derived from the Orangun et al. (1977) regression database and refined through the ACI 408R-03 reliability analysis. The general expression for development length is:

$$l_d = \frac{3 f_y \psi_t \psi_e \psi_s}{40\lambda \sqrt{f'_c} \times \frac{c_b + K_{tr}}{d_b}} \times d_b \tag{2.22}$$

where ψ_t is the reinforcement location factor (1.3 for top bars; 1.0 otherwise), ψ_e is the epoxy coating factor (1.5 for coated bars with cover $< 3d_b$ or clear spacing $< 6d_b$; 1.2 for other coated bars; 1.0 for uncoated), ψ_s is the bar size factor (0.8 for No. 6 (19 mm) and smaller; 1.0 for larger), λ accounts for lightweight concrete (0.75 for all-lightweight; 0.85 for sand-lightweight; 1.0 for normalweight), and $(c_b + K_{tr})/d_b$ is the confinement term, limited to a maximum of 2.5, where c_b is the smaller of the distance from bar centre to nearest concrete surface and half the centre-to-centre bar spacing, and $K_{tr} = 40A_{tr}/sn$ is the transverse reinforcement index.

Table 25.4.2.3 of ACI 318-19 simplifies this expression into a tabulated format for two common design cases (a) bars with sufficient cover and spacing (or with minimum stirrups) and (b) other cases as reproduced in Table 1.3.1 below:

Table 2.3.6.1: Simplified development length for deformed bars in tension (adapted from ACI 318-19 Table 25.4.2.3)

Condition	Clear spacing $\geq d_b$, cover $\geq d_b$, stirrups $\geq$ minimum	Clear spacing $\geq 2d_b$, cover $\geq d_b$ (no stirrups required)	**Other cases**
$d_b \leq 19$ mm (No. 6 bar and smaller)	$l_d = \frac{f_y \psi_t \psi_e}{25\lambda\sqrt{f'_c}} \times d_b$	$l_d = \frac{f_y \psi_t \psi_e}{25\lambda\sqrt{f'_c}} \times d_b$	$l_d = \frac{3 f_y \psi_t \psi_e}{50\lambda\sqrt{f'_c}} \times d_b$
$d_b \geq 22$ mm (No. 7 bar and larger)	$l_d = \frac{3 f_y \psi_t \psi_e}{50\lambda\sqrt{f'_c}} \times d_b$	$l_d = \frac{3 f_y \psi_t \psi_e}{50\lambda\sqrt{f'_c}} \times d_b$	$l_d = \frac{9 f_y \psi_t \psi_e}{100\lambda\sqrt{f'_c}} \times d_b$

Adapted from ACI 318-19 Table 25.4.2.3. All expressions produce l_d in mm when f_y and f'_c are in MPa and d_b is in mm. Minimum $l_d = 300$ mm.

The ACI 318-19 provisions apply to bars with yield strengths up to 550 MPa (80 ksi). For higher-grade bars, ACI 318 currently requires that development length be determined through alternative methods, such as the ACI 408R approach or testing, because the code provisions have not been validated beyond this range. This gap directly parallels the situation for SENSE600 bars in Australian practice prior to the Foster et al. (2024) experimental program.

The ACI Committee 408 report (ACI 408R-03) provides important analytical context for the ACI 318 provisions. The report argues that bond is a structural property, not a simple material property, and that empirical development length equations must reflect confinement, bar geometry, and realistic splice lengths. ACI 408R-03 concludes that the concrete contribution to bond is better represented by $f_c'^{0.25}$ than by $f_c'^{0.5}$, and it rejects the 0.8 bar-size factor for smaller bars as unjustified once realistic development lengths are considered. These recommendations have not yet been adopted in ACI 318-19, but they are reflected in the Li et al. (2012) composite bond strength formula (where $f_c'^{0.25}$ is used) and are relevant to the calibration of the SENSE600 development length equation.

2.3.7 Foster et al. (2024) - SENSE600 Development Length Equation

Foster et al. (2024) represent the most recent and directly applicable contribution to the analytical development length literature for the Australian context. Building on the Andreasen and Nielsen (1989) plasticity-based anchorage model reviewed in Chapter 1, and validated against 54 pull-out tests on InfraBuild SENSE600 bars from 9.8 to 36.5 mm in diameter at a concrete strength of approximately 32.8 MPa, they proposed the following development length equation for Grade 600 SENSE600 bars:

$$L_{sy.tb} = \frac{270 k_1 k_3 d_b}{k_2 \sqrt{f'_c}} \geq 31.6 k_1 d_b \tag{2.23}$$

where k_1,k_2 and k_3 are as defined in AS 3600:2018. This equation adopts the existing AS 3600 modification factor framework to facilitate integration with Australian code practice, with the numerical coefficient 270 (compared to the AS 3600 coefficient of $0.5 \times f_{sy} = 0.5 \times 500 = 250$ for 500 MPa bars at benchmark conditions) reflecting the higher yield strength of the SENSE600 bar.

The analytical significance of the equation is considerable. Foster et al. demonstrated experimentally that an embedment of $16.2d_b$ is sufficient to develop the nominal yield strength of 600 MPa (with a factor of safety of 1.4 relative to the ϕ-reduced stress), while $23d_b$ consistently causes bar fracture, confirming that the design equation provides an appropriate margin between yield development and bar fracture. Direct comparison with AS 3600:2018 requirements for equivalent-capacity Grade 500 bars showed that smaller-diameter Grade 600 SENSE600 bars can replace Grade 500 bars without requiring longer absolute development lengths, maintaining similar factors of safety against bond failure and bar splitting.

However, Foster et al. explicitly acknowledged that the equation was derived from tests at a single concrete strength ($f_c\prime \approx 33MPa$) and a single transverse reinforcement configuration and does not capture the variation in bond performance with concrete strength, cover geometry, and bar size that a complete bond model would provide. This limitation is the central analytical gap addressed by the present thesis.

2.4 Analytical Work on Development Length

The empirical and code-based development length expressions reviewed in the previous section are fundamentally regression models: they distil large experimental datasets into compact formulae calibrated to a specific range of material grades, bar geometries, and concrete strengths. While this approach is practically convenient and has served the industry reliably for 500 MPa class reinforcement, it carries an inherent limitation when bar grade, rib geometry, or concrete strength fall outside the calibration range, there is no mechanistic basis for extrapolation. The development of a bond model and development length equation for SENSE600 bars confronts precisely this limitation. Meeting it requires an analytical framework capable of predicting bond behaviour from first principles, representing the geometry of the bar-matrix interface, the stress state of the surrounding material, and the physical conditions under which failure initiates.

2.4.1 The Volkersen Shear-Lag Model - Analytical Baseline

The earliest and most enduring analytical approach to force transfer in adhesively bonded and embedded rod systems is the shear-lag model proposed by Volkersen (1938). The model treats the embedded rod and its surrounding matrix as two elastic

adherends connected by a shear-only interface layer and derives the shear stress distribution along the bond length from a differential equilibrium argument. For a cylindrical bonded assembly, structurally analogous to a reinforcing bar embedded in concrete, Gustafsson and Serrano (2001) extended Volkersen's formulation to express the pull-out load capacity as:

$$\frac{P_f}{2\pi r L f_v^A} = \frac{G_f \omega E_s A_s}{\pi r L f_v^{2A}} \left(\frac{\tanh(\omega L)}{1 + E_s A_s / E_t A_t} \right) \tag{2.24}$$

where G_f is the fracture energy, ${f_v}^A$ is the bond interface shear strength, G_A the shear modulus of the interface layer, r_s and r_A the rod and interface radii, E_s and E_t the rod and matrix elastic moduli, and A_s and A_t the respective cross-sectional areas (Gustafsson and Serrano, 2001).

The shear stress concentration is highest at the loaded end and diminishes exponentially toward the free end, consistent with the experimental observation that bond failure initiates at the loaded end and propagates along the embedment (Alavi et al., 2025).

Despite its widespread adoption, the Volkersen model has well-documented limitations. Alavi et al. (2025) validated the model against pull-out tests on glued-in rod (GiR) timber joints across embedment lengths of 80–360 mm and found that the Volkersen model consistently overestimated pull-out capacity by margins of 35–122% relative to experimental failure loads. Three physical deficiencies account for this overestimation. First, the model ignores all deformation of the adherends in the radial direction, meaning the radial (splitting) component of the interface stress, responsible for splitting failure in both GiR joints and reinforced concrete, is entirely absent from the formulation (Nemes and Lachaud, 2009; Alavi et al., 2025). Second, the influence of adherend deformation and the non-linearity of the adhesive are ignored, making the model applicable only to joints with a uniform overlap length and relatively thin adherends or adhesives (Alavi et al., 2025). Third, the model does not consistently yield accurate results when the load levels associated with different failure modes are not well separated; that is, when pull-out and splitting failure occur at similar load levels, because a single shear-only interface representation cannot capture the mode interaction (Alavi et al., 2025). The Volkersen model therefore provides a useful analytical baseline and comparative benchmark but cannot serve as the foundation for a rigorous development length model that must capture the pull-out/splitting mode transition.

2.4.2 Elasticity-Based Cylindrical Stress Models

2.4.2.1 Nemes and Lachaud (2009) - Refined Cylindrical Assembly Model

A significant advance over the Volkersen shear-lag approach was made by Nemes and Lachaud (2009), who developed a refined analytical model for the stress distribution in cylindrical adhesively bonded assemblies based on the variational principle of minimum complementary energy. Their model formulates all stress components, axial (σ_{zz}), circumferential ($\sigma_{\theta\theta}$), radial (σ_{rr}), and shear (τ_{rz}), within the inner rod,

adhesive layer, and outer matrix as functions of the axial stress $\sigma_{zz}^{(1)}(z)$ in the inner tube, using equilibrium equations in a cylindrical coordinate system (z, r, θ) and boundary conditions at each material interface (Nemes and Lachaud, 2009). Table 1 of Nemes and Lachaud (2009) shows that prior models — including their own earlier formulation, had assumed the radial stress to be zero; their refined model was the first in the cylindrical assembly literature to explicitly include σ_{rr} as a non-trivial stress component.

Nemes and Lachaud (2009) assumed a linear variation of radial stress in each material:

$$\sigma_{rr}^{(1)} = \alpha_1 \left[r - r_i \right] \tag{2.25}$$

$$\sigma_{rr}^{(c)} = \beta = \text{constant} \tag{2.26}$$

$$\sigma_{rr}^{(2)} = \alpha_2 \left[r - r_e \right] \tag{2.27}$$

where r_i and r_e are the inner and outer radii of the adhesive layer, and α_1, α_2, and β are constants determined from boundary conditions. The axial stress in the inner rod is then related to the radial stress through the equilibrium equations, allowing for a complete resolution of the stress field in all three materials.

Substituting these radial stress expressions into the equilibrium equations and the potential energy integral, and applying variational calculus, yields a fourth order governing ordinary differential equation for $\sigma_{zz}^{(1)}(z)$ (Nemes and Lachaud, 2009):

$$E \, \frac{d^4 \sigma_{zz}}{dz^4} \; + \; (B - C) \, \frac{d^2 \sigma_{zz}}{dz^2} \; + \; A \, \sigma_{zz} \; + \frac{D}{2} \; + \frac{\alpha_1 k}{2} \; = \; 0 \tag{2.28}$$

where A, B, C, D and E are integration constants depending on the elastic moduli, Poisson's ratios and radii of all three materials, and k, h and m depend on the load and dimensional specifications of the assembly Nemes and Lachaud (2009). The general solution takes the form $\sigma_{zz}(z) = \sum_i B_i \exp(\omega_i z) + \text{constant}$, with integration constants determined from the loading boundary conditions at $z = 0$ and $z = L$.

The key result from the comparative study of Nemes and Lachaud (2009) was that incorporating radial stress has an important influence particularly on the distribution of the circumferential (hoop) stress $\sigma_{\theta\theta}$ in both the inner tube and the adhesive layer while having a more modest effect on the shear stress τ_{rz} distribution. Since hoop stress is the stress component directly responsible for splitting failure in both bonded rod systems and reinforced concrete, it is the hoop stress that drives longitudinal cracking through the cover in Tepfers' (1973) thick-walled cylinder model. This finding has a direct and important implication: any analytical bond model that ignores radial stress will misrepresent the component of the stress field most critical to the pull-out/splitting mode transition. The Nemes and Lachaud (2009) framework therefore provide the first analytical model suitable as a basis for predicting whether splitting or pull-out governs in a bonded cylindrical assembly.

2.4.2.2 Alavi, Valipour and Bradford (2025) — Extended Model for GiR Timber Joints

Alavi, Valipour, and Bradford (2025) built directly on the (Nemes and Lachaud 2009) framework to develop a comprehensive elasticity-based analytical model for single glued-in rod (GiR) timber joints subjected to pull-pull, pull-push, rod-pull, and wood-pull loading configurations. GiR joints, comprising a steel rod bonded into a timber element by adhesive, are structurally analogous to deformed reinforcing bars in concrete: in both systems a cylindrical rod embedded in a matrix material transfers axial force through interfacial shear and rib bearing. The analytical insights and solution methodology developed for GiR joints are therefore directly transferable to the reinforcing bar bond problem (Alavi, Valipour, and Bradford 2025).

The model formulated the governing equations in a cylindrical coordinate system, treating the steel bar as isotropic and the timber as transversely isotropic (with distinct longitudinal E_l and transverse E_t elastic moduli, shear modulus G_{12}, and Poisson's ratio ν_{12}). A key improvement over Nemes and Lachaud (2009) was the adoption of a non-linear radial stress distribution (Alavi, Valipour, and Bradford 2025):

$$\sigma_r^s(r) = \Lambda_1 r^2 \tag{2.29}$$

$$\sigma_r^A = \Lambda_2 \tag{2.30}$$

$$\sigma_r^t(r) = \Lambda_3 \left(r^2 - r_t^2\right) \tag{2.31}$$

where Λ_i are constants determined from radial stress continuity at the rod–adhesive and adhesive–timber interfaces (Alavi, Valipour, and Bradford 2025). Substituting these into the equilibrium equations yields the shear and hoop stress components in each material analytically as functions of the first and second derivatives of $\sigma_z^s(z)$. Introducing all stress components into the strain energy expression and applying the variational principle $\delta U = 0$ produces a fourth-order governing ODE for $\sigma_z^s(z)$ with coefficients Ξ_1 through Ξ_7 that incorporate the transversely isotropic timber properties (Alavi, Valipour, and Bradford 2025). The general solution is:

$$\sigma_z^s(z) = B_1 e^{\omega_1 z} + B_2 e^{-\omega_1 z} + B_3 e^{\omega_2 z} + B_4 e^{-\omega_2 z} + \wp \tag{2.32}$$

where ω_1 and ω_2 are the characteristic roots of the ODE and $\wp$ is a particular solution depending on material properties, geometry and loading conditions. The four integration constants B_1–B_4 are determined by applying boundary conditions corresponding to each of the four loading configurations (Alavi, Valipour, and Bradford 2025). Extending the analytical solution to all four loading configurations simultaneously was a key contribution, as prior analytical models had only addressed pull–pull and pull–push conditions.

The GiR joint stiffness is derived directly from the analytical solution via conservation of energy: setting external work equal to internal strain energy,

$$k_s = \frac{f}{\delta_z^s} = \frac{f^2}{2U} \tag{2.33}$$

yielding a closed-form expression for joint stiffness without the need for finite element analysis. Validation against detailed FE simulations showed that the non-linear radial stress formulation significantly improved accuracy relative to models assuming zero or linear radial stress. The proposed model achieved a stiffness prediction within 14% of experimental values across rod radii of 4–6 mm and embedment lengths of 80–360 mm, whereas the Volkersen model overestimated failure loads by 35–122% and the EN1995-1-1 code provisions grossly underestimated both failure load and stiffness (Alavi, Valipour, and Bradford 2025).

The analytical framework of Alavi, Valipour, and Bradford (2025)) is directly relevant to SENSE600 bond modelling for three reasons. First, the cylindrical geometry, isotropic steel rod, and force-transfer mechanism are structurally identical to those of a deformed reinforcing bar in concrete. Second, the non-linear radial stress formulation correctly captures the hoop stress that governs splitting failure — the dominant design concern for SENSE600 bars in slabs without transverse reinforcement. Third, the closed-form solution provides a basis for analytically deriving development length as a function of material properties and geometry, rather than empirical regression.

2.4.3 Fracture Mechanics Failure Criteria

2.5 Background on SENSE600 Bars and the Australian Context

2.5.1 Introduction of 600 MPa Reinforcement in Australia

The introduction of 600 MPa Class N reinforcing steel into Australian construction practice represents the latest step in the progressive evolution of reinforcement grade, following the transition from 400 to 500 MPa in the 1990s. In early 2024, following amendments to AS 3600:2018 and AS/NZS 4671:2019, 600 MPa Class N bars became available for structural use in Australia. The impetus for this development was primarily sustainability: by reducing bar cross-sectional area for the same tensile capacity, 600 MPa bars enable material savings of up to 17% and embodied carbon reductions of approximately 39% per tonne of steel placed, attributed to both lower mass and the use of recycled steel in electric arc furnace production (Foster et al., 2024).

The commercial product that has driven this development in Australia is the SENSE600 bar, manufactured by InfraBuild Australia. SENSE600 bars are produced in a range of diameters scaled to deliver equivalent force capacity ($A_s f_{sy}$) to standard 500 MPa bars of the next larger nominal size. The bars feature a proprietary optimised rib pattern, with increased relative rib area relative to standard 500 MPa Class N bars, designed to ensure that the smaller diameter SENSE600 bar achieves bond and development length performance equivalent to, or better than, the larger 500 MPa bar it replaces.

2.5.2 AS/NZS 4671:2019 and Bar Geometry Requirements

AS/NZS 4671:2019 specifies the mechanical and geometric requirements for reinforcing steel bars for structural use in Australia and New Zealand, including minimum yield strength, tensile strength, elongation at fracture, and relative rib area. For Class N bars, the minimum relative rib area $f_{R,min}$ increases with bar diameter, reflecting the greater embedment depth required for larger bars and the importance of maintaining adequate mechanical interlock. The SENSE600 bar's rib pattern was designed to exceed the minimum f_R requirement, providing a positive bond performance margin over the code minimum. Ng and McGregor (2024) confirmed through measurement that the SENSE600 bar meets and exceeds the AS/NZS 4671:2019 relative rib area requirements across all tested diameters.

2.5.3 Ng and McGregor (2024) - Stage I Experimental Program

Ng and McGregor (2024) reported the Stage I experimental investigation of bond performance for InfraBuild SENSE600 bars, comprising 18 pull-out tests on bars with nominal equivalent diameters of 14.6, 21.9, and 29.2 mm (corresponding to 500 MPa equivalent sizes of 16, 24, and 32 mm). Tests were conducted at embedment lengths ranging from $13d_b$ to $32d_b$ in 25 MPa concrete confined by 5.4 mm diameter helical reinforcement. The key outcome was that an embedment length of approximately $22d_b$ was required to achieve the code criterion of 0.2 mm free-end slip at the nominal yield stress of 600 MPa, while $32d_b$ consistently produced bar fracture, confirming full anchorage. The improved rib pattern was shown to deliver bond performance equivalent to standard 500 MPa bars of the equivalent force capacity, validating the design concept.

The Stage I program, while important as proof-of-concept, was limited by its use of a single concrete strength (nominally 25 MPa), a single transverse reinforcement configuration, and only three bar diameters. These limitations motivated the more comprehensive Stage II program reported by Foster et al. (2024) and underscore the need for a broader experimental database from which a generalised bond model can be derived.

2.5.4 Foster et al. (2024) - Stage II Experimental Program and Development Length Equation

Foster, Ng, McGregor, and Valipour Goudarzi (2024) presented the Stage II experimental program, extending the dataset to 54 pull-out specimens across six bar sizes (nominal diameters 9.8 to 36.5 mm), all tested at an embedment length of 20db in concrete with a mean compressive strength of 32.8 MPa, confined by 5.4 mm helical reinforcement. All specimens achieved or exceeded the nominal yield strength of 600 MPa at 20db, with stresses at 0.2 mm free-end slip ranging from 671 to 784 MPa — all above the mean 600 MPa yield threshold. At failure, bar stresses were at or near the ultimate tensile strength, confirming that 20db represents the transition point between bond failure and bar fracture for this configuration.

A plasticity-based analytical model, adapted from the Andreasen and Nielsen (1989) upper- and lower-bound plasticity framework, was calibrated against the combined Stage I and II dataset and showed good agreement with the experimental results. The model characterises anchorage strength as:

$$\sigma_b = \eta \ f_c \left(\frac{l_b}{d_b} \right) \tag{2.34}$$

where η is a non-dimensional parameter dependent on bar rib geometry, concrete efficiency, and confinement level. Based on the experimental program and analytical model, Foster et al. (2024) proposed the development length equation seen above. The equation adopts the existing AS 3600 framework to facilitate code integration and was validated at $f_c \approx 33MPa$.

The authors explicitly acknowledged the key limitation of the proposed equation: it is based on a single concrete strength and a single confinement configuration. A validated bond model capable of predicting development length across the full range of concrete strengths (25–65 MPa) and cover conditions encountered in Australian practice, and from which the development length equation can be derived analytically, remains an outstanding research need. This is the central gap that the present thesis aims to close.

2.5.5 Ng and McGregor (2024) - Changes to Australian Standards

In a complementary paper, Ng and McGregor (2024) reviewed the changes to AS 3600:2018 and AS/NZS 4671:2019 required to accommodate higher-strength reinforcing steels, situating the SENSE600 development within the broader context of international trends toward 600 MPa and higher-grade reinforcement. They documented the technical rationale for the equivalent force capacity substitution approach in which a 600 MPa bar of diameter d_b replaces a 500 MPa bar of diameter d_b and discussed its implications for development length, lap splice design, and structural detailing. The paper reinforced the conclusion that while the substitution approach is technically sound as a first approximation, purpose-built bond model and development length provisions for SENSE600 bars would provide a more rigorous and flexible design basis.

2.6 Summary and Research Gap

2.6.1 Summary

2.6.2 Research Gap

Chapter 3

Methodology

3.1 The bond pullout test

The configuration modelled throughout this chapter is the concentric pull-out test in its *pull–push* form, the most widely used test for bond between a deformed bar and concrete.

A single bar is cast centrally in a concrete prism or cylinder and protrudes from one face. The bar is gripped outside the specimen and loaded in direct tension, while the concrete bears against a stiff reaction plate at the *same* face. The bar is therefore pulled and the concrete pushed, from which the configuration takes its name. The specimen is self-equilibrated: at every section the tensile force remaining in the bar is balanced by an equal compressive force in the surrounding concrete, and no external restraint acts on the far end.

The bonded length L is not the full depth of the specimen. Plastic or PVC sleeves are bonded to the bar over the remainder of its embedded length so that bond develops only over a controlled zone, which allows L to be varied independently of specimen size. RC 6 specifies a bonded length of $5d_b$ with the bond-free length placed adjacent to the loaded face, so that the bonded zone lies outside the compression cone spreading from the reaction plate. Instrumentation comprises a load cell in line with the bar and displacement transducers measuring slip; free-end (unloaded-end) slip is normally preferred to loaded-end slip because it excludes the elastic extension of the bar between the grip and the specimen face.

Two consequences of this arrangement matter for the interpretation of the model and of the comparisons in Chapter [X].

The bearing reaction places the concrete surrounding the bonded length in *axial compression*, and the resulting transverse confinement suppresses splitting. Pull-out tests therefore return higher bond strengths, and are more likely to produce a pull-out rather than a splitting failure, than arrangements in which the concrete is in tension—a lapped splice, a beam-end test, or the pull–pull configuration treated by Alavi, Valipour, and Bradford (2025). Capacities derived from pull-out data are

consequently an upper bound on what is available in a tension member, and the model inherits that bias.

The short bonded length prescribed by RC 6 is chosen deliberately so that the bond stress is approximately *uniform* over the bonded zone, allowing the measured capacity to be reported as a single average bond strength $\tau_u = F_u/(\pi d_b L)$. The present model makes no such assumption: it resolves the full distribution of bond stress along z, and Section 3.9 shows that the distribution is strongly non-uniform once L exceeds a few diameters. The two are consistent at short bonded lengths and diverge as L grows, which is the central reason for sweeping L/d_b rather than working at a single standardised length.

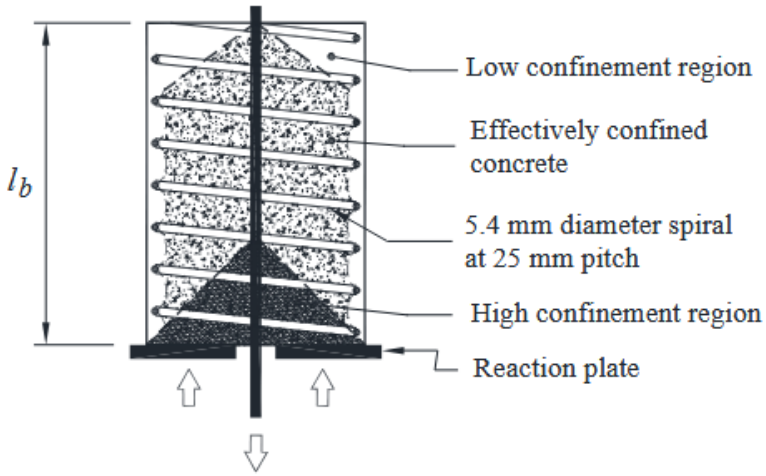


Figure 3.1.0.1: Bond pullout test specimen: bar, bonded length L, reaction plate and clearance hole.

3.1.1 Idealisation of the specimen

The model represents this specimen as follows. A cubic or prismatic specimen is replaced by an equivalent concentric cylinder of outer radius r_e, chosen to preserve the concrete cross-sectional area or the minimum cover, whichever the comparison requires; a cylinder of 150 mm diameter is used here. The debonded zones carry no shear and are therefore outside the domain: the model's $z = 0$ is the start of the bonded length and $z = L$ its end, not the specimen faces. The reaction plate enters only as the boundary condition $\sigma_z^c(0) = -q$ of Eq. (3.19), which transmits the full bar force into the concrete in bearing at the loaded end of the bonded zone.

This last point is an approximation worth stating plainly. In the standard specimen the bond-free length separates the bonded zone from the plate, so the bearing stress has partly dispersed before it reaches $z = 0$; in the model the two are coincident. The idealisation is therefore slightly conservative with respect to the hoop stress at the loaded end, where the dispersal would relieve some of the splitting demand.

3.2 Three-layer model and assumptions

The model predicts the elastic stress field in a single deformed bar bonded in a concrete cylinder, and from it the governing failure mode, the load-carrying capacity, the elastic bond stiffness and the slip at failure. It follows the variational procedure introduced by Nemes and Lachaud (2009) for cylindrical adhesively bonded joints and extended by Alavi, Valipour, and Bradford (2025) to glued-in rod (GiR) timber connections, with the timber adherend replaced by concrete and the adhesive layer reinterpreted as a thin *rib layer* representing the mechanically interlocked annulus immediately surrounding the bar.

The specimen is idealised as three concentric, axisymmetric, linearly elastic cylinders over the embedded length L:

Zone	Radial extent	Properties
Bar	$r_i < r < r_s$	$E_s,\ \nu_s,\ G_s$
Rib layer	$r_s < r < r_r$	$E_r,\ \nu_r,\ G_r$
Concrete	$r_r < r < r_e$	$E_c,\ \nu_c,\ G_c$

Table 3.2.0.1: Zones of the three-layer idealisation.

with $r_i = 0$ for a solid bar. All three materials are taken as isotropic, $G = E/[2(1+\nu)]$. The rib layer is presently assigned the concrete properties, so its role is geometric—it carries the discontinuity in shear and hoop stress between bar and concrete—and it provides the single parameter, $t_{\text{rib}} = r_r - r_s$, through which rib geometry can later be introduced.

The analysis adopts the hypotheses of Nemes and Lachaud (2009): axisymmetry suppresses $\tau_{r\theta}$ and $\tau_{z\theta}$; the axial stress in the rib layer is neglected, $\sigma_z^r = 0$; the axial stresses in the bar and the concrete are functions of z alone; and the radial stresses vary with r alone. The entire field is expressed in terms of one unknown function, the bar axial stress $\sigma(z) \equiv \sigma_z^s(z)$, and one unknown scalar, the radial-stress parameter α_1.

With $A_s = \pi(r_s^2 - r_i^2)$ and $A_c = \pi(r_e^2 - r_r^2)$, an applied load F produces a bar stress $f = F/A_s$ at the loaded end and a nominal bearing stress $q = F/A_c$ on the concrete face.

3.3 Assumed stress field

Axial and radial equilibrium of an axisymmetric element,

$$\frac{\partial}{\partial r}\left[r\sigma_r^{(i)}\right] + \frac{\partial}{\partial z}\left[r\tau_{rz}^{(i)}\right] = \sigma_\theta^{(i)}, \qquad \frac{\partial}{\partial r}\left[r\tau_{rz}^{(i)}\right] + \frac{\partial}{\partial z}\left[r\sigma_z^{(i)}\right] = 0, \tag{3.1}$$

following Eqs. (6)–(7) of Nemes and Lachaud (2009). The radial stresses are assumed to vary *linearly* and to vanish on the two free surfaces, $r = r_i$ and $r = r_e$, after Eq. (8) of Nemes and Lachaud (2009):

$$\sigma_r^s = \alpha_1(r - r_i), \qquad \sigma_r^r = \text{const.}, \qquad \sigma_r^c = \alpha_2(r - r_e). \tag{3.2}$$

Continuity of σ_r across both interfaces—a consequence of momentum conservation at a closed interface, as noted in Eq. (13) of Alavi, Valipour, and Bradford (2025)—fixes the rib-layer constant and α_2:

$$\sigma_r^r = \alpha_1(r_s - r_i), \qquad \alpha_2 = \frac{\alpha_1(r_s - r_i)}{r_r - r_e}. \tag{3.3}$$

The linear form (3.2) is Nemes and Lachaud's; Alavi, Valipour, and Bradford (2025) replace it with a quadratic variation, $\sigma_r^s = \Lambda_1 r^2$ and $\sigma_r^t = \Lambda_3(r^2 - r_t^2)$ (their Eq. 6), which they show improves agreement with FE near the loaded end. The linear form is retained here because it leaves a single unknown α_1 and keeps every radial integral elementary; the quadratic refinement is identified in Section [X] as the first candidate improvement.

Integrating (3.1) subject to (3.2)–(3.3) and to continuity of τ_{rz} at both interfaces gives the complete field in terms of $\sigma(z)$, reproducing Eqs. (12)–(13), (15)–(16) and (19)–(20) of Nemes and Lachaud (2009).

Bar, $r_i < r < r_s$**.**

$$\sigma_z^s = \sigma, \qquad \tau_{rz}^s = \frac{r_i^2 - r^2}{2r}\,\sigma', \qquad \sigma_\theta^s = \frac{r_i^2 - r^2}{2}\,\sigma'' + \alpha_1(2r - r_i). \tag{3.4}$$

Rib layer, $r_s < r < r_r$**.**

$$\sigma_z^r = 0, \qquad \tau_{rz}^r = \frac{r_i^2 - r_s^2}{2r}\,\sigma', \qquad \sigma_\theta^r = \alpha_1(r_s - r_i) + \frac{r_i^2 - r_s^2}{2}\,\sigma''. \tag{3.5}$$

Concrete, $r_r < r < r_e$**.**

$$\tau_{rz}^c = \frac{(r_e^2 - r^2)(r_i^2 - r_s^2)}{2r\,(r_e^2 - r_r^2)}\,\sigma', \qquad \sigma_\theta^c = \frac{(r_i^2 - r_s^2)(r_e^2 - r^2)}{2(r_e^2 - r_r^2)}\,\sigma'' + \alpha_2(2r - r_e). \tag{3.6}$$

The concrete axial stress follows from assembly equilibrium, the rib layer carrying no axial load. For the self-equilibrated pull–push specimen the bar force is reacted entirely by the concrete at every section, so (cf. Eq. (11b) of Alavi, Valipour, and Bradford (2025))

$$\sigma_z^c = -\frac{(r_s^2 - r_i^2)\,\sigma}{r_e^2 - r_r^2} = -\frac{A_s}{A_c}\,\sigma. \tag{3.7}$$

Two structural features of (3.4)–(3.7) are used repeatedly below. The shear stress is proportional to σ' and the hoop stress to σ'' plus a term linear in α_1, so both separate into a function of r times a function of z; and every component is linear in the applied load, which is what makes the capacity scaling of Section 3.8 and the sweep of Section 3.9 possible.

Continuity of σ_r and τ_{rz} at $r = r_s$ and $r = r_r$ is verified symbolically before the energy integration, as a check on (3.2)–(3.7).

3.4 Strain energy

The strain energy of each layer is obtained by integrating the complementary energy density over its annulus and along the bonded length, after Eq. (21) of Nemes and Lachaud (2009) and Eq. (17) of Alavi, Valipour, and Bradford (2025):

$$U^s = \pi \int_0^L \int_{r_i}^{r_s} \left[\frac{\sigma_\theta^{s2}}{E_s} + \frac{\sigma_z^{s2}}{E_s} - \frac{2\nu_s}{E_s}\,\sigma_z^s \sigma_\theta^s + \frac{\tau_{rz}^{s2}}{G_s} \right] r \,\mathrm{d}r \,\mathrm{d}z, \tag{3.8}$$

$$U^r = \pi \int_0^L \int_{r_s}^{r_r} \left[\frac{\sigma_\theta^{r2}}{E_r} + \frac{\tau_{rz}^{r2}}{G_r} \right] r \,\mathrm{d}r \,\mathrm{d}z, \tag{3.9}$$

$$U^c = \pi \int_0^L \int_{r_r}^{r_e} \left[\frac{\sigma_\theta^{c2}}{E_c} + \frac{\sigma_z^{c2}}{E_c} - \frac{2\nu_c}{E_c}\,\sigma_z^c \sigma_\theta^c + \frac{\tau_{rz}^{c2}}{G_c} \right] r \,\mathrm{d}r \,\mathrm{d}z, \tag{3.10}$$

and the total strain energy of the assembly is

$$U = U^s + U^r + U^c. \tag{3.11}$$

Substituting the assumed field (3.4)–(3.7) into (3.8)–(3.10) makes it possible to write the strain energy in terms of σ_z^s and its derivatives alone, as in Eq. (22) of Nemes and Lachaud (2009):

$$U = \pi \int_0^L \Gamma \,\mathrm{d}z, \tag{3.12}$$

in which the integrand is

$$\begin{aligned} \Gamma = {} & A\,\sigma_z^{s2} + B\,\sigma_z^s \frac{\mathrm{d}^2\sigma_z^s}{\mathrm{d}z^2} + C\left(\frac{\mathrm{d}\sigma_z^s}{\mathrm{d}z}\right)^2 + \tilde{D}\,\sigma_z^s \\ & + E\left(\frac{\mathrm{d}^2\sigma_z^s}{\mathrm{d}z^2}\right)^2 + \tilde{F}\,\frac{\mathrm{d}^2\sigma_z^s}{\mathrm{d}z^2} + \tilde{K}, \end{aligned} \tag{3.13}$$

and

$$\tilde{D} = D + \alpha_1 k, \qquad \tilde{F} = F + \alpha_1 h, \qquad \tilde{K} = K + \alpha_1^2 m. \tag{3.14}$$

The coefficients A–K and h, k, m are functions of the radii and the elastic constants only, and carry no dependence on z or on the applied load. They are obtained by expanding the energy density and taking partial derivatives with respect to σ_z^s and its derivatives, evaluated at zero.

Their closed forms are lengthy, involving the logarithms $\ln(r_r/r_s)$ and $\ln(r_e/r_r)$ returned by the integrals, and are set out in full in Appendix [X]. The absence of a term linear in α_1 in $\tilde{K}$ is specific to the pull–push configuration, where (3.7) contains no constant contribution from the applied load.

3.5 Governing differential equation and its solution

Requiring the strain energy to be stationary, $\delta U = 0$, and applying the Euler–Lagrange equation of the functional (3.12) gives the governing equation, Eq. (26) of Nemes and Lachaud (2009):

$$E\,\frac{\mathrm{d}^4\sigma_z^s}{\mathrm{d}z^4} + (B-C)\,\frac{\mathrm{d}^2\sigma_z^s}{\mathrm{d}z^2} + A\,\sigma_z^s + \frac{\tilde{D}}{2} = 0. \tag{3.15}$$

This is a linear, fourth-order, constant-coefficient ordinary differential equation and is solved in closed form. Its characteristic roots are

$$\lambda_{1,2}^2 = \frac{-(B-C) \pm \sqrt{(B-C)^2 - 4AE}}{2E}, \tag{3.16}$$

equivalent to $\varpi_{1,2}$ in Eq. (21) of Alavi, Valipour, and Bradford (2025), and the general solution is

$$\sigma_z^s(z) = C_1 e^{\lambda_1 z} + C_2 e^{-\lambda_1 z} + C_3 e^{\lambda_2 z} + C_4 e^{-\lambda_2 z} - \frac{\tilde{D}}{2A}. \tag{3.17}$$

No numerical integration, discretisation or shooting is used at any stage; the solution is analytical throughout, and the only numerical operations are the evaluation of (3.17) and the stress averaging of Section 3.8.

3.6 Boundary conditions and the radial-stress parameter

3.6.1 Boundary conditions

Four conditions determine $C_1 \ldots C_4$. The bonded length terminates at free surfaces, so the interface shear vanishes there,

$$\tau_{rz}^s(z=0) = \tau_{rz}^s(z=L) = 0, \tag{3.18}$$

The two axial conditions express the pull–push arrangement through (3.7): the concrete carries the full bearing stress at the loaded face and is unstressed at the free end,

$$\sigma_z^c(0) = -q, \qquad \sigma_z^c(L) = 0, \tag{3.19}$$

equivalently $\sigma_z^s(0) = f$ and $\sigma_z^s(L) = 0$, i.e. complete transfer of the bar force to the concrete over the embedded length. These are the pull–push conditions of Eq. (22b) of Alavi, Valipour, and Bradford (2025). The four conditions are assembled into a linear system and solved symbolically, so $C_1 \ldots C_4$ are obtained as explicit functions of the geometry, the moduli, the load and α_1.

3.6.2 Determination of α_1

α_1 is the one remaining unknown and is fixed by making the energy stationary with respect to it, $\partial U/\partial\alpha_1 = 0$. Since α_1 enters Γ only through $\tilde{D}$, $\tilde{F}$ and $\tilde{K}$, differentiating (3.12)–(3.14) gives

$$2m\alpha_1 L + \int_0^L \left(k\,\sigma_z^s + h\,\frac{\mathrm{d}^2\sigma_z^s}{\mathrm{d}z^2} \right) \mathrm{d}z = 0, \tag{3.20}$$

in which the second term integrates to $h\,[\mathrm{d}\sigma_z^s/\mathrm{d}z]_0^L$ and vanishes identically by the boundary condition (3.18), leaving

$$2m\alpha_1 L + \int_0^L k\,\sigma_z^s\,\mathrm{d}z = 0. \tag{3.21}$$

Equation (3.21) is implicit in appearance, since σ_z^s was itself constructed from α_1, but the dependence is affine in both places it arises—the particular solution $-(D + \alpha_1 k)/2A$, and the constants $C_1 \ldots C_4$, whose linear system carries that particular solution on its right-hand side. The field therefore splits exactly into its α_1-independent and α_1-linear parts,

$$\sigma_z^s(z) = \sigma_0(z) + \alpha_1\,\sigma_1(z), \qquad \sigma_0 = \sigma_z^s|_{\alpha_1=0}\,, \qquad \sigma_1 = \frac{\partial\sigma_z^s}{\partial\alpha_1}, \tag{3.22}$$

both functions of z alone. Substituting (3.22) into (3.21) and using the linearity of the integral,

$$2m\alpha_1 L + \int_0^L k\,\sigma_0(z)\,\mathrm{d}z + \alpha_1 \int_0^L k\,\sigma_1(z)\,\mathrm{d}z = 0, \tag{3.23}$$

with α_1 taken outside the second integral because σ_1 does not depend on it. The two definite integrals are

$$I_0 = k\int_0^L \sigma_z^s|_{\alpha_1=0}\,\mathrm{d}z, \qquad I_1 = k\int_0^L \frac{\partial\sigma_z^s}{\partial\alpha_1}\,\mathrm{d}z, \tag{3.24}$$

where k comes outside because it depends only on the radii and the elastic constants. Neither integral contains α_1, so (3.23) is one linear equation in the one unknown and closes the problem exactly, without iteration:

$$2m\alpha_1 L + I_0 + \alpha_1 I_1 = 0 \qquad \Longrightarrow \qquad \alpha_1 = -\,\frac{I_0}{2mL + I_1}. \tag{3.25}$$

Substituting (3.25) into (3.17) and then into (3.4)–(3.7) yields the complete stress field. All symbolic manipulation uses exact rational arithmetic, and material values are substituted only at this stage at 50 significant digits: the boundary-value system is built on $e^{\pm\lambda z}$, whose two branches differ in magnitude by $e^{2\lambda L}$ across the bonded length, so its condition number grows exponentially with λL and double precision is exhausted well before the longest case in the sweep.

3.7 Bond Stiffness

The stress field of Section 3.6 also fixes the stiffness of the anchorage, which governs behaviour at service load as the criteria of Section 3.8 govern it at ultimate. Following Section 3 of Alavi, Valipour, and Bradford (2025), the displacement of the loaded end is recovered from the strain energy rather than by integrating the strain field. Loading is monotonic and the assembly linearly elastic, so the external work done by the applied load F as it is brought up from zero equals the strain energy stored,

$$U = \frac{F\,\delta}{2}, \tag{3.26}$$

whence the axial displacement of the loaded end is

$$\delta = \frac{2U}{F}, \qquad U = \pi \int_0^L \Gamma \,\mathrm{d}z, \tag{3.27}$$

with Γ from (3.13) evaluated on the closed-form solution (3.17). The integrand is a quadratic form in sums of exponentials, so U follows from the same elementary integrals as (3.24) and no quadrature is required. The bond stiffness is the ratio of load to displacement,

$$k_s = \frac{F}{\delta} = \frac{F^2}{2U}. \tag{3.28}$$

Equation (3.28) does not depend on the applied load. The energy coefficients D, F and K of (3.14) all vanish for the pull–push configuration, so Γ is a homogeneous quadratic form in σ_z^s, its derivatives and α_1, every one of which is proportional to the load. Hence, U scales as the square of the load, the ratio $F^2/2U$ is unchanged by it, and k_s is a property of the geometry and the moduli alone, computed once per case alongside the capacity. The displacement at failure follows directly,

$$\delta_u = \frac{F_u}{k_s}, \tag{3.29}$$

with F_u from (3.36).

Two limitations bear on the comparison with test data in Chapter [X]. First, δ is the displacement of the loaded end relative to the bearing face and includes the elastic extension of the bar and the compression of the concrete; it is neither the relative slip across the interface nor the free-end slip usually reported in pull-out tests, and is comparable only with a measured loaded-end displacement. Second, the model is linearly elastic with no bond degradation or interface softening represented, so k_s is the *initial* stiffness and (3.29) extrapolates it linearly to the failure load. Since a measured bond–slip curve softens progressively, the predicted slip at failure is a lower bound on the measured value and k_s an upper bound on the secant stiffness at peak load; the comparison in Chapter [X] is accordingly made against the initial branch of the measured response wherever the test data permit it.

3.8 Failure criteria

Two failure modes are considered, both adapted from the six-mode framework of Alavi, Valipour, and Bradford (2025); modes III–VI of that work are specific to the anisotropy of timber and have no concrete counterpart.

3.8.1 Mode I: yielding of the bar

Adapting Eq. (27a) of Alavi, Valipour, and Bradford (2025), with the peak bar stress occurring at the loaded end where $\sigma = f$,

$$\varphi_{\mathrm{I}} = \frac{\sigma_z^s}{f_{sy}} - 1, \qquad F_{\mathrm{I}} = f_{sy} A_s. \tag{3.30}$$

3.8.2 Mode II: cracking of the concrete

The quadratic interaction of Eq. (28) of Alavi, Valipour, and Bradford (2025), there written for the adhesive, is applied to the hoop and shear stresses in the rib layer and the surrounding concrete:

$$\varphi_{\mathrm{II}} = \left(\frac{\bar{\sigma}_\theta}{f_t}\right)^2 + \left(\frac{\bar{\tau}_{rz}}{f_v}\right)^2 - 1, \qquad f_t = f_v = 0.5\sqrt{f'_c}. \tag{3.31}$$

Only tensile hoop stress drives splitting, so $\bar{\sigma}_\theta$ is replaced by $\max(\bar{\sigma}_\theta, 0)$; compressive hoop makes no contribution. Failure in either mode corresponds to $\varphi = 0$.

The strength f_t is the *mean* uniaxial tensile strength of the concrete. AS 3600 gives the lower characteristic uniaxial tensile strength as $f'_{ct} = 0.36\sqrt{f'_c}$, with the mean and upper characteristic values obtained by multiplying it by 1.4 and 1.8 respectively ("Reinforced Concrete Basics" 2021), so that

$$f_t = 1.4 \times 0.36\sqrt{f'_c} = 0.5\sqrt{f'_c}. \tag{3.32}$$

The mean rather than the characteristic value is appropriate here because the model is compared against the measured capacities of individual specimens rather than used in design: since both strengths in (3.31) are squared, adopting the lower characteristic value would reduce every predicted capacity by exactly the factor 1.4. Setting $f_v = f_t$ is a simplification, made because no independent shear strength is available for plain concrete and because failure under the combined state of (3.31) is in any case a tensile separation mechanism; a distinct f_v could be introduced without altering the formulation.

3.8.3 The theory of critical distances

Applying (3.31) pointwise to an elastic field predicts cracking as soon as the peak stress touches the strength, which grossly underestimates brittle capacity because the stress concentration at the loaded end is both severe and highly localised. Alavi, Valipour, and Bradford (2025) address the same difficulty by averaging $\sigma_{\theta\theta}$ and τ_{rz} over a fracture length of 20 mm. Here the generalisation of that device due to Taylor (1999) and Taylor (2010) is adopted. The Theory of Critical Distances unifies several non-local criteria distinguished only by the geometry over which the elastic field is averaged: the Point Method samples at a single location $l_{\mathrm{ch}}/2$ ahead of the stress raiser, the Line Method averages along a line of length $2\,l_{\mathrm{ch}}$, and the Area Method averages over a semicircle of radius $1.32\,l_{\mathrm{ch}}$ (Taylor 2010). The Area Method is adopted here because it samples the field in both the axial and radial directions and is therefore the least sensitive to the strongly biaxial character of the stress state at the bar surface. The stresses entering (3.31) are accordingly the means over a semicircular area of radius

$$R = 1.32\,l_{\mathrm{ch}}, \tag{3.33}$$

centred on the point being assessed, where the coefficient 1.32 is Taylor's calibration of the semicircular averaging area against the Point and Line Methods for a sharp crack (Taylor 1999; Taylor 2010), and l_{ch} is the critical distance, taken as 20 mm after Alavi, Valipour, and Bradford (2025). The semicircle lies in the (z, r) plane and sits on the bar surface, since no material exists below $r = r_s$.

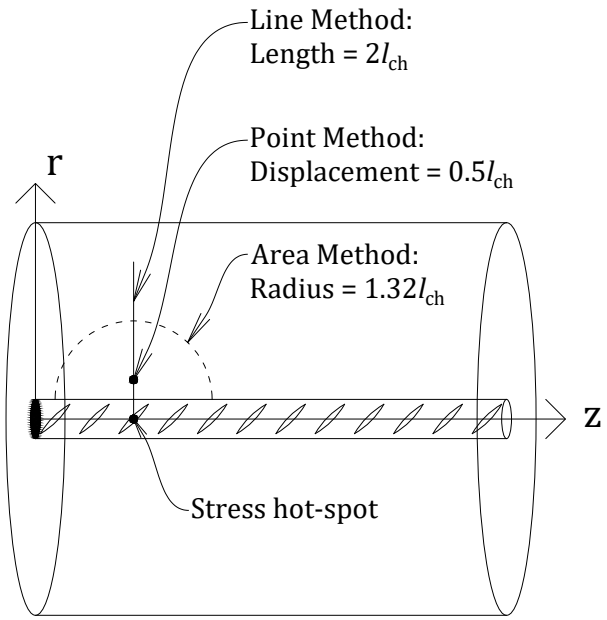


Figure 3.8.3.1: The three averaging methods of the Theory of Critical Distances, shown at a stress hot-spot on the bar surface in the (z, r) plane.

The procedure is summarised in Figure 3.8.3.2.

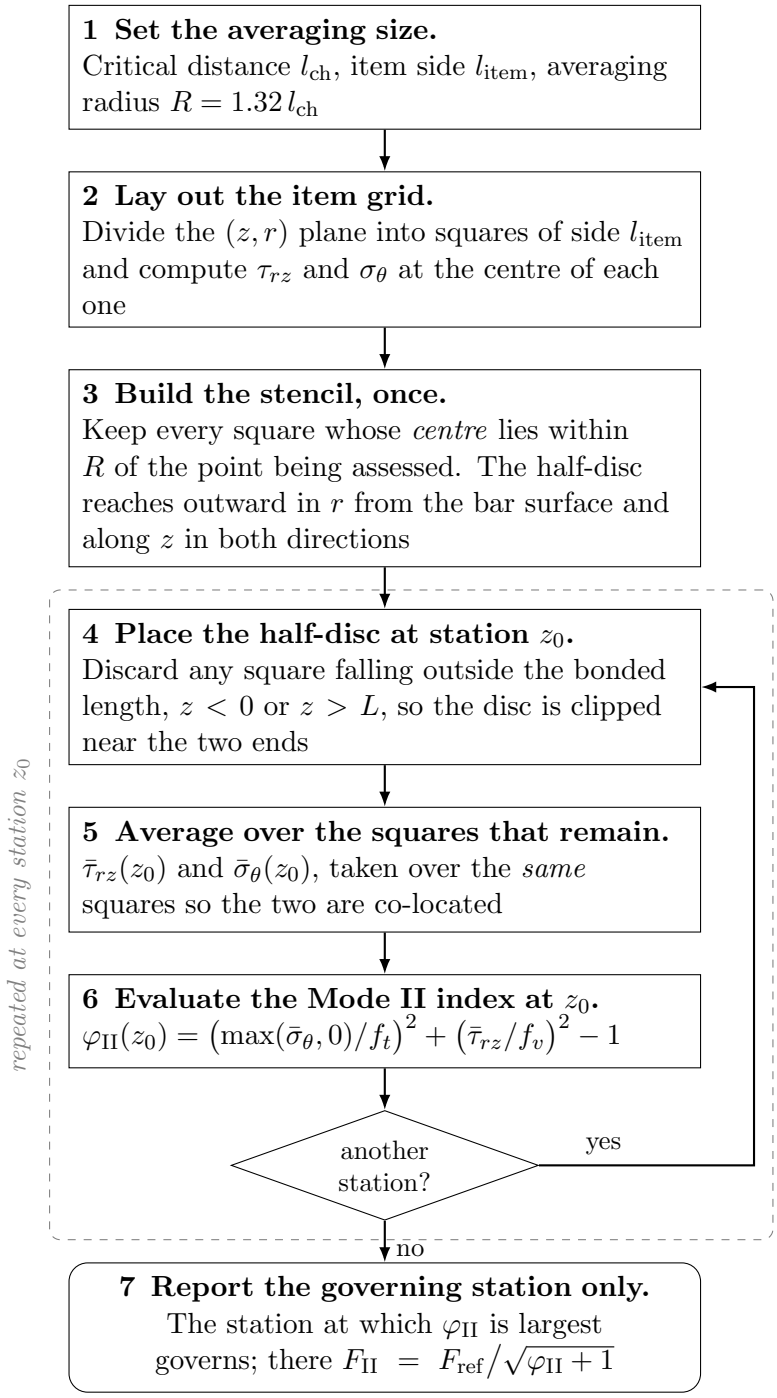


Figure 3.8.3.2: Evaluation of the Mode II index by Taylor's Area Method. Steps 1–3 depend only on the geometry and are carried out once; steps 4–6 are repeated at every axial station, and only the governing station is reported.

3.8.4 Predicted capacity

The averaged stresses of Section 3.8.3 come from a solution built at one particular load, whereas what is wanted is the load at which (3.31) is first satisfied. Because the formulation is linear elastic, the second follows from the first by substitution, with no iteration and no second solve.

Let the field be evaluated at an arbitrary reference load F_{ref}, and let $\bar{\sigma}_{\theta,0}$ and $\bar{\tau}_{rz,0}$ be the two area-averaged stresses obtained there, at the station that maximises the index. The load enters the boundary value problem only through the inhomogeneous condition $\sigma_z^c(0) = -q$ with $q = F/A_c$; the governing equation (3.15), the stationarity condition (3.21) and the other three boundary conditions are homogeneous. Doubling the load therefore doubles $\sigma_z^s(z)$ and α_1, and with them every component of (3.4)–(3.7), so that

$$\bar{\sigma}_\theta(F) = \bar{\sigma}_{\theta,0}\,\frac{F}{F_{\mathrm{ref}}}, \qquad \bar{\tau}_{rz}(F) = \bar{\tau}_{rz,0}\,\frac{F}{F_{\mathrm{ref}}}. \tag{3.34}$$

Substituting (3.34) into the failure condition $\varphi_{\mathrm{II}} = 0$ gives

$$\begin{gathered}\left(\frac{\bar{\sigma}_{\theta,0}}{f_t}\,\frac{F}{F_{\mathrm{ref}}}\right)^2 + \left(\frac{\bar{\tau}_{rz,0}}{f_v}\,\frac{F}{F_{\mathrm{ref}}}\right)^2 = 1,\\ \left(\frac{F}{F_{\mathrm{ref}}}\right)^2\left[\left(\frac{\bar{\sigma}_{\theta,0}}{f_t}\right)^2 + \left(\frac{\bar{\tau}_{rz,0}}{f_v}\right)^2\right] = 1,\end{gathered} \tag{3.35}$$

the load ratio being common to both terms. Solving for the load gives the Mode II capacity,

$$\boxed{F_{\mathrm{II}} = \frac{F_{\mathrm{ref}}}{\sqrt{\left(\dfrac{\bar{\sigma}_{\theta,0}}{f_t}\right)^2 + \left(\dfrac{\bar{\tau}_{rz,0}}{f_v}\right)^2}}} \tag{3.36}$$

and the governing capacity is the smaller of the two modes, $F_u = \min\left(F_{\mathrm{I}},\, F_{\mathrm{II}}\right)$.

Three properties worth noting. The square root is there because the index grows with the *square* of the load: doubling the load doubles both averaged stresses, and since each is squared the index is multiplied by four, so an index of four is brought back to one by halving the load — a factor of $1/\sqrt{4}$, not $1/4$. Mode I is linear in the load, which is why (3.30) carries no square root.

The reference load cancels, since $\bar{\sigma}_{\theta,0}$ and $\bar{\tau}_{rz,0}$ are themselves proportional to it. Any convenient value may be used, and repeating a case with F_{ref} doubled must return an identical capacity, which serves as a check on the implementation.

The governing station is fixed by the geometry, because the factor $(F/F_{\mathrm{ref}})^2$ in (3.35) is common to every axial station. It can be identified once, at the reference load, rather than re-established at each trial load, which is what makes the single solve legitimate.

3.9 Calibration curves

Sections 3.3–3.8 analyse one specimen. To produce design curves the analysis is repeated over a grid of bar diameters and anchorage lengths by means of three nested loops.

Loop 1 — bar diameter. d_b is varied over $\{10, 12, 16, 20\}$ mm, with the rib thickness t_{rib} and the cylinder diameter held constant, so that r_s, r_r, A_s, A_c, f and q are reset at each step. The coefficients A, B, C, E, h, k, m of (3.12)–(3.14) and the roots λ_1, λ_2 of (3.16) depend only on the radii and the moduli and contain neither L nor the load, so they are evaluated *once per diameter* rather than once per case. The Area Method stencil (Taylor 1999; Taylor 2010) likewise depends only on the bar radius and the mesh, and is built here. Mode I is independent of L and is also evaluated at this level, $F_{\mathrm{I}} = f_{sy} A_s$.

Loop 2 — anchorage length. The sweep is driven by the slenderness ratio rather than by the physical length, $L/d_b = 2, 4, \ldots, 30$, with $L = (L/d_b)\, d_b$, so that every bar size is sampled at the same normalised anchorage and the resulting curves are directly comparable. For each length the boundary conditions (3.18)–(3.19) are imposed, $C_1 \ldots C_4$ are recovered from the symbolic solution of Section 3.6.1, α_1 follows from (3.25) after Nemes and Lachaud (2009) and Alavi, Valipour, and Bradford (2025), and the rib and concrete shear and hoop fields are rebuilt. The item grid of Section 3.8.3 is laid out over the current length and both stresses are sampled at every item centre.

Loop 3 — axial station. At each station z_0 the clipped half-disc is assembled from the stencil, $\bar{\tau}_{rz}$ and $\bar{\sigma}_\theta$ are formed over the admitted items, and the Mode II index of (3.31) is evaluated; the running maximum over z is retained. The capacity then follows from (3.36) and the governing mode is recorded.

The three loops, and what is reset at each level, are shown in Figure 3.9.0.1.

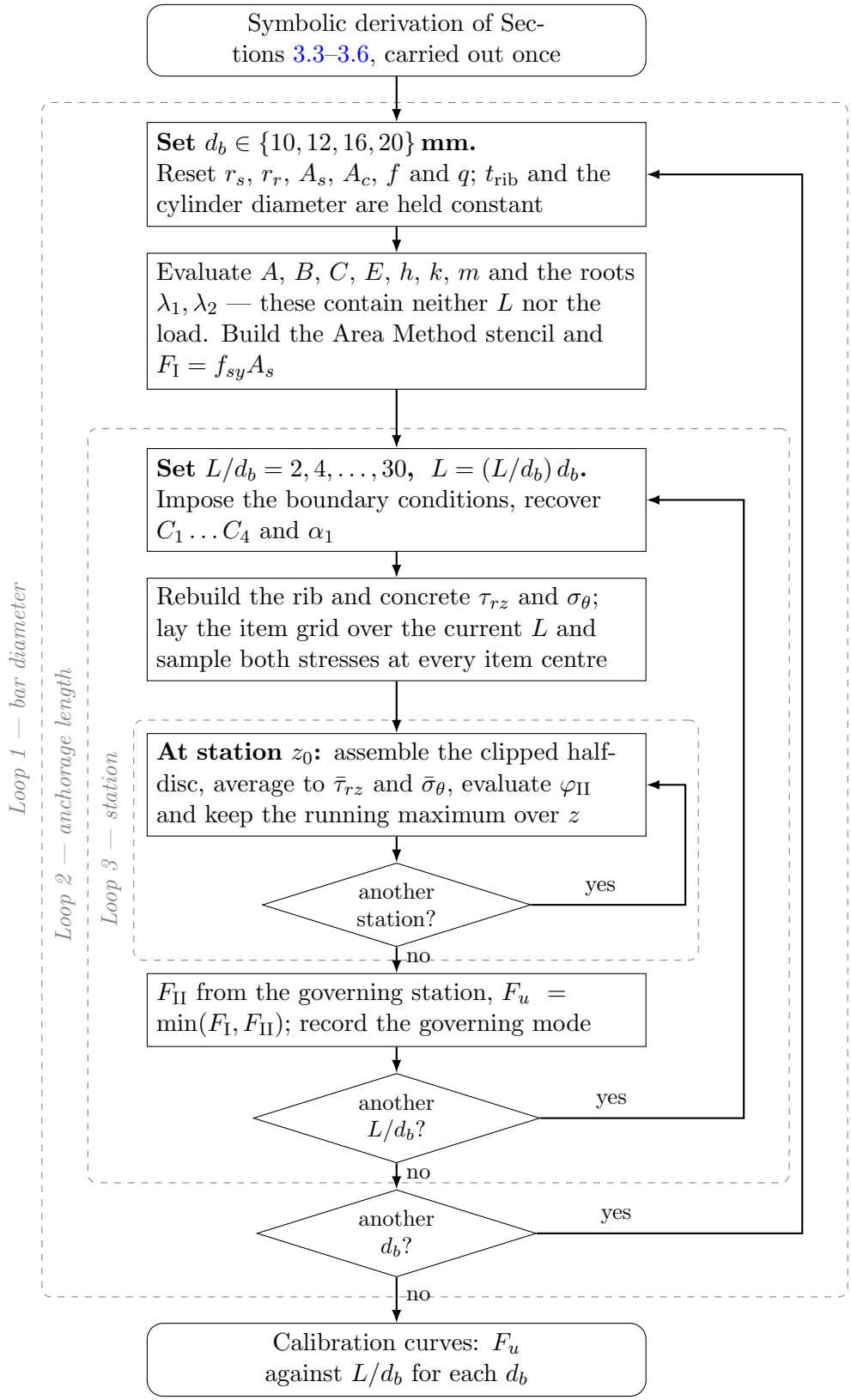


Figure 3.9.0.1: Structure of the calibration sweep. The symbolic derivation is performed once; the coefficients, roots and stencil are reset only when the bar diameter changes; the boundary-value solve is repeated for each anchorage length; and only the averaging and index evaluation run at every axial station.

Two properties of the formulation make this affordable. First, the L-independence of the coefficients removes the dominant symbolic cost from the inner loops. Second, linear elasticity makes a single solve at F_{ref} sufficient for each case, by (3.36). The working precision is reduced from 50 to 40 significant digits for the sweep, which remains sufficient for the exponential basis at the longest case considered, $L = 600\,\text{mm}$.

Table 3.9.0.1: Parameter matrix for the calibration sweep: force at failure and governing failure mode for increasing bonded length across four bar diameters. Pull–push configuration, $f'_c = 32\,\text{MPa}$, $f_{sy} = 550\,\text{MPa}$, $E_s = 200\,\text{GPa}$, $E_c = 30.1\,\text{GPa}$, $l_{\text{ch}} = 20\,\text{mm}$, $R = 1.32\,l_{\text{ch}} = 26.4\,\text{mm}$, $l_{\text{item}} = 2\,\text{mm}$, $t_{\text{rib}} = 0.5\,\text{mm}$, concrete cylinder diameter 150 mm.

Bar diameter d_b (mm)	**Normalised length** L/d_b	**Bonded length** L (mm)	**Force at failure** F_u (kN)	**Failure mode**
10	2	20		
10	4	40		
10	...	...		
10	30	300		
12	2	24	Force at failure output,	
12	...	...	failure mode output and	
12	30	360	description, for each of the	
16	2	32	60 combinations	
16	...	...	(4 diameters × 15 lengths)	
16	30	480		
20	2	40		
20	...	...		
20	30	600		

Each row of Table 3.9.0.1 is one closed-form solution of (3.17) followed by one Area Method evaluation, so the sweep comprises 60 independent solutions. The resulting capacities, the governing mode in each case and the two normalised forms of the calibration curve—F_u against L/d_b, and the mean bond stress at failure $\tau_u = F_u/(\pi d_b L)$ against L/d_b—are presented and discussed in Chapter [X]. The complete numerical output is tabulated in Appendix [X].

Chapter 4

Validation of the Proposed Method

4.1 Approach

The formulation of Chapter 3 is assessed in two stages, which answer different questions and are kept separate throughout.

Verification asks whether the implementation solves the equations it is intended to solve. It is carried out in Section 4.2 by reproducing the benchmark of Nemes and Lachaud (2009), the source of the variational procedure itself: the same geometry, the same material properties and the same loading configuration are supplied to the present code, and the resulting stress distributions are compared with those published. Since both are analytical solutions of the same boundary value problem, agreement tests the implementation and nothing else; disagreement would indicate an error in the coding of the energy integrals, the boundary conditions or the closure for α_1, rather than a deficiency of the model.

Two benchmarks are used, chosen so that between them they exercise every branch of the formulation: Nemes and Lachaud (2009) in Section 4.2 and Hassanieh et al. (2018) in Section 4.3. Section 4.4 sets out what the two together establish.

Validation asks whether the equations describe reality, and requires measured data. It is carried out in Chapter 8 against the database of bond tests assembled from the literature.

4.2 Verification against Nemes and Lachaud (2009)

4.2.1 Benchmark configuration

Nemes and Lachaud (2009) analyse a tubular adhesive joint loaded in tension at both ends, which in the terminology of Section 3.1 is the *pull–pull* configuration (shown in Figure 4.2.1.1).

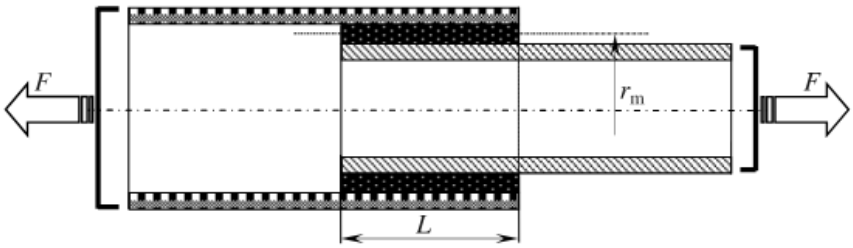


Figure 4.2.1.1: Diagram of the adhesively bonded joint from Nemes and Lachaud (2009).

The three concentric layers of their assembly map directly onto those of Section 3.2: the inner aluminium tube takes the place of the bar, the adhesive layer that of the rib layer, and the outer carbon/epoxy tube that of the concrete. The geometry and material properties are those of their Table 2 and are reproduced in Table 4.2.1.1.

Table 4.2.1.1: Benchmark parameters, after Table 2 of Nemes and Lachaud (2009).

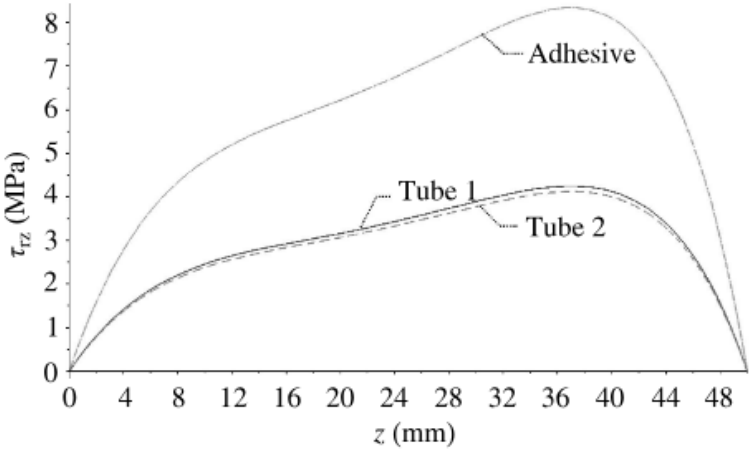
	Inner tube Aluminium ALU 7075	Adhesive Araldite Redux 340 SP	Outer tube Carbon/epoxy G0969/M18
Inner radius (mm)	$r_i = 44.8$	$r_{ic} = 47.8$	$r_{ec} = 48.0$
Outer radius (mm)	$r_{ic} = 47.8$	$r_{ec} = 48.0$	$r_e = 50.0$
E (MPa)	72 000	2 700	44 080
G (MPa)	27 692	1 000	16 640
ν	0.30	0.35	0.325
Overlap length $L = 50$ mm; applied stress $f = 100$ MPa on the inner tube			

Two features of the benchmark make it a searching test of the implementation. The adherends are stiff relative to a very thin and very compliant interlayer, $r_{ec} - r_{ic} = 0.2$ mm with $E_c/E_1 = 0.0375$, so the shear-lag behaviour is pronounced; and the inner tube is hollow, $r_i \neq 0$, which exercises the general form of the radial integrals rather than the solid-bar limit used throughout Chapter 3. The outer tube is quoted as orthotropic, but the values given satisfy $G_{xy} = E_x/[2(1+\nu_{xy})]$ to within 0.04 %, so the layer is isotropic for the purposes of this calculation and the isotropic form of the energy density of Section 3.4 applies without modification.

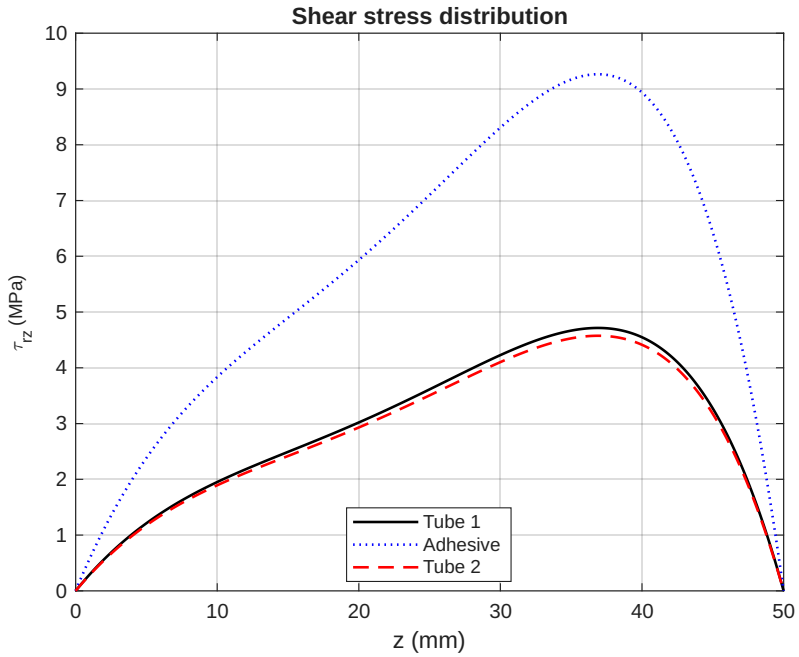
No parameter was adjusted or calibrated. The code of Chapter 3 was run with `setup = 'pull-pull'` and the values of Table 4.2.1.1 substituted directly.

4.2.2 Shear stress distribution

Figure 4.2.2.1 compares the computed interfacial shear stress with Figure 3(b) of Nemes and Lachaud (2009). The agreement is close in every respect that characterises the distribution.



(a)



(b)

Figure 4.2.2.1: Interfacial shear stress along the joint. (a) Figure 3(b) of Nemes and Lachaud (2009); (b) the present model, run with the parameters of Table 4.2.1.1 and no adjustment.

Table 4.2.2.1: Shear stress τ_{rz} at the benchmark load, present model against Figure 3(b) of Nemes and Lachaud (2009). Tube values are taken at the mid-thickness of the layer, the adhesive value at the interface.

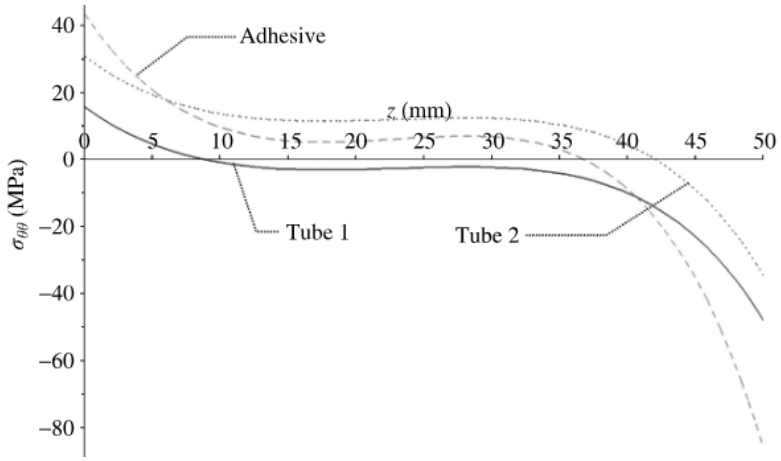
	Present model	Nemes and Lachaud (2009)	Difference
Peak τ_{rz}, inner tube (MPa)	4.72	≈ 4.25	+11%
Peak τ_{rz}, adhesive (MPa)	9.26	≈ 8.4	+10%
Peak τ_{rz}, outer tube (MPa)	4.57	≈ 4.1	+12%
Location of peak, z (mm)	36.9	≈ 37	—
Ratio adhesive / inner tube	1.97	≈ 1.98	−0.6%

The computed distribution vanishes at both free ends, as required by (3.18); rises steeply from the unloaded end; and peaks at $z = 36.9\,\mathrm{mm}$, against approximately 37 mm read from the published figure. The marked asymmetry of the distribution, which is the distinguishing feature of the model that retains σ_{rr} — Figure 3(a) of the same work, computed with $\sigma_{rr} = 0$, is nearly symmetric with a broad plateau — is reproduced. The ratio of the adhesive peak to the adherend peak, 1.97 against 1.98, is matched to better than one per cent, which is a stringent test because that ratio is fixed by the radial distribution of τ_{rz} through (3.4)–(3.6) and is independent of the magnitude of the solution.

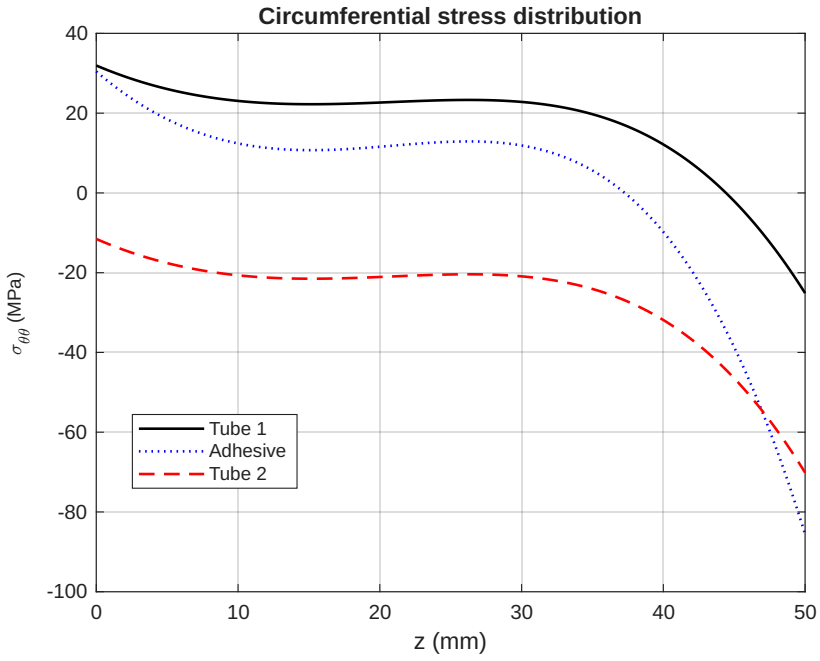
The absolute magnitudes are uniformly about ten per cent higher than the published values. The uniformity is the significant observation: all three curves are scaled by the same factor, to within the precision with which the figures can be read, so the difference is a single multiplicative factor on $\mathrm{d}\sigma_z^s/\mathrm{d}z$ rather than a distortion of the solution. A structural error — a mis-stated boundary condition, a missing term in the energy, an incorrect closure for α_1 — would alter the shape, the peak location or the inter-layer ratios, and none of these is affected.

4.2.3 Circumferential stress distribution

The comparison of hoop stress with Figure 4(b) of Nemes and Lachaud (2009), shown in Figure 4.2.3.1 and tabulated in Table 4.2.3.1, is less conclusive, for a reason that is a property of the quantity rather than of either calculation.



(a)



(b)

Figure 4.2.3.1: Circumferential stress along the joint. (a) Figure 4(b) of Nemes and Lachaud (2009); (b) the present model, run with the parameters of Table 4.2.1.1 and no adjustment.

In the adhesive the agreement is good: the computed value at the loaded end is 30.5 MPa against approximately 30 MPa, and the interior plateau is 12.8 MPa against approximately 11 MPa. The adhesive is the one layer for which the comparison is unambiguous, because σ_θ^r of (3.5) is independent of r and the thinness of the layer therefore removes any question of where the value is taken.

Table 4.2.3.1: Circumferential stress σ_θ at the benchmark load, present model against Figure 4(b) of Nemes and Lachaud (2009). Tube values are taken at the mid-thickness of the layer, the adhesive value at the interface.

σ_θ (MPa)	Present model	Nemes and Lachaud (2009)	Difference
Adhesive at $z = 0$	+30.5	$\approx +30$	+2%
Adhesive, interior plateau	+12.8	$\approx +11$	+17%
Inner tube at $z = 0$	+32.0	$\approx +15$	†
Outer tube at $z = 0$	−11.5	$\approx +41$	†
Inner tube, across the wall	+16.4 to +48.0	not stated	—

† Not comparable: the radius at which the published adherend curves are evaluated is not stated, and the computed value varies by a factor of three across the wall of the inner tube.

In the two adherends the computed curves differ from the published ones in magnitude and, over part of the range, in sign. The hoop stress in those layers is dominated by the term linear in α_1 of (3.4) and (3.6), which varies strongly across the thickness: in the inner tube the computed value at the loaded end ranges from 16.4 MPa on the bore to 48.0 MPa at the adhesive interface, a factor of three across 3 mm of wall. The published figure does not state the radius at which each curve is evaluated, so the two sets of curves are not necessarily being compared at the same point. Given that sensitivity, the difference in the adherend hoop stresses cannot be attributed to the implementation on the available evidence, and no claim is made here either way.

4.2.4 Discussion

Two observations follow.

The model reproduces the benchmark in every feature that is independent of absolute scale: the shape of the shear distribution, its asymmetry, the vanishing of the interfacial shear at both ends, the location of the peak to within 0.1 mm, and the ratio of adhesive to adherend shear to better than one per cent. These are the features that test the energy integration, the boundary conditions and the determination of α_1, and they are reproduced without adjustment.

The residual difference in magnitude, uniformly of order ten per cent, is consistent with a difference in the treatment of a single scalar rather than with an error in the formulation. Several possible explanations cannot be separated on the available information: the published figures may have been produced with a slightly different value of the applied stress or of one elastic constant than appears in their Table 2; the radii at which the curves are plotted are not stated; and the figures themselves can be read only to perhaps five per cent. It is noted that Nemes and Lachaud

(2009) report in their conclusions a peak adhesive shear of approximately eight per cent of the applied load, consistent with their Figure 3(b), against 9.3 % here.

The most significant of these is that Nemes and Lachaud (2009) state the governing differential equation but not its solution. Neither the general form adopted, nor the constants of integration, nor the manner in which the four boundary conditions were imposed is given, so the step from the fourth-order equation to the plotted curves had to be reconstructed here and cannot be checked against the source. That step is also the natural place for a uniform difference to arise. It fixes the magnitude of σ_z^s and hence of every stress derived from it, whereas the shape of the distribution and the ratios between the layers are fixed by the assumed stress field and are insensitive to it. A difference confined to the solution of the governing equation would therefore scale all three curves by the same factor and leave the peak location unmoved, which is what is observed.

On this evidence the implementation is taken to be a correct realisation of the variational formulation, and is carried forward to the concrete application of Chapter 3 with the configuration switched to pull–push.

4.3 Verification against Hassanieh et al. (2018)

4.3.1 Rationale for a second benchmark

Section 4.2 establishes that the implementation reproduces the formulation it was built from. That is a necessary check but a narrow one: a single benchmark can be matched by a code that happens to be correct only for the particular combination of geometry, material symmetry and root type that the benchmark contains. A second, independent benchmark was therefore selected to exercise the parts of the formulation that Nemes and Lachaud (2009) does not.

textcitehassaniehGluedinrodTimberJoints2018 analyse glued-in rod timber joints using the same variational framework but with three substantive differences, each of which tests a different part of the implementation:

- *Their stress field is simpler.* They take the radial and tangential stresses to be equal, $\sigma_r = \sigma_\theta$, and obtain both from radial equilibrium instead of assuming a form for them. Their model therefore has no radial-stress parameter α_1, and no second minimisation to determine it, as Section 3.6.2 does here: the whole field follows from σ_z^s alone. Reproducing their results therefore exercises the energy integration and the solution of the governing equation on their own, apart from the α_1 machinery that the first benchmark tests.
- *Their rod is solid rather than hollow,* $r_i = 0$. The inner adherend of Nemes and Lachaud (2009) is a tube, so its radial integrals never reach the axis. A solid rod is what Chapter 3 assumes throughout, so this benchmark tests the geometry the thesis model actually uses.
- *Their outer adherend is directional.* Timber is transversely isotropic: it has one stiffness along the grain and a much lower one across it. Both of Nemes and

Lachaud's adherends are isotropic, with the same stiffness in every direction, so this is the only benchmark that tests the code on a material whose properties depend on direction.

A further advantage is that Hassanieh et al. (2018) publish the coefficients of their energy functional in closed form. The present code obtains those coefficients by integrating the strain energy across the three layers symbolically, so the published expressions provide a direct check on that integration, independent of any comparison of plotted curves.

4.3.2 Benchmark configuration

The case selected is that of their Figure 4: a 12 mm steel rod glued into glulam over an embedment length of 360 mm and loaded in pull–pull to 28.8 kN, forty per cent of the measured ultimate capacity. Parameters are given in Table 4.3.2.1.

Table 4.3.2.1: Benchmark parameters, after Hassanieh et al. (2018). Quantities marked † are not stated in the source and are assumed.

	Rod	Adhesive	Timber (glulam)
Inner radius (mm)	0	$a = 6$	$b = 7$
Outer radius (mm)	$a = 6$	$b = 7$	$c = 60$
E (MPa)	200 000†	3 000	$E_1 = 13\,000$, $E_2 = E_1/30$
G (MPa)	—	—	$G = E_1/16$
ν	0.30†	0.20	$\nu_{12} = \nu_{13} = 0.30$†
Embedment $L = 360$ mm;	$F = 28.8$ kN, hence $f = F/\pi a^2 = 254.6$ MPa		

The hole is 2 mm oversize, giving a 1 mm glue line, and the half-width of the glulam for a 12 mm rod is 60 mm. As in Section 4.2, no parameter was adjusted.

4.3.3 Results

The roots of the characteristic equation come out real and distinct,

$$w_1 = 0.0637\,\mathrm{mm}^{-1}, \qquad w_2 = 0.0136\,\mathrm{mm}^{-1}, \tag{4.1}$$

so the stress decays away from each end over characteristic lengths of $1/w_1 = 15.7$ mm and $1/w_2 = 73.6$ mm.

This is a second respect in which the benchmark complements the first. The roots of Section 4.2 are a complex conjugate pair, which makes the stress oscillate as it decays; these are real, which makes it decay smoothly without oscillation. Which of the two occurs depends on the sign of the discriminant in (3.16), and the solution is written differently in each case, so between them the two benchmarks test both.

Three of the checks that follow need no reading of a published figure at all, and so carry more weight than any curve comparison.

Interface continuity. The radial and shear tractions are continuous across both interfaces, $r = a$ and $r = b$. The difference is exactly zero, not merely small, confirming that the assumed stress field has been written and substituted correctly.

Energy coefficients. Hassanieh et al. (2018) publish two of their energy coefficients as algebraic expressions, $A = \frac{1}{2}a^2/E^{(1)} - \frac{1}{2}a^4/[E_1^{(2)}(b^2 - c^2)]$ and $D = a^4 f/[E_1^{(2)}(b^2 - c^2)]$. The present code arrives at the same two numbers by integrating the complementary energy symbolically, without using their expressions:

$$A = 1.040\,37 \times 10^{-4}, \qquad D = -7.1491 \times 10^{-3}, \tag{4.2}$$

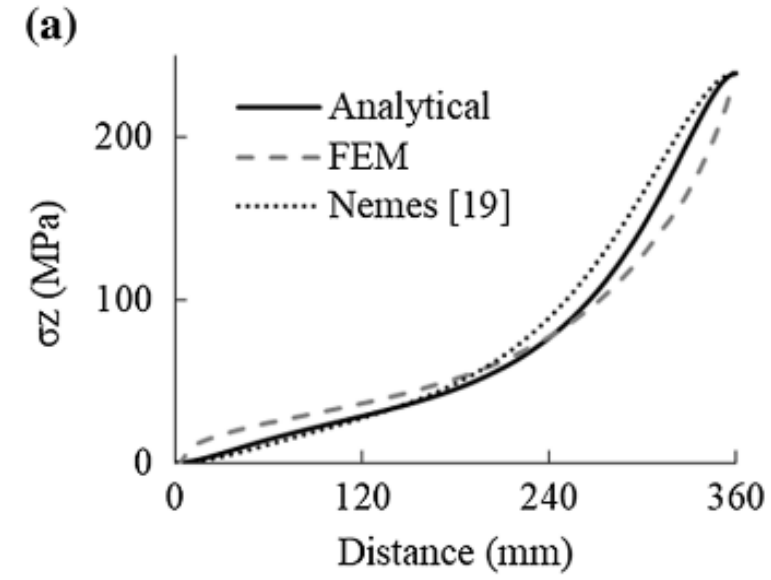
each agreeing with the published closed form to a ratio of 1.000000. This is the strongest single check in either benchmark, because it compares two numbers directly rather than two curves read off a figure, and because it tests the radial integration on its own.

Boundary conditions. The rod carries no axial stress at the free end and the full applied stress at the loaded end: 0.00 MPa at $z = 0$ and 254.65 MPa at $z = L$, against the imposed $f = 254.6$ MPa. The distribution between those two values is compared with their Figure 4(a) in Figure 4.3.3.1.

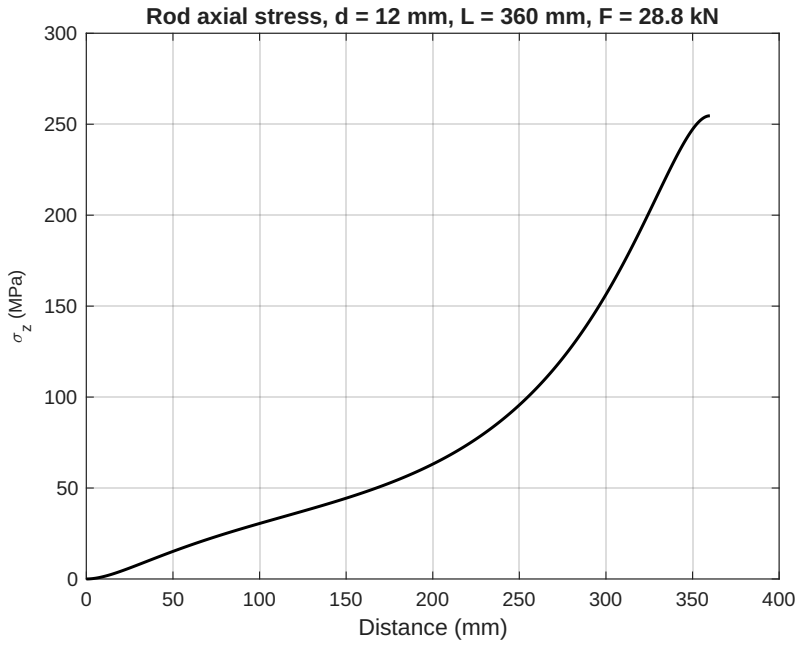
The stress distributions themselves are compared with Figure 4 of the source in Table 4.3.3.1 and Figures 4.3.3.1 to 4.3.3.3.

Table 4.3.3.1: Present code against Figure 4 of Hassanieh et al. (2018), evaluated on the glue line.

	Present code	Hassanieh et al. (2018)	Difference
σ_z in the rod at $z = L$ (MPa)	254.65	254.6	0.0%
Peak τ_{rz} (MPa)	5.09	≈ 4.8	+6%
Location of peak, z (mm)	329	≈ 340	−3%
σ_θ at $z = 0$ (MPa)	+1.03	$\approx +1$	—
σ_θ at $z = L$ (MPa)	−5.78	≈ -4.5	+28%
Joint stiffness K_s (kN/mm)	201.0	112.7	+78%

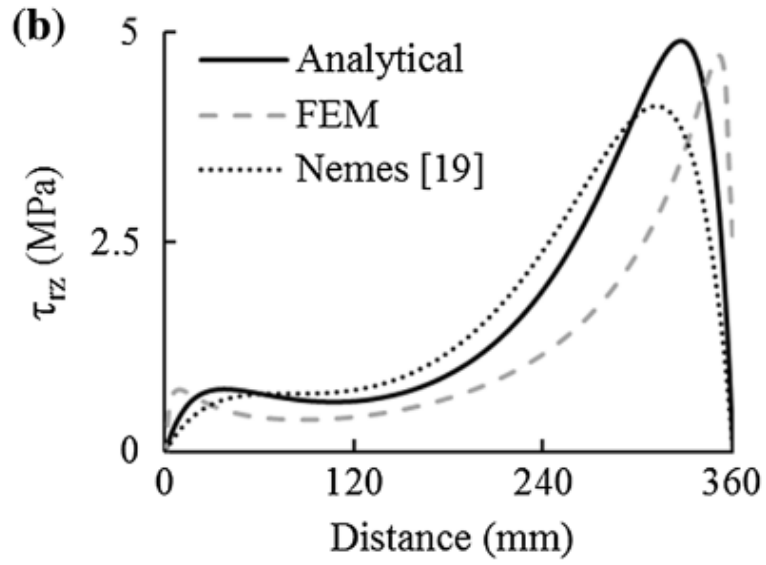


(a)

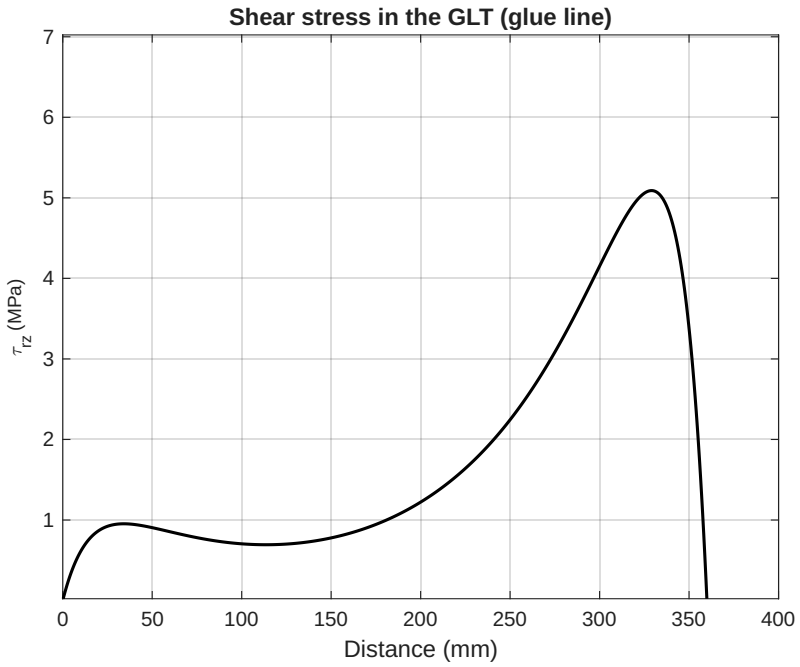


(b)

Figure 4.3.3.1: Axial stress in the rod along the embedment. (a) Figure 4(a) of Hassanieh et al. (2018); (b) the present model. The stress rises from zero at the free end to the full applied 254.6 MPa at the loaded end, slowly over the first half of the embedment and steeply over the last quarter. That shape is the load transfer concentrating near the loaded end, and it is what the shear distribution of Figure 4.3.3.2 differentiates.

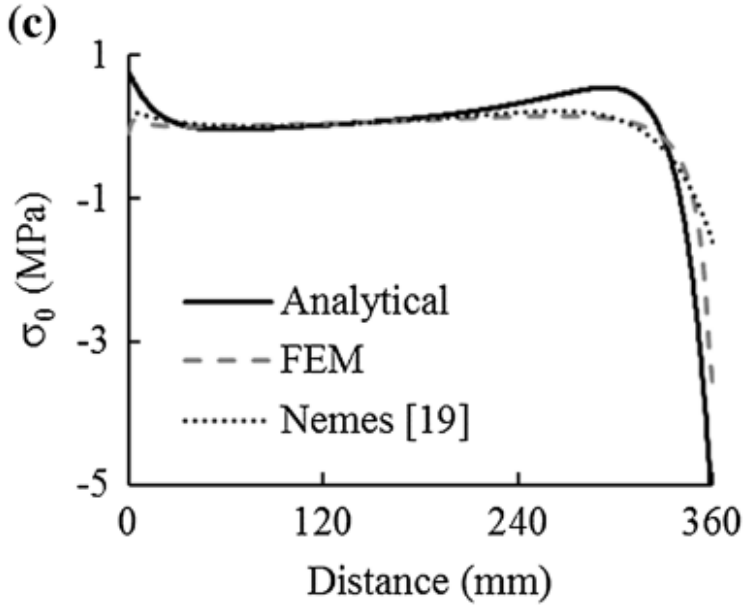


(a)

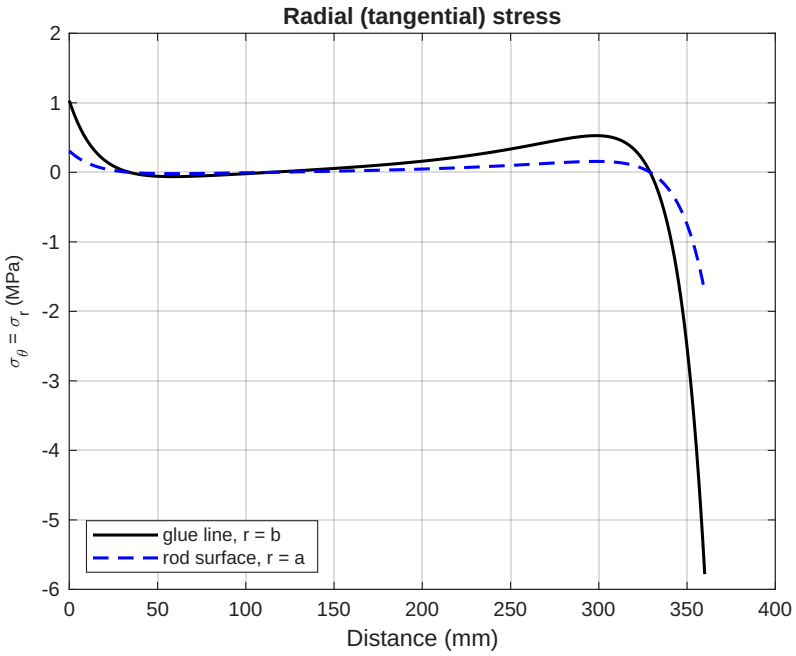


(b)

Figure 4.3.3.2: Shear stress on the glue line. (a) Figure 4(b) of Hassanieh et al. (2018); (b) the present model, run with the parameters of Table 4.3.2.1 and no adjustment. Both show the same three-part profile: a local maximum about 30 mm from the free end, a shallow trough through the middle of the bond line, and the true peak close to the loaded end. Comparison is with their analytical curve; the finite element and Nemes curves on their figure are shown for their own purposes.

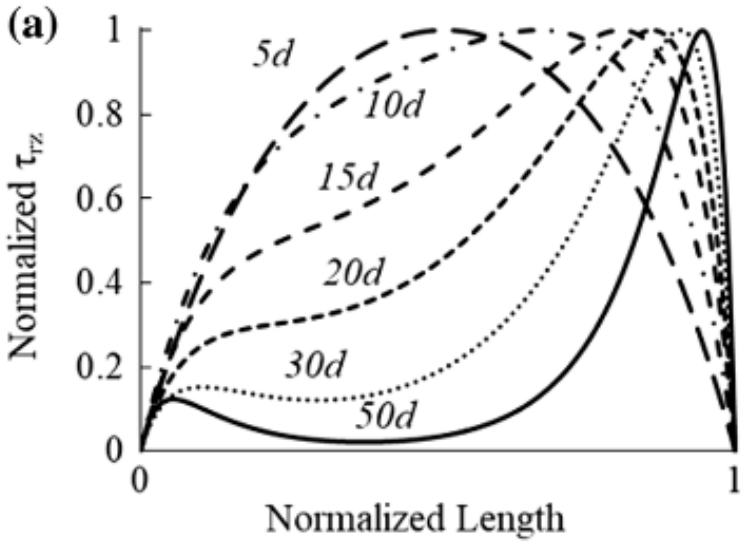


(a)

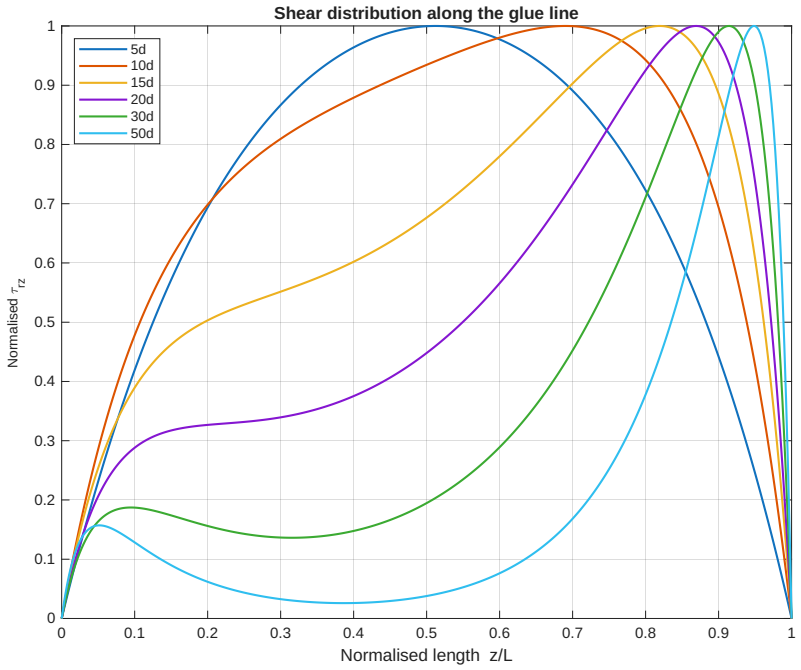


(b)

Figure 4.3.3.3: Tangential stress along the embedment, equal to the radial stress under the assumption of Hassanieh et al. (2018). (a) Their Figure 4(c); (b) the present model. The stress is tensile at the free end, passes through a near-zero plateau, recovers to a small positive value near $z = 300\,\mathrm{mm}$ and then turns sharply compressive over the last tenth of the embedment. Panel (b) is drawn at two radii, the glue line $r = b$ and the rod surface $r = a$, which the single published curve does not distinguish; the two coincide over most of the length and separate only where the stress is changing fastest.



(a)



(b)

Figure 4.3.3.4: Normalised shear distribution along the glue line for embedment lengths from $L/d = 5$ to 50, each curve scaled to its own maximum. (a) Figure 8(a) of Hassanieh et al. (2018); (b) the present model. At $L/d = 5$ the peak lies near mid-length; as the embedment lengthens it migrates to the loaded end and a trough opens up over the middle of the joint, until at $L/d = 50$ the interior carries almost no shear at all. This comparison is of a trend across six embedment lengths rather than of a single curve.

Both distributions come out the right shape. The shear stress (Figure 4.3.3.2) rises from zero at the free end to a small local maximum near $z = 30\,\mathrm{mm}$, drops back to a shallow trough around mid-length, then climbs to its true peak close to the loaded end and returns to zero there. Their Figure 4(b) shows the same three stages in the same places. The tangential stress (Figure 4.3.3.3) is tensile at the free end, stays close to zero through the middle, and turns sharply compressive over the last tenth of the embedment, as in their Figure 4(c).

Neither shape is arbitrary. In a joint this long the load is transferred almost entirely within a short distance of each end, so the middle of the bond line carries very little. That is what produces both the trough in the shear and the compression at the loaded end.

The model also reproduces how the distribution changes as the embedment gets longer. This is a harder test, because it compares a trend rather than a single curve. Figure 4.3.3.4 repeats the calculation for L/d from 5 to 50 alongside their Figure 8(a). At $L/d = 5$ the shear peaks at mid-length. As the embedment lengthens the peak moves towards the loaded end and the middle of the joint carries less and less, until at $L/d = 50$ the shear there is only three per cent of its peak. The roots in (4.1) explain why: the stress decays over a length of about $1/w_2 = 73.6\,\mathrm{mm}$, so once the embedment is several times that, the two ends no longer influence one another.

4.3.4 Discussion

Agreement on the stress field is good; agreement on the stiffness is not. The two should be read separately.

The stress distributions match in shape everywhere, and in magnitude to within six per cent for the shear. The tangential stress at the loaded end differs by twenty-eight per cent, but that figure deserves little weight. The curve falls almost vertically over the last few millimetres, so reading it a millimetre further along the bond line changes the value substantially, and the published figure cannot be read there to better than about twenty per cent.

Three places in the source needed interpretation before the model could be run. Each is recorded because anyone reproducing this work will meet the same difficulty, not because any of them changes the conclusion.

Their Eq. (10). This gives the tangential stress in the adhesive, but it does not match the rod at the interface: substituting $r = a$ returns zero, where the rod gives $\frac{1}{4}a^2\,\mathrm{d}^2\sigma_z^{(1)}/\mathrm{d}z^2$. The surrounding text says the adhesive expressions come from making the stresses continuous at $r = a$ and $r = b$, and their Eq. (9) for the shear is exactly a straight line between the two adherend values. The tangential stress was therefore taken as the same kind of straight line, which does meet both interfaces.

Their Eq. (26). This gives the displacement, and cannot be correct as printed: the leading factor adds a stress to an area, which have different units. The displacement was obtained instead by integrating the axial strain $\varepsilon_z^{(1)} = (\sigma_z^{(1)} - 2\nu^{(1)}\sigma_\theta^{(1)})/E^{(1)}$

along the bonded length. The tangential term drops out of that integral, because the interfacial shear vanishes at both ends, leaving $\delta = \int_0^L \sigma_z^{(1)}\,\mathrm{d}z \,/\, E^{(1)}$. This gives 0.143 mm at the benchmark load and hence $K_s = 201\,\mathrm{kN\,mm^{-1}}$, against the $112.7\,\mathrm{kN\,mm^{-1}}$ they report.

The stiffness difference cannot be blamed on the stress field. The same field reproduces their Figure 4(a), and the rod axial stress is precisely what the displacement integrates, so an error in the field would show up in the axial stress as well. The difference therefore lies either in their displacement expression or in what their reported displacement measures — whether, for instance, it includes deformation of the timber, or of the free length of rod between the two embedded ends of the pull–pull specimen. That cannot be settled from the published information, so the stiffness comparison is not relied upon in what follows.

4.4 What the two benchmarks establish

Taken together the two verifications support a stronger conclusion than either does alone, because the features they share are the implementation and the features they do not share are the model.

The two formulations differ in every respect that a coding error could hide behind. One carries a radial-stress parameter determined by a second minimisation and the other has no such parameter; one has a hollow inner adherend and the other a solid one; one is isotropic throughout and the other has a transversely isotropic outer adherend; one produces complex conjugate characteristic roots and an oscillatory solution, the other real roots and a monotonic one; the overlap of the first is three shear-transfer lengths and the embedment of the second is twenty-four. What the two share is the chain of operations that Chapter 3 implements: an assumed stress field reduced to one scalar function, radial integration of the complementary energy, extraction of the coefficients of a quadratic functional, the resulting fourth-order equation, its closed-form solution, and the imposition of four boundary conditions. That chain is reproduced in both cases without adjustment, which is evidence about the chain rather than about either particular model.

Four results are independent of how accurately a published figure can be read, and these carry most of the weight.

- Interface continuity of the radial and shear tractions returns exactly zero in both benchmarks, confirming the assumed field and its substitution.
- The energy coefficients A and D computed here by symbolic integration match the closed forms published by Hassanieh et al. (2018) to machine precision, confirming the radial integration and the coefficient extraction.
- The imposed boundary conditions are recovered exactly in both cases: the interfacial shear vanishes at both ends and the adherend axial stresses take their prescribed end values.
- The vanishing of D, F and K for the pull–push configuration, noted in Sec-

tion 3.4 as a consequence of (3.7) containing no constant load term, is reproduced independently in the Appendix of Hassanieh et al. (2018), where the same three coefficients vanish for the same configuration. Since the present implementation derives this rather than assuming it, the agreement confirms the treatment of the loading configuration — the one feature of the thesis model that neither benchmark exercises directly, because both are posed in pull–pull.

Of the quantities that must be read from published figures, the shear distribution agrees in shape, in peak location and in the ratio between layers in both cases, and in magnitude to within six per cent for Hassanieh et al. (2018) and about ten per cent for Nemes and Lachaud (2009). Residual differences of that order are consistent with the precision of the source figures and with parameters not stated in the sources, and in neither case do they affect the shape, the peak location or the inter-layer ratios, which are the features a formulation error would disturb.

Two limitations are stated for completeness. Both exercises are verification and not validation: they compare the implementation against other analytical solutions, not against measurement, and they therefore say nothing about whether the underlying assumptions describe a real bonded joint. That question is addressed in Chapter [X]. And neither benchmark exercises the failure criteria of Section 3.8 or the Area Method averaging of Section 3.8.3, which post-process the stress field and are verified separately in Section [X].

Subject to those limitations, the stress field computed by the implementation of Chapter 3 is taken to be correct, and the model is carried forward to the concrete application with the configuration switched to pull–push.

Chapter 5

Taylor's Theories of Critical Distances

The Mode II criterion of (3.31) compares a hoop stress and a shear stress against the tensile and shear strengths of the concrete. Applied pointwise to the elastic field of Chapter 3, it predicts cracking as soon as the peak of that field touches the strength. That peak is both severe and highly localised: the stress concentration at the loaded end decays over a short fraction of the bonded length, so a pointwise criterion is governed by a state of stress that exists almost nowhere in the specimen. The predicted capacity is accordingly far below anything measured, and it falls further as the field is resolved more finely, because nothing in the criterion fixes the scale over which the stress is to be judged. That behaviour is the signature of a failure criterion with no length scale in it.

The Theory of Critical Distances (TCD) supplies that length scale. Developed by Taylor (2010) from the earlier non-local proposals of Neuber and Peterson, it asserts that failure is controlled not by the peak elastic stress but by the elastic stress sampled over a finite region of material, whose size is the critical distance l_{ch}: a property of the material rather than of the geometry.

The theory admits three standard forms, distinguished only by the geometry over which the elastic field is sampled: the Point Method, the Line Method and the Area Method. All three were implemented in the code and compared against one another. This chapter sets out the theory and the implementation of each in turn, and then examines how far each one's prediction depends on the discretisation it requires rather than on the mechanics it represents.

The worked examples that follow all use the same specimen, evaluated in the pull-push arrangement: a 12 mm bar at 150 mm embedment in a 150 mm concrete cylinder of $f_c' = 32\,\mathrm{MPa}$, so that $f_t = f_v = 0.5\sqrt{f_c'} = 2.828\,\mathrm{MPa}$ and $F_{\mathrm{I}} = f_{sy}A_s = 72.38\,\mathrm{kN}$, with a critical distance of $l_{\mathrm{ch}} = 20\,\mathrm{mm}$ and every stress evaluated at a reference load of $F_{\mathrm{ref}} = 50\,\mathrm{kN}$. The full parameter set is listed in Table 5.2.1.1, where the same specimen is carried into the sensitivity study.

5.1 Three methods of Critical Distances

5.1.1 Area Method

Theory

The Area Method replaces the stress at a point by its mean over a semicircle of radius $R = 1.32\, l_{\mathrm{ch}}$ centred on that point, the flat edge of the semicircle lying on the free surface. For the bond problem the semicircle lies in the (z, r) plane and sits on the bar surface, since no material exists below $r = r_s$.

The average is formed by dividing that half-disc into small squares of side l_{item} and representing each square by the stress at its centre. A square counts towards the average at station z_0 if its centre lies within R of (z_0, r_s). Writing N for the number of squares that qualify and (z_k, r_k) for their centres,

$$\bar{\sigma}_\theta(z_0) = \frac{1}{N}\sum_{k=1}^{N} \sigma_\theta(z_k, r_k), \qquad \bar{\tau}_{rz}(z_0) = \frac{1}{N}\sum_{k=1}^{N} \tau_{rz}(z_k, r_k). \tag{5.1}$$

Both stresses are averaged over the same squares, so the pair entering (3.31) is co-located by construction. Taking the stress at the centre of each square is the midpoint rule, so (5.1) approaches the true mean over the semicircle as l_{item} is reduced; how quickly it does so is the subject of Section 5.2.2.

The count N is not the same at every station. Away from the ends the half-disc is complete, but as z_0 approaches the loaded end part of it falls outside the bonded length. Those squares do not exist, so they are not counted and the divisor is reduced accordingly: the half-disc is progressively clipped, as shown in Figure 5.1.1.1. This matters because the governing station in this model almost always lies at the loaded end, so the average that decides the predicted capacity is usually taken over a clipped half-disc rather than a complete one.

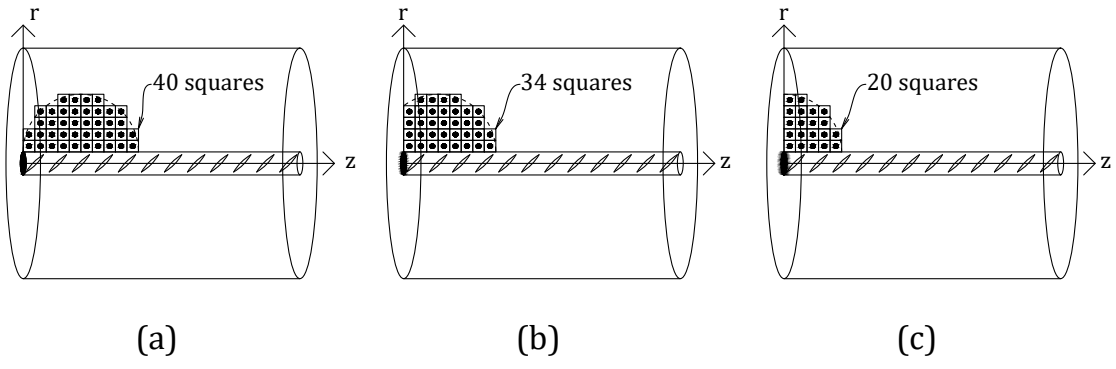


Figure 5.1.1.1: Area Method average at three stations along the bar. The half-disc of radius $R = 1.32\, l_{\mathrm{ch}}$ is tiled with squares of side l_{item}, and a square counts only if its centre lies inside it. (a) Complete region away from the ends; (b), (c) clipped as the station approaches the loaded end, where part of the region falls outside the bonded length. Counts are illustrative.

Averaging in both the axial and the radial direction makes the Area Method the only one of the three that responds to the axial gradient of the field, which at the loaded end is far steeper than the radial gradient.

Implementation

The solver follows the procedure set out in Chapter 3 and summarised there in Figure 3.8.3.2: the averaging size is fixed, the (z, r) plane is tiled with squares, the set of squares admitted to the half-disc is built once as a stencil, and that stencil is then placed at each station in turn, clipped where it overruns the bonded length, averaged, and converted to the Mode II index, with only the governing station reported.

Two features of that implementation bear on what follows. Each square is evaluated from the closed-form expressions of whichever layer its radius falls in, the rib layer for $r \le r_r$ and the concrete beyond, so no interpolation across the interface is involved and the average remains exact in the sense that every sample is an exact value of the analytical field. And the accuracy of (5.1) is controlled by the single parameter l_{item}, which fixes both the resolution of the half-disc and the spacing of the stations along the bar; it is the quantity refined in Section 5.2.2.

Worked example

With $l_{\text{item}} = 2\,\text{mm}$ and $R = 1.32 \times 20 = 26.4\,\text{mm}$, the tiling holds 75 stations along the bar and 14 radii outward from its surface, and a complete half-disc contains 273 squares. Sweeping (5.1) along the bar, the stresses at the maximum φ_{II} are

$$\bar{\tau}_{rz} = 8.818\,\text{MPa}, \qquad \bar{\sigma}_{\theta} = 6.818\,\text{MPa},$$

so that

$$\varphi_{\text{II,max}} = \left(\frac{6.818}{2.828}\right)^2 + \left(\frac{8.818}{2.828}\right)^2 - 1 = 5.81 + 9.72 - 1 = 14.53,$$

and substituting into (3.31),

$$F_{\text{II}} = \frac{F_{\text{ref}}}{\sqrt{\varphi_{\text{II,max}} + 1}} = \frac{50}{\sqrt{14.53 + 1}} = 12.69\,\text{kN}.$$

Since $F_{\text{II}} < F_{\text{I}} = 72.38\,\text{kN}$, Mode II governs and the predicted capacity is 12.69 kN, a mean bond stress of $F_u/(\pi d_b L) = 2.24\,\text{MPa}$. Both squared terms are substantial, the shear contributing 63% and the hoop 37%: the governing state is genuinely biaxial, and a criterion written on either component alone would be unconservative by a quarter or more.

5.1.2 Line Method

Theory

The Line Method replaces the stress at a point by its mean along a straight line of length $2\,l_{\mathrm{ch}}$ drawn from that point into the material. Here the line is drawn radially outward from the bar surface, so that at station z

$$\bar{\sigma}_\theta(z) = \frac{1}{2\,l_{\mathrm{ch}}} \int_{r_s}^{r_s+2l_{\mathrm{ch}}} \sigma_\theta(z, r)\ \mathrm{d}r, \tag{5.2}$$

and identically for $\bar{\tau}_{rz}$. The method carries no axial averaging at all: the value reported at z depends only on the field in the plane z.

Implementation

The method is the same device as the Area Method, reduced from a half-disc of squares to a single row of steps along one radius, and is carried out as follows.

1. **Set the averaging length.** The line runs radially outward from the bar surface over a distance $2\,l_{\mathrm{ch}}$, from r_s to $r_s + 2\,l_{\mathrm{ch}}$. The solver checks that it stays inside the specimen, $r_s + 2\,l_{\mathrm{ch}} \le r_e$, which for a 150 mm cylinder limits l_{ch} to 34.5 mm.

2. **Divide the line into steps.** It is cut into n_{item} equal steps of width $\Delta r = 2\,l_{\mathrm{ch}}/n_{\mathrm{item}}$.

3. **Place a sample at the centre of each step,** at $r_i = r_s + \left(i - \frac{1}{2}\right)\Delta r$ for $i = 1 \ldots n_{\mathrm{item}}$.

4. **Evaluate both stresses at every sampling radius,** taking τ_{rz} and σ_θ from the closed-form expressions of whichever layer the radius falls in: the rib layer for $r_i \le r_r$, the concrete beyond.

5. **Average the samples** to obtain the two values entering the index at that section,

$$\bar{\sigma}_\theta(z) \approx \frac{1}{n_{\mathrm{item}}} \sum_{i=1}^{n_{\mathrm{item}}} \sigma_\theta(z, r_i), \qquad \bar{\tau}_{rz}(z) \approx \frac{1}{n_{\mathrm{item}}} \sum_{i=1}^{n_{\mathrm{item}}} \tau_{rz}(z, r_i). \tag{5.3}$$

   Both are averaged over the same radii, so the pair is co-located by construction.

6. **Repeat at every station** on a uniform grid of $n_z = 801$ points spanning $0 \le z \le L$. This axial grid is a separate discretisation from the sampling along the line and is held fixed throughout.

7. **Take the worst station.** The phi value is calculated at each z and the largest value governs, giving the capacity from (3.31).

Sampling the step centres rather than the two ends of the line keeps every sample strictly inside the averaging region. In particular the bar surface itself is never sampled, which matters because the shear there is larger, and the hoop stress less representative of the material around it, than anywhere else on the line. It is also the more accurate choice: halving the step divides the error by four, whereas a rule that includes both ends of the line only halves it, because the two end points then carry more weight than they should.

Five steps are enough. Because the hoop stress varies almost linearly along the line, the rule converges quickly: raising n_{item} from 5 to 400 moves the predicted capacity by only 0.1%. The quadrature along the line is therefore not a meaningful source of error in this problem, and $n_{\mathrm{item}} = 5$ is used throughout.

Worked example

With $l_{\mathrm{ch}} = 20\,\mathrm{mm}$ and $n_{\mathrm{item}} = 5$, the line runs from $r_s = 6\,\mathrm{mm}$ to $46\,\mathrm{mm}$ in steps of $\Delta r = 8\,\mathrm{mm}$ and is sampled at the step centres, $r = 10,\ 18,\ 26,\ 34,\ 42\,\mathrm{mm}$. The index peaks at the loaded end, $z_{\mathrm{cr}} = 0$, where the shear boundary condition $\tau_{rz} = 0$ holds exactly and the hoop stress is at its maximum. The five samples are listed in Table 5.1.2.1.

Table 5.1.2.1: Line Method sampling at the governing station $z = 0$, reference specimen of Table 5.2.1.1. The shear vanishes identically because $\tau_{rz} = 0$ is imposed at both ends of the bonded length.

r (mm)	τ_{rz} (MPa)	σ_θ (MPa)
10	0	24.224
18	0	23.197
26	0	21.614
34	0	19.473
42	0	16.775
mean	0	21.057

Hence

$$\varphi_{\mathrm{II,max}} = \left(\frac{21.057}{2.828}\right)^2 + 0 - 1 = 54.42, \qquad F_{\mathrm{II}} = \frac{50}{\sqrt{54.42 + 1}} = 6.72\,\mathrm{kN}.$$

Figure 5.1.2.1 shows the construction at that section: the two stress components plotted against radius, the five sampled values marked on them, and the resulting average drawn as a level.

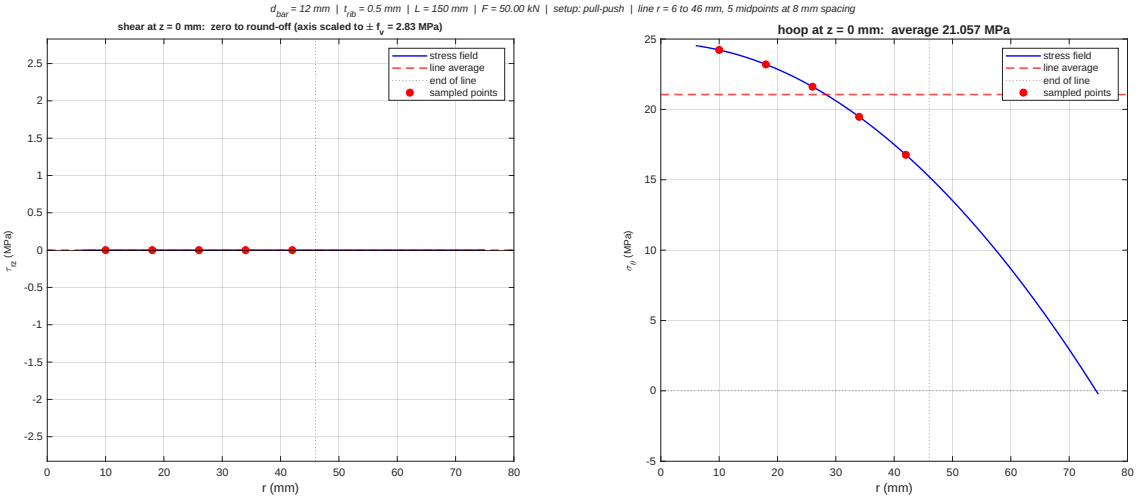


Figure 5.1.2.1: The Line Method average at the governing section $z = 0$. Markers are the five step centres at $r = 10,\ 18,\ 26,\ 34,\ 42\,\text{mm}$, the dashed line their mean, and the dotted line the end of the averaging line at 46 mm. Left: the shear, forced to zero at this section by the end boundary condition, plotted against $\pm f_v$. Right: the hoop stress, whose mean is 21.06 MPa.

5.1.3 Point Method

Theory

The Point Method performs no averaging at all. The stress entering (3.31) is the elastic stress evaluated at a single point a distance $l_{\text{ch}}/2$ from the stress raiser, measured here radially outward from the bar surface:

$$\bar{\sigma}_\theta(z) = \sigma_\theta(z,\ r_{\text{pt}})\,, \qquad \bar{\tau}_{rz}(z) = \tau_{rz}(z,\ r_{\text{pt}})\,, \qquad r_{\text{pt}} = r_s + \tfrac{1}{2} l_{\text{ch}}. \tag{5.4}$$

The offset preserves the non-local character of the criterion, so the capacity still carries a length scale; what is lost is any smoothing, so the prediction inherits whatever the sampling radius happens to pass through.

Implementation

There is no quadrature, no stencil and no averaging window to choose: the two closed-form expressions are evaluated once at $r = r_{\text{pt}}$ and substituted directly into (3.31). As in the Line Method the layer expressions are selected by radius, and the solver halts if $r_{\text{pt}} > r_e$.

Worked example

With $l_{\text{ch}} = 20\,\text{mm}$ the sampling radius is $r_{\text{pt}} = 16\,\text{mm}$. The index again peaks at $z_{\text{cr}} = 0$, where the shear vanishes and $\sigma_\theta = 23.506\,\text{MPa}$, giving

$$\varphi_{\text{II,max}} = \left(\frac{23.506}{2.828}\right)^2 - 1 = 68.07, \qquad F_{\text{II}} = \frac{50}{\sqrt{68.07 + 1}} = 6.02\,\text{kN}.$$

5.1.4 Comparison of the three methods

Table 5.1.4.1 collects the three predictions.

Table 5.1.4.1: The three critical-distance methods applied to the reference specimen of Table 5.2.1.1, $l_{\mathrm{ch}} = 20\,\mathrm{mm}$. Stresses and $\varphi_{\max}$ are at the production discretisation; the last column is the converged value from Section 5.2.

Method	$\bar{\tau}_{rz}$ (MPa)	$\bar{\sigma}_{\theta}$ (MPa)	$\varphi_{\max}$	F_{II} (kN)	converged (kN)
Point Method	0	23.51	68.07	6.02	6.02
Line Method ($n_{\mathrm{item}} = 5$)	0	21.06	54.42	6.72	6.72
Area Method ($l_{\mathrm{item}} = 2\,\mathrm{mm}$)	8.82	6.82	14.53	12.69	12.51

The Point and Line Methods agree closely with one another, while the Area Method is roughly twice either of them: the spread between the three converged predictions is 77% of their mean. That is a very large discrepancy for three methods calibrated to be identical, and the reason is specific to the bond problem.

The three methods were calibrated on the field ahead of a sharp crack, which decays the same way in every direction; sampling it along a line, over a semicircle or at a single point therefore gives the same answer. The bond field has no such symmetry. Along the bar the hoop stress collapses: at the Point Method's sampling radius it falls from 23.51 MPa at the loaded end to 2.14 MPa only 15 mm in, and reverses sign before 20 mm. Across the radius it barely changes, dropping 4% over the first 10 mm. The direction a method averages in therefore decides its answer. The Area Method averages along the bar as well as across it, so most of the hoop peak is averaged away: its worst section moves off the loaded end, the shear term returns, and the capacity doubles. The Point and Line Methods average only across the radius, where there is almost nothing to average away.

This is why the Area Method is adopted for the remainder of this work. It is the only one of the three that samples the axial gradient, and the axial gradient is where the physics of the bond problem lies: the stress concentration at the loaded end is a z-direction feature, not an r-direction one. A further consequence is that the Point and Line Methods are largely insensitive to the embedment length, since the state they sample at $z = 0$ is governed by the end condition alone, whereas the Area Method's average responds to how quickly the field decays into the bar.

The three methods also differ in how the governing mechanism is selected. For the Area Method it depends on l_{ch}: at 5 mm the half-disc is too small to reach the end hoop peak and the interior shear maximum at $z = 18.5\,\mathrm{mm}$ governs, whereas for $l_{\mathrm{ch}} \geq 10\,\mathrm{mm}$ the end region does. The Point and Line Methods select $z = 0$ for every l_{ch} tested. The Area Method is thus the only one in which the competition between the shear mechanism along the bar and the splitting mechanism at the loaded end is resolved by the model rather than fixed in advance.

5.2 Sensitivity Analysis

Each of the three methods requires a discretisation, and each discretisation introduces an error that has nothing to do with the mechanics. The Area Method must tile its half-disc; the Line Method must place quadrature points along its line; all three must sweep a grid of stations along the bar. This section quantifies that error by refining each method's own discretisation until its answer stops moving, and reporting how far every coarser run sits from that converged value.

5.2.1 Reference specimen

Every result in this chapter, both the worked examples of Section 5.1 and the convergence study that follows, uses the single configuration of Table 5.2.1.1, evaluated in the pull-push arrangement. It is a 12 mm bar embedded 150 mm into a 150 mm concrete cylinder, so the embedment is $12.5\,d_b$ and the cover is more than six bar diameters: short enough that the stress distribution stays close to elastic up to failure, and thick enough that the outer boundary does not control the hoop stress at the bar surface.

Table 5.2.1.1: Reference specimen used throughout this chapter.

Quantity	Symbol	Value
Bar diameter	d_b	12 mm
Rib layer thickness	t_{rib}	0.5 mm
Concrete cylinder diameter	d_{cyl}	150 mm
Bonded length	L	150 mm
Bar modulus, Poisson ratio	$E_s,\ \nu_s$	200 GPa, 0.3
Bar yield stress	f_{sy}	500 MPa
Concrete modulus, Poisson ratio	$E_c,\ \nu_c$	30.1 GPa, 0.2
Concrete strength	f'_c	32 MPa
Tensile, shear strength	$f_t = f_v = 0.5\sqrt{f'_c}$	2.828 MPa
Reference load	F_{ref}	50 kN
Critical distance	l_{ch}	20 mm
Bar stress at F_{ref}	$f = F_{\text{ref}}/A_s$	442.1 MPa
Mode I capacity	$F_{\text{I}} = f_{sy}A_s$	56.55 kN

Three conventions govern the study that follows. First, l_{ch} and all specimen parameters are held at the values of Table 5.2.1.1, so every difference reported here is numerical rather than physical. Second, each method is measured against its *own* finest run rather than against the other methods: the gap between the three converged values quantified in Section 5.1.4 is a modelling choice, not a discretisation error, and mixing the two would make both unreadable. Third, nothing is re-derived between runs. The symbolic solution of Chapter 3 is obtained once and the four stress expressions compiled to numerical functions once, so that refining a mesh changes only how often those closed-form expressions are evaluated, never what they are.

5.2.2 Mesh Size vs Pullout Load

Area Method

Refinement means raising the number of steps n_{item} along the line. The results are in Table 5.2.2.2.

Table 5.2.2.1: Line Method: convergence of the predicted Mode II capacity with the number of steps along the averaging line. Line length 40 mm; each run averages the stress at the n_{item} step centres, as in (5.3).

n_{item}	step (mm)	F_{II} (kN)	error (%)
1	40.0	6.5432	−2.685
2	20.0	6.6776	−0.685
5	8.0	6.7163	−0.110
10	4.0	6.7218	−0.027
20	2.0	6.7232	−0.007
50	0.8	6.7235	−0.002
100	0.4	6.7237	—

The rule converges from below and at second order: each halving of the step divides the error by four. Convergence is from below because the samples never reach the bar surface, where the hoop stress is highest, so every finite step underestimates the mean slightly. In absolute terms the error is negligible almost immediately: two steps are already within 0.7% of the converged 6.7237 kN and the production setting of five is within 0.11%. Beyond about twenty steps the remaining differences are smaller than the rounding in the fourth decimal place. The quadrature along the line is therefore not a meaningful source of error in this problem, because the radial variation it integrates is close to linear.

Line Method

Refinement means raising the number of steps n_{item} along the line. The results are in Table 5.2.2.2.

Table 5.2.2.2: Line Method: convergence of the predicted Mode II capacity with the number of steps along the averaging line. Line length 40 mm; each run averages the stress at the n_{item} step centres, as in (5.3).

n_{item}	step (mm)	F_{II} (kN)	error (%)
1	40.0	6.5432	−2.685
2	20.0	6.6776	−0.685
5	8.0	6.7163	−0.110
10	4.0	6.7218	−0.027
20	2.0	6.7232	−0.007
50	0.8	6.7235	−0.002
100	0.4	6.7237	—

The rule converges from below and at second order: each halving of the step divides the error by four. Convergence is from below because the samples never reach the bar surface, where the hoop stress is highest, so every finite step underestimates the mean slightly. In absolute terms the error is negligible almost immediately: two steps are already within 0.7% of the converged 6.7237 kN and the production setting of five is within 0.11%. Beyond about twenty steps the remaining differences are smaller than the rounding in the fourth decimal place. The quadrature along the line is therefore not a meaningful source of error in this problem, because the radial variation it integrates is close to linear.

Point Method

The Point Method averages nothing, so it carries no quadrature error at all. Its only discretisation is the set of sections at which the index is formed while searching for the governing one, and for this specimen that search is exact at every spacing tested: refining it by a factor of sixty-four returns $F_{\mathrm{II}} = 6.016\,31$ kN every time.

The reason is that the φ value peaks at the loaded end and every grid begins at the loaded end, so the governing section is sampled however coarse the spacing is. This is not particular to one critical distance. Across the range $l_{\mathrm{ch}} = 5$ mm to 30 mm the hoop stress at the loaded end dominates the index and the interior shear maximum never governs, so the governing section stays at $z = 0$ throughout. The Point Method is therefore mesh independent for this problem, and the spacing along the bar can be as coarse as convenience allows.

That independence follows from where the maximum lies rather than from anything in the method, and it would be lost if the governing section were interior, as it is for the Area Method at small critical distances. A grid deliberately offset by half a spacing, so that the loaded end is not sampled, confirms as much: the predicted capacity then falls from 7.03 kN to 6.03 kN as the spacing is refined, halving its error at each step, because with no averaging to flatten the peak the error depends entirely on how close the nearest section falls to it. The configurations examined in this work do not produce that situation, but it is the reason the axial spacing of the other two methods is held fixed rather than treated as free.

Summary

Table 5.2.2.3 collects the converged values and the coarsest mesh that reaches 1% of each.

Table 5.2.2.3: Converged Mode II capacities and the coarsest discretisation within 1% of each.

Method	refinement parameter	coarsest within 1%	converged F_{II} (kN)
Area	l_{item}	1 mm	12.508
Line	n_{item}	2	6.724
Point	section spacing	any spacing tested	6.016

At any reasonable mesh the discretisation error of each method is an order of magnitude smaller than the 77% spread between the methods themselves. The choice of averaging geometry therefore dominates the choice of mesh by a wide margin, and the calibration of Chapter [X] is a calibration of that choice together with l_{ch}, not of the numerics.

5.2.3 Mesh Size vs Computation Time

Cost is measured as the time to produce one value of F_{II} for a given method and mesh: building the sampling grid, evaluating the two stress fields on it, averaging, forming the index and taking the worst station. The one-off symbolic derivation and the compilation of the stress expressions are excluded, because they are shared by all three methods and would swamp the comparison. Each case is repeated until at least a fixed interval of wall clock has elapsed and the total divided by the repeat count, so that the measurement is not limited by timer resolution.

Table 5.2.3.1: Computation time per evaluation of F_{II} against mesh refinement, with the discretisation error of Section 5.2.2 alongside. Times measured on an Intel Core i9-12900H (20 logical cores, 2.5 GHz, 32 GB RAM) running MATLAB R2026b under Windows 11.

Method	Mesh	Items per station	error (%)	time (ms)
Area	$l_{\mathrm{item}} = 4\,\mathrm{mm}$	69	+3.85	0.29
Area	$l_{\mathrm{item}} = 2\,\mathrm{mm}$	273	+1.43	0.55
Area	$l_{\mathrm{item}} = 1\,\mathrm{mm}$	1092	+0.56	2.67
Area	$l_{\mathrm{item}} = 0.75\,\mathrm{mm}$	1947	+0.43	5.70
Area	$l_{\mathrm{item}} = 0.5\,\mathrm{mm}$	4383	+0.29	21.93
Area	$l_{\mathrm{item}} = 0.25\,\mathrm{mm}$	17516	—	139.79
Line	$n_{\mathrm{item}} = 2$	2	−0.685	0.23
Line	$n_{\mathrm{item}} = 10$	10	−0.027	0.31
Line	$n_{\mathrm{item}} = 50$	50	−0.002	0.39
Line	$n_{\mathrm{item}} = 100$	100	—	0.50
Point	For all values	1	—	0.041

Three features of Table 5.2.3.1 matter for the way the model is used.

The Area Method's cost grows as l_{item}^{-3}. Halving the item side multiplies the work eightfold, because the number of items inside the half-disc grows as l_{item}^{-2} while the number of stations at which that average must be formed grows as l_{item}^{-1}. The measured cost bears this out: refining from 1 mm to 0.25 mm raises the time from 2.67 ms to 139.79 ms, a factor of 52 against the 64 predicted, and the times rise monotonically across the whole range. Combined with the first-order convergence of Section 5.2.2, cubic cost growth means the Area Method's cost rises as the inverse cube of the accuracy demanded: cutting the discretisation error by ten costs a thousandfold in time.

The Line and Point Methods are effectively free by comparison. The Line Method's cost is almost flat below $n_{\text{item}} = 50$ because it is dominated by the fixed axial grid rather than by the sampling along the line, and the Point Method's cost is simply that of evaluating two closed-form expressions at every section. Both remain below 0.6 ms at every mesh tested, more than two orders of magnitude below the Area Method's finest run.

Most importantly, the accuracy actually required is cheap in every case. The coarsest mesh within 1% of the converged value costs 2.67 ms for the Area Method against 139.79 ms for the finest run, a saving of a factor of 52; for the Line Method the corresponding figures are 0.23 ms against 0.50 ms. Running each method at its finest tested mesh buys at most 0.6 percentage points of accuracy for one to two orders of magnitude in time.

This matters because the model is not run once. The calibration of l_{ch} in Chapter [X] evaluates the Mode II capacity across a database of specimens, and the parametric study of Chapter [X] sweeps bar diameter, embedment length and concrete strength, so a single capacity evaluation is executed thousands of times. At $l_{\text{item}} = 1\,\text{mm}$ a sweep of 10 000 cases costs under half a minute, against roughly 23 min at 0.25 mm, for a difference of less than one percent in every predicted capacity. The working value $l_{\text{item}} = 1\,\text{mm}$ is adopted on that basis.

These times exclude the symbolic solution itself, which is obtained once per specimen configuration and dominates the total runtime of the solver. On the evidence of Table 5.2.3.1 the discretisation of the critical-distance averaging is not a bottleneck in this model at any mesh accurate enough to use.

Chapter 6

Parametric Study of Factors Influencing Bond

The preceding chapters established the model and showed that it reproduces the stress fields of two published analytical solutions. This chapter puts it to work on the question it was built to answer: which properties of reinforced concrete affect anchorage, control its strength, its stiffness and the slip at which it fails, and by how much.

Ten factors are examined, each varied about a common baseline while the rest are held fixed. For every case the chapter reports the predicted cracking load, the mean bond stress, the bond stiffness and the slip, together with the two normalised stress components that identify how the failure criterion is being satisfied. The outcome is a ranking of the factors by influence, a decomposition of the concrete-strength effect into its strength and stiffness parts, and an identification of the one assumption in the formulation that is both unverified and consequential.

The findings established here are carried into Chapter 7, which assembles them into calibration curves and inverts those curves to give a development length expression.

6.1 Approach

6.1.1 Baseline specimen

Every sweep changes one property at a time and holds the rest at the baseline of Table 6.1.1.1: a 12 mm deformed bar embedded 150 mm into a 150 mm diameter cylinder of 32 MPa concrete. This is the geometry of the pull-out specimens used for comparison in Chapter [X] and the case used for the mesh study of Chapter 5.

Table 6.1.1.1: Baseline specimen for the parametric study.

Property	Symbol	Value
Bar diameter	d_b	12 mm
Embedment length	L	150 mm
Cylinder diameter	d_{cyl}	150 mm
Rib layer thickness	t_{rib}	0.5 mm
Concrete strength	f'_c	32 MPa
Concrete modulus	E_c	30 100 MPa
Concrete Poisson's ratio	ν_c	0.20
Steel modulus	E_s	200 GPa
Steel yield stress	f_{sy}	550 MPa
Tensile and shear strength	$f_t = f_v$	$0.5\sqrt{f'_c} = 2.83$ MPa
Critical distance	l_{char}	20 mm
Item size	l_{item}	2 mm
Loading configuration	—	pull–push

The concrete modulus follows Table 3.1.2 of Standards Australia (2018) and is therefore tied to f'_c wherever the strength is varied. Section 6.3.2 breaks that tie deliberately, and doing so resolves a discrepancy with the bond literature that would otherwise be unexplained.

6.1.2 Quantities reported

Four outputs are reported for every case. The Mode II cracking load F_{II} comes from the Area Method of Chapter 5 through the quadratic interaction index of (3.36). The mean bond stress

$$\tau_u = \frac{F_u}{\pi d_b L} \tag{6.1}$$

removes the trivial growth of capacity with bonded area and is the form in which experimental bond correlations are reported. The bond stiffness k_s of (3.28) and the slip δ_u of (3.29) come from the same solved stress field at no additional cost. Both are *initial* values: the formulation is linearly elastic, so (3.29) extrapolates the initial tangent out to the cracking load. A measured bond–slip response softens progressively well before its peak, so throughout this chapter δ_u is a lower bound on measured slip and k_s an upper bound on the stiffness at peak load. Finally, the two normalised stresses at the governing station,

$$\hat{\sigma} = \left.\frac{\bar{\sigma}_\theta}{f_t}\right|_{F_u}, \qquad \hat{\tau} = \left.\frac{\bar{\tau}_{rz}}{f_v}\right|_{F_u}, \qquad \hat{\sigma}^2 + \hat{\tau}^2 = 1, \tag{6.2}$$

record how the failure criterion is being satisfied, and so identify whether a case is failing in hoop tension or in interfacial shear.

6.1.3 Factors considered and the threshold for influence

Ten factors are swept, in two groups: geometric (embedment length, bar diameter at fixed L and at fixed L/d_b, outer radius, rib layer thickness) and material (concrete strength with E_c tied, concrete modulus with f'_c fixed, shear-to-tensile strength ratio, Poisson's ratio, rib layer modulus, steel modulus).

Influence is quantified throughout by the exponent n of a power law $F_{II} \propto x^n$ fitted across each swept range, because an exponent is independent of the range chosen while a percentage change is not. A factor is described as *negligible* where $|n| < 0.05$, as *moderate* for $0.05 \leq |n| \leq 0.5$, and as *dominant* above that. 1

6.1.4 Isolating a mechanism by removing it

Two of the results below are obtained not by varying a property but by breaking a link that is normally fixed. Concrete strength and modulus rise together in any real concrete, so the sweep in Section 6.3.1 can only report their combined effect; Section 6.3.2 separates them by holding the strength constant and varying the modulus alone, which is not a physically realisable concrete but is a perfectly admissible numerical experiment. The same logic identifies the cause of the non-monotonic cover response in Section 6.2.3. Where a mechanism is suspected, the most direct test is to remove it and see whether the behaviour disappears.

6.1.5 Factors outside the scope of the model

Two factors that literature treats as essential are absent from the model. Transverse confinement enters Orangun, Jirsa, and Breen (n.d.) through K_{tr} and Zuo, Leon, et al. (n.d.) through the area and spacing of ties, but no transverse steel is modelled here: the cylinder is plain concrete with no spiral. The omission matters least at for large cover, where the concrete ring confines the bar on its own, and most at small cover where splitting governs, the same regime in which Section 6.2.3 finds the averaging method least reliable.

Rib geometry, which Ng, McGregor, et al. (n.d.) exploit to shorten development lengths for Grade 600 bars, is absent for the same kind of reason: the bar modelled is plain, its interlock carried by a smeared rib layer rather than by ribs of a given height, spacing and face angle. Only that layer's thickness and modulus can be varied, and Sections 6.2.4 and 6.3.4 test both.

The two are not equally fundamental. A spiral is axisymmetric once smeared into an equivalent radial stiffness, so confinement could be added to the present formulation; discrete ribs are break the axial symmetry it depends on, so rib geometry could not. Both are returned to in Section 6.5.

6.2 Geometric factors

6.2.1 Embedment length

Lengthening the anchorage raises the predicted cracking load, but only up to a point. Table 6.2.1.1 shows the baseline bar embedded between 50 and 300 mm: the load rises steeply to 100 mm and then stops, gaining 1.1 % over the remaining 200 mm.

Table 6.2.1.1: Effect of embedment length ($d_b = 12$ mm).

L (mm)	L/d_b	F_{II} (kN)	τ_u (MPa)	k_s (kN/mm)	δ_u (mm)	$\hat{\sigma}$	$\hat{\tau}$
50	4.17	9.310	4.939	537.0	0.01734	0.629	0.778
75	6.25	12.054	4.263	621.9	0.01938	0.622	0.783
100	8.33	12.597	3.341	630.7	0.01997	0.616	0.788
125	10.42	12.668	2.688	631.3	0.02007	0.613	0.790
150	12.50	12.688	2.244	631.2	0.02010	0.612	0.791
200	16.67	12.709	1.686	631.2	0.02014	0.610	0.793
250	20.83	12.722	1.350	631.1	0.02016	0.609	0.793
300	25.00	12.731	1.126	631.1	0.02017	0.608	0.794

Three features of this table matter.

The mean bond stress falls, but the bond does not weaken. τ_u drops from 4.94 to 1.13 MPa across the range, which follows from how the quantity is defined rather than from any change in the concrete. The load itself has stopped growing: Figure 6.2.1.1 shows it flattening by about 100 mm, with a gain of only 1.1 % as the embedment then triples. Since $\tau_u = F_u/(\pi d_b L)$, a constant load divided by a growing length must fall as $1/L$.

The stiffness levels off at the same embedment. By $L = 125$ mm, it reaches 631 kN mm^{-1}, up from 537; it is then flat to within 0.03 %. This rules out an awkward objection. The averaging disc has a radius of 26.4 mm and sits at the loaded end, so the failure criterion sees only the first 27 mm of the bar, and the capacity might stop growing simply because the criterion cannot look further along. The stiffness answers that, because it is computed from the energy stored over the *whole* embedded length and uses nothing from the criterion. Were the flattening a blind spot, the capacity would level off while the stiffness kept rising. Both flatten together, so load transfer is genuinely complete by 100 mm.

The slip at failure is constant. It holds at 0.0201 mm for every embedment beyond 100 mm. This follows from the two results above, since $\delta_u = F_u/k_s$ and both level off, but it is worth stating because it is directly testable: first cracking is predicted at the same loaded-end displacement however long the anchorage. The constancy holds along this axis only: Section 6.2.2 shows δ_u falling as $d_b^{-0.43}$ with bar diameter, so this is a stress criterion that happens not to depend on length, not a slip criterion in disguise.

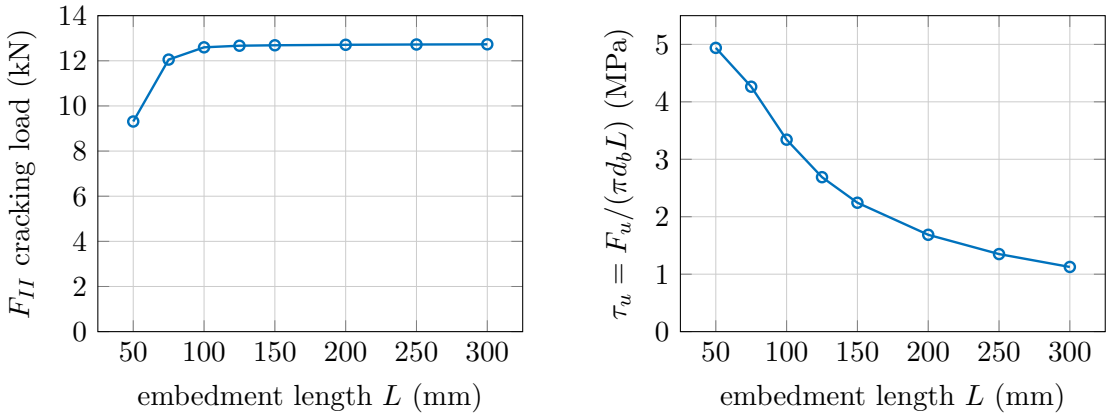


Figure 6.2.1.1: Effect of embedment length on the baseline specimen (d_b = 12 mm, f'_c = 32 MPa, r_e = 75 mm). (a) The predicted cracking load levels off by about 100 mm, gaining only 1.1 % over the remaining 200 mm. (b) The mean bond stress continues to fall as $1/L$, because a load that has stopped growing is being divided by a length that has not.

6.2.2 Bar diameter

Bar diameter is the most influential geometric factor. Table 6.2.2.1 varies it with the embedment held at the baseline, and Table 6.2.2.2 repeats the sweep at a constant slenderness of $L/d_b = 10$ so that bar size and slenderness can be separated.

Table 6.2.2.1: Effect of bar diameter at fixed embedment length (L = 150 mm).

d_b (mm)	L/d_b	F_{II} (kN)	τ_u (MPa)	k_s (kN/mm)	δ_u (mm)	$\hat{\sigma}$	$\hat{\tau}$
8	18.75	8.826	2.341	365.0	0.02418	0.501	0.866
10	15.00	10.760	2.283	494.1	0.02178	0.562	0.828
12	12.50	12.688	2.244	631.2	0.02010	0.612	0.791
16	9.38	16.443	2.181	924.0	0.01780	0.690	0.724
20	7.50	19.983	2.120	1233.0	0.01621	0.747	0.665
24	6.25	23.256	2.056	1550.1	0.01500	0.790	0.614

Table 6.2.2.2: Effect of bar diameter at constant slenderness ($L/d_b = 10$, so the embedment varies with the bar).

d_b (mm)	L (mm)	F_{II} (kN)	τ_u (MPa)	k_s (kN/mm)	δ_u (mm)	$\hat{\sigma}$
8	80	8.770	4.362	364.9	0.02403	0.506
10	100	10.722	3.413	494.0	0.02170	0.565
12	120	12.662	2.799	631.3	0.02006	0.614
16	160	16.456	2.046	923.9	0.01781	0.689
20	200	20.067	1.597	1233.0	0.01627	0.745
24	240	23.447	1.296	1550.6	0.01512	0.787

Cracking capacity grows almost in proportion to bar diameter. Fitting a power law to each table (Appendix A.2) gives $F_{II} \propto d_b^{0.886}$ at fixed L and $d_b^{0.898}$ at fixed L/d_b, so

$$F_{II} \propto d_b^{0.89}, \tag{6.3}$$

close to direct proportionality (Figure 6.2.2.1). Capacity is not proportional to the bonded area $\pi d_b L$: at constant slenderness that area grows ninefold while the capacity grows only 2.67-fold. Load transfer is confined to a zone near the loaded end that does not lengthen with the bar (Section 6.2.1), so most of the added interface carries almost nothing.

Demand grows faster. The yield force $A_s f_{sy}$ goes as d_b^2 against the $d_b^{0.89}$ of (6.3), so doubling the bar multiplies capacity by 1.85 and demand by four. This is the chapter's most consequential result, and the mechanical reason anchoring large high-grade bars is the binding constraint identified by Ng and McGregor (n.d.).

The failure mode shifts with bar size. Squaring the last two columns of Table 6.2.2.1 gives each mechanism's share of the failure criterion, the two summing to one. At $d_b = 8$ mm the split is 75 % shear to 25 % hoop; by 24 mm it has reversed to 38 and 62 %, crossing near 17 mm. Shear is a pull-out failure and hoop tension a splitting failure, so the model predicts small bars pulling out and large bars splitting, the transition reported by Li et al. (2012) and one it was not fitted to reproduce.

Stiffness grows faster than strength. The bond stiffness rises as $k_s \propto d_b^{1.32}$, faster than the capacity, so the slip at failure falls as the difference of the exponents, $\delta_u \propto d_b^{-0.43}$. Against a free bar of the same size the anchorage is stiffer by a falling margin, 5.45 times at 8 mm against 2.57 at 24 mm, because only the loaded end is ever stretched and that region is relatively shorter for a small bar.

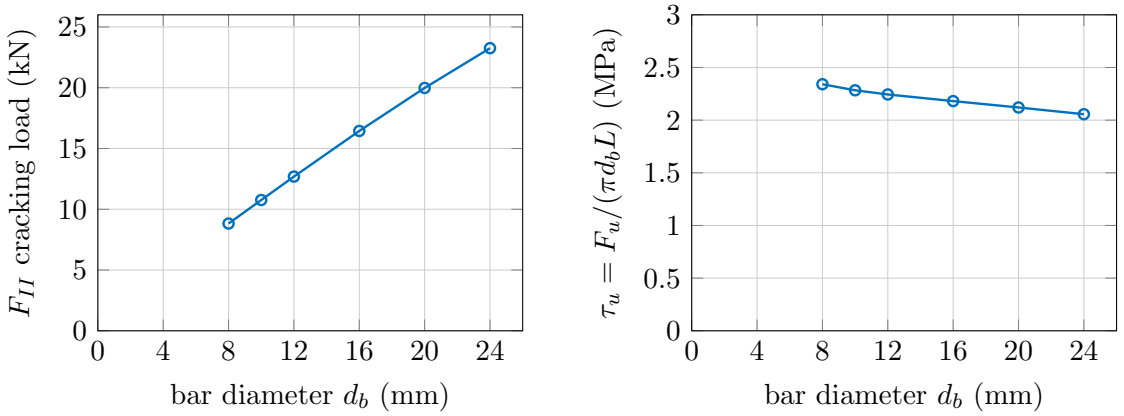


Figure 6.2.2.1: Effect of bar diameter at fixed embedment ($L = 150$ mm, $f'_c = 32$ MPa, $r_e = 75$ mm). (a) The cracking load rises almost in proportion to the diameter, following $d_b^{0.89}$. (b) The mean bond stress is nearly constant, falling only 12 % over a threefold increase in bar size, which is the same result seen another way.

6.2.3 Cover ratio

Cover is the dominant geometric variable in every empirical bond model: it appears as c/d_b in Orangun, Jirsa, and Breen (n.d.), as $(c+0.5d_b)$ in Zuo and Darwin (2000), and Hadi (2008) isolates it by casting the same bars in two cylinder sizes. Here it enters through the outer radius r_e and is reported as

$$\frac{c}{d_b} = \frac{r_e - d_b/2}{d_b}. \tag{6.4}$$

Table 6.2.3.1: Effect of concrete cover ($d_b = 12\,\text{mm}$, $L = 150\,\text{mm}$). The first two rows are not comparable with the rest; see the text.

r_e (mm)	c/d_b	radial items (of 14 needed)	F_{II} (kN)	τ_u (MPa)	k_s (kN/mm)	δ_u (mm)	$\hat{\sigma}$
25	1.58	9†	10.087	1.784	826.0	0.01221	0.252
30	2.00	12†	11.393	2.015	808.4	0.01409	0.348
40	2.83	17	11.079	1.959	761.9	0.01454	0.489
50	3.67	22	11.282	1.995	717.4	0.01573	0.548
60	4.50	27	11.768	2.081	678.9	0.01734	0.582
75	5.75	34	12.688	2.244	631.2	0.02010	0.612
90	7.00	42	13.696	2.422	593.0	0.02310	0.631
110	8.67	52	15.091	2.669	552.2	0.02733	0.648

† The averaging disc of radius $R_{avg} = 26.4\,\text{mm}$ does not fit inside the concrete and is clipped at the outer face, so these two cases average over less material than the rest and are not directly comparable.

Two points must be set aside. The load in Table 6.2.3.1 does not rise steadily: it climbs from 10.09 to 11.39 kN between $r_e = 25$ and 30 mm, then *falls* to 11.08 kN at 40 mm. Since the only thing changing is the amount of concrete, that dip needs explaining, and the cause turns out to be numerical rather than physical.

The Area Method averages the stress over a half-disc of radius 26.4 mm centred on the bar. In the two smallest specimens that disc does not fit inside the concrete, so it is cut short at the outer face and the criterion averages over a smaller and more highly stressed region than elsewhere. The hoop share confirms it: 0.25 in the first row against 0.49 in the first uncut row, so those two cases are evaluating a different stress state, not simply a smaller one. They are shown as open markers in Figure 6.2.3.1, and the solver now warns whenever the disc is cut.

Setting them aside, the remaining six points rise steadily and give

$$F_{II} \propto \left(\frac{c}{d_b}\right)^{0.281}, \qquad R^2 = 0.943, \tag{6.5}$$

a 36 % gain as the cover ratio rises from 2.83 to 8.67. Fitting all eight would have given 0.199 with $R^2 = 0.857$: the two cut cases pull the exponent down by nearly a third.

The cover effect is weaker than measurements show. An exponent of 0.28 is modest beside the cover terms of Orangun, Jirsa, and Breen (n.d.), and the reason is built into the method. Concrete more than 26.4 mm from the bar lies outside the averaging disc, so it is never sampled directly; it reaches the stresses that decide failure only indirectly, by stiffening the surrounding ring and redistributing the field. That indirect route is the weaker of the two, which is why the model understates the benefit of cover whenever the cover exceeds the disc radius, here $c/d_b \gtrsim 2.2$.

Cover buys strength but costs stiffness. Across the six admissible points the capacity rises by 36 % while the stiffness *falls* by 28 % and the slip at failure almost doubles. A thicker ring of concrete carries lower stress and spreads the load transfer over a longer part of the bar, so more of the bar stretches and the loaded end moves further. In design terms, extra cover improves resistance to splitting but does nothing for stiffness in service, and slightly worsens it. The hoop share rises from 0.49 to 0.65 over the same range, so more cover also pushes the predicted failure towards splitting.

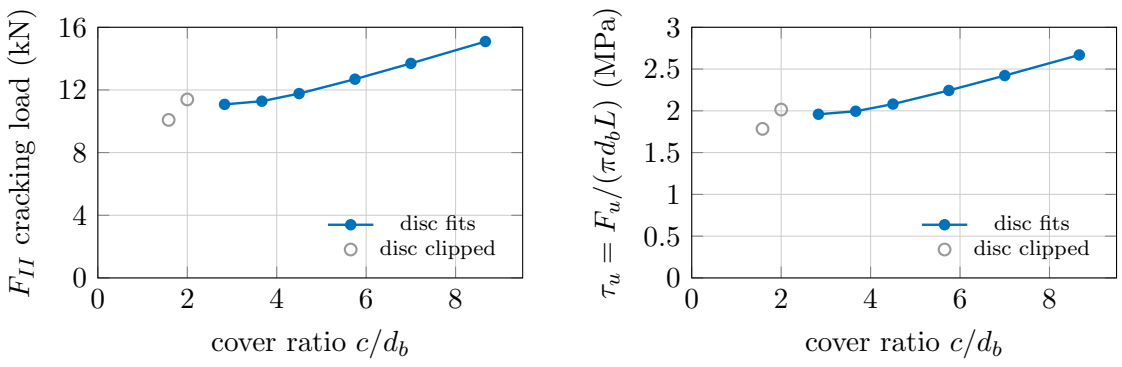


Figure 6.2.3.1: Effect of concrete cover ($d_b = 12\,\text{mm}$, $L = 150\,\text{mm}$, $f'_c = 32\,\text{MPa}$) on (a) the cracking load and (b) the mean bond stress. The two open markers are the cases in which the averaging disc does not fit inside the concrete and is cut short at the outer face; they are excluded from the fit of (6.5). The remaining six points rise steadily.

6.2.4 Rib layer thickness

The rib layer is the least physically grounded part of the idealisation. It stands in for the mechanical interlock between the lugs and the surrounding concrete, and in the implementation it is an annulus assigned the same moduli as the concrete. Rib geometry measurably changes real bond behaviour (Section 6.1.5), so a layer that smears it away needs its sensitivity established rather than assumed.

Table 6.2.4.1: Sensitivity to the assumed rib layer thickness.

t_{rib} (mm)	F_{II} (kN)	τ_u (MPa)	k_s (kN/mm)	δ_u (mm)
0.25	12.691	2.244	631.5	0.02010
0.50	12.688	2.244	631.2	0.02010
0.75	12.685	2.243	631.0	0.02010
1.00	12.680	2.242	630.7	0.02010
1.50	12.676	2.242	630.2	0.02012
2.00	12.672	2.241	629.6	0.02013

An eightfold change in the assumed thickness moves the capacity by 0.15 % and the stiffness by 0.30 %. The rib layer is a harmless numerical device: it supplies the kinematic continuity the three-layer formulation requires without influencing the answer. Whatever limitations the model has, they do not arise from this parameter.

6.3 Material factors

6.3.1 Concrete strength

Concrete strength enters the model twice: through the strengths $f_t = f_v = 0.5\sqrt{f'_c}$ that normalise the interaction index, and through the modulus E_c that determines how stress is distributed in the first place. Table 6.3.1.1 varies both together, as they vary together in a real concrete.

Table 6.3.1.1: Effect of concrete strength, with E_c following Standards Australia (2018) Table 3.1.2.

f'_c (MPa)	E_c (MPa)	F_{II} (kN)	τ_u (MPa)	k_s (kN/mm)	δ_u (mm)
20	24000	10.770	1.905	577.7	0.01864
25	26700	11.642	2.059	602.5	0.01932
32	30100	12.688	2.244	631.2	0.02010
40	32800	13.815	2.443	652.3	0.02118
50	34800	15.170	2.683	667.1	0.02274
65	37400	16.925	2.993	685.4	0.02469
80	39600	18.459	3.264	700.1	0.02637

A power law fitted across the range gives

$$\tau_u \propto (f'_c)^{0.390}, \qquad R^2 = 0.998. \tag{6.6}$$

This sits between the two values the literature offers. If capacity were controlled by strength alone it would follow $f_t \propto \sqrt{f'_c}$ and the exponent would be 0.50; the descriptive equations of Orangun, Jirsa, and Breen (n.d.) normalise by $\sqrt{f'_c}$ on exactly that reasoning. Against this, Zuo and Darwin (2000) found that splice strength without confining reinforcement is better characterised by $(f'_c)^{1/4}$, and Zuo, Leon, et al. (n.d.) adopted the quarter power.

Harajli (2004) reconciles the two by showing that the appropriate exponent depends on the development length: the quarter power collapses normal- and high-strength data for $L/d_b \gtrsim 17$–20 while under-representing the effect of f'_c at shorter lengths. The present baseline, at $L/d_b = 12.5$, lies below that threshold, so an exponent above 0.25 is what Harajli's finding would lead one to expect. Section 6.3.2 accounts for the remainder of the gap below 0.50.

Figure 6.3.1.1 shows the trend, and three further points follow from it.

The return on extra strength diminishes. Going from 20 to 25 MPa buys 0.174 kN of capacity for each megapascal added, but going from 65 to 80 MPa buys only 0.102 kN, a fall of 41 %. Any exponent below one behaves this way, and it is the gentle flattening of the curves in Figure 6.3.1.1.

The exponent is not quite constant. Fitting between each pair of successive points, rather than across the whole range at once, gives a value that climbs steadily from 0.349 at the bottom to 0.418 at the top. Equation (6.6) is therefore a good approximation over the range tested rather than an exact description, and Section 6.3.2 identifies the cause.

Stiffness benefits far less than strength. The bond stiffness follows $k_s \propto (f'_c)^{0.137}$, so quadrupling the concrete strength raises the capacity by 71 % but the stiffness by only 21 %. The slip at failure follows from those two, since $\delta_u = F_u/k_s$, and grows by 41 %: a stronger concrete carries more load without becoming proportionally stiffer, so it stretches further before it cracks.

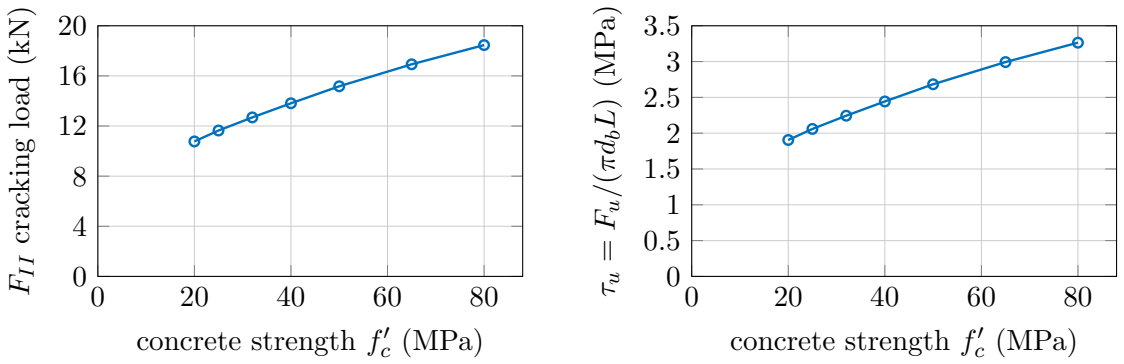


Figure 6.3.1.1: Effect of concrete strength, with E_c following Standards Australia (2018) Table 3.1.2 ($d_b = 12\,\text{mm}$, $L = 150\,\text{mm}$, $r_e = 75\,\text{mm}$). (a) The predicted cracking load and (b) the mean bond stress, both rising as $(f'_c)^{0.390}$ and flattening gently as the strength increases. Because d_b and L are fixed throughout this sweep the two differ only by the constant bonded area.

6.3.2 Concrete stiffness

Because the elastic modulus E_c and the strength f'_c were tied in the previous sweep their separate contributions cannot be read from it. Table 6.3.2.1 breaks the tie by varying the modulus at constant strength, so that f_t and f_v are fixed and only the stress distribution changes.

Table 6.3.2.1: Effect of concrete modulus at constant strength ($f'_c = 32$ MPa, so $f_t = f_v$ unchanged).

E_c (MPa)	F_{II} (kN)	τ_u (MPa)	k_s (kN/mm)	δ_u (mm)
24000	13.623	2.409	577.7	0.02358
27000	13.125	2.321	605.2	0.02169
30100	12.688	2.244	631.2	0.02010
33000	12.334	2.181	653.8	0.01886
36000	12.012	2.124	675.7	0.01778
40000	11.640	2.058	702.7	0.01656

The capacity *falls* as the modulus rises, following $\tau_u \propto E_c^{-0.308}$. The reason is that a stiffer ring of concrete takes up more of the load close to the bar, which is exactly where the averaging disc looks. At a given applied force the stresses there are therefore higher, so the failure criterion is satisfied at a lower load. The *bond* stiffness itself rises, as it must, following $k_s \propto E_c^{0.384}$.

Section 6.3.1 found the capacity rising as $(f'_c)^{0.390}$, and the elastic modulus is why that exponent is not 0.500. Strength enters the criterion as $f_t = f_v = 0.5\sqrt{f'_c}$, so on its own it would make the capacity rise as the square root of f'_c. But a stronger concrete is also a stiffer one, and stiffness works the other way, as this sweep shows. The two partly cancel, costing roughly a fifth of that half. The quarter power of Zuo and Darwin (2000) is on this reading not an anomaly but a consequence of the two rising together. Appendix A.2 sets out the arithmetic.

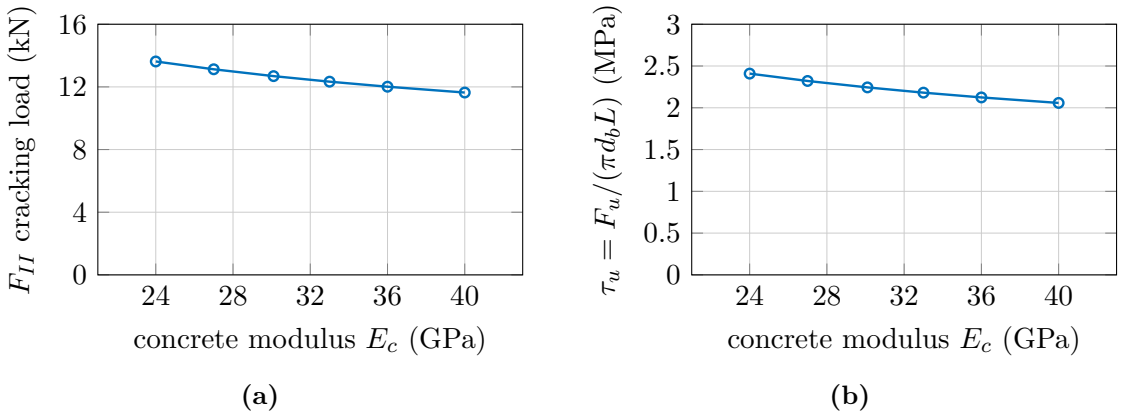


Figure 6.3.2.1: Effect of concrete modulus at constant strength ($f'_c = 32$ MPa, $d_b = 12$ mm, $L = 150$ mm, $r_e = 75$ mm): (a) cracking load, (b) mean bond stress. Both *fall* as the concrete stiffens, because f_t and f_v are held constant and only the stress distribution changes.

6.3.3 Tensile-to-shear strength ratio

Throughout this work the tensile and shear strengths have been set equal. That equality is an assumption rather than a measurement, and because the two enter the interaction index separately,

$$\Phi_{II} = \left(\frac{\bar{\sigma}_\theta}{f_t}\right)^2 + \left(\frac{\bar{\tau}_{rz}}{f_v}\right)^2 - 1, \tag{6.7}$$

it must be established how much of the predicted capacity rests on it.

Table 6.3.3.1: Effect of the shear-to-tensile strength ratio ($f_t = 2.83\,\text{MPa}$ held).

f_v/f_t	f_v (MPa)	F_{II} (kN)	τ_u (MPa)	$\hat{\sigma}$	$\hat{\tau}$
0.50	1.41	7.480	1.323	0.361	0.933
0.75	2.12	10.406	1.840	0.502	0.865
1.00	2.83	12.688	2.244	0.612	0.791
1.25	3.54	14.415	2.549	0.695	0.719
1.50	4.24	15.709	2.778	0.757	0.653
2.00	5.66	17.418	3.080	0.840	0.543

The dependence is strong near the baseline: halving the assumed shear strength cuts the predicted capacity by 41 % and doubling it raises the capacity by 37 %. This is the largest material effect in the chapter.

It is not, however, a simple scaling, and the last two columns show why. At the baseline ratio the shear term carries 63 % of the criterion against the hoop term's 37 %, so failure is shear-controlled and the model responds strongly to f_v. As f_v rises the shear term shrinks and hoop tension takes over above $f_v/f_t \approx 1.30$, after which extra shear strength buys very little. In the limit of an infinitely strong interface the capacity would reach only 20.7 kN, 63 % above the baseline, because the concrete would split first. Appendix A.2 gives a closed form that reproduces every case in Table 6.3.3.1 to better than 0.1 %; a fitted power law is accurate only to 6.5 % and should not be used outside the range tested.

The bond stiffness does not change at all across this sweep, as it should not, because it comes from the elastic solution and the strengths do not enter there.

Measuring f_v for the concretes in the test database is therefore the cheapest available improvement to the model, and should come before any further calibration.

6.3.4 Factors of minor influence

Three further factors were swept and are reported together because none of them changes the answer appreciably. Table 6.3.4.1 gives the range covered in each case and the resulting sensitivities.

Table 6.3.4.1: The three remaining factors, each swept over five values about the baseline. Changes are quoted across the range shown.

Factor	range swept	$n(F_{II})$	$n(k_s)$	ΔF_{II}	Δk_s
Steel modulus E_s (GPa)	185–215	+0.309	+0.615	4.8 %	9.7 %
Poisson's ratio ν_c	0.15–0.25	+0.022	−0.043	1.1 %	−2.2 %
Rib modulus E_r/E_c	0.25–4.0	−0.008	+0.013	−2.2 %	3.7 %

A moderate exponent need not mean a moderate effect. The steel modulus carries an exponent of 0.309, which by the classification of Section 6.1 is moderate and comparable to that of the concrete modulus. It nevertheless moves the capacity by less than 5 %, because E_s is not a quantity that varies: reinforcing steel is 200 GPa to within a few per cent whatever its grade, and the range swept here already spans more variation than a designer would ever encounter. The exponent and the percentage change are answering different questions, and this is the one factor in the study where they point opposite ways.

Poisson's ratio and the rib modulus are negligible. Both exponents are below the 0.05 threshold. The rib modulus is the stronger test of the two: changing it by a factor of sixteen, from a quarter of the concrete modulus to four times it, moves the capacity by 2.2 %. Taken with the thickness result of Section 6.2.4, this establishes that neither property of the rib layer is silently controlling the predictions, which matters because the layer is an idealisation rather than a measured feature of the specimen.

6.4 Synthesis

6.4.1 Ranking the factors

Table 6.4.1.1 collects every sweep, ranked by the magnitude of the capacity exponent, and sets the capacity and stiffness sensitivities side by side.

Table 6.4.1.1: Sensitivity of capacity and stiffness to every factor, ranked by $|n(F_{II})|$. Here n is the exponent of a power law $F_{II} \propto x^n$ fitted across each swept range, as defined in Section 6.1; the cover entries exclude the two clipped cases of Table 6.2.3.1.

Factor	$n(F_{II})$	$n(k_s)$	class	direction
Bar diameter d_b	+0.886	+1.318	dominant	together
Strength ratio f_v/f_t †	+0.615	0	dominant	capacity only
Concrete strength f'_c	+0.390	+0.137	moderate	together
Steel modulus E_s	+0.309	+0.615	moderate	together
Concrete modulus E_c	−0.308	+0.384	moderate	*opposed*
Cover ratio c/d_b	+0.281	−0.289	moderate	*opposed*
Embedment length L	+0.131	+0.064	moderate	together
Poisson's ratio ν_c	+0.022	−0.043	negligible	—
Rib modulus E_r/E_c	−0.008	+0.013	negligible	—
Rib thickness t_{rib}	−0.001	−0.001	negligible	—

6.4.2 Capacity and stiffness are not governed alike

Three patterns appear in Table 6.4.1.1.

Most factors move capacity and stiffness in the same direction, and for those there is nothing further to say: a change that strengthens the anchorage also stiffens it.

Two factors move them in *opposite* directions. Raising the concrete modulus stiffens the anchorage by 22 % while reducing its capacity by 15 %; increasing the cover raises the capacity by 36 % while reducing the stiffness by 28 %. Both say the same thing differently: capacity is governed by the averaged stress near the bar, whereas stiffness is governed by the compliance of the whole load path, and a change that redistributes stress away from the bar surface helps one and hurts the other. For design this means cover should not be read as a single measure of anchorage quality. Extra cover improves resistance to splitting, which is the role it plays in bond design, but it does not improve behaviour at working load: on these results the anchorage becomes slightly more flexible as cover grows, and the slip at first cracking almost doubles across the range tested.

One factor moves capacity alone. The strength ratio f_v/f_t enters only the failure criterion and not the elastic solution, so it leaves k_s exactly unchanged. This is useful for calibration: the stiffness provides a check on the elastic model that is completely independent of the failure criterion, so a model that predicts stiffness well but capacity poorly localises the error to the criterion rather than to the stress field.

6.4.3 Where the uncertainty sits

Influence and uncertainty are not the same thing. Bar diameter is the most influential factor in the study, but it is also known exactly for any real anchorage, so however strongly it moves the capacity it adds nothing to the uncertainty of a prediction. The strength ratio f_v/f_t is the opposite case. It is the next most influential factor, and it has never been measured for the concretes in the test database, so its value is a choice rather than a datum. Across the range swept here that choice alone moves the predicted capacity by a factor of 2.3.

Every other property in the study was measured: cover, concrete strength, the three moduli, Poisson's ratio and the rib dimensions. Taken together across their swept ranges they move the capacity by about a factor of two. A single untested assumption therefore carries as much uncertainty as every measured property of the specimen put together, which is the honest summary of the model's present standing. Reducing that exposure is a matter of measuring f_v rather than assuming it.

6.5 Conclusions

The parametric study establishes the following.

Cracking capacity scales almost in proportion to bar diameter. Both diameter sweeps return $F_{II} \propto d_b^{0.89}$ with $R^2 > 0.999$, agreeing to within 1.4 % although one holds the embedment fixed and the other the slenderness. Since the force required to yield a bar grows as d_b^2, anchoring large bars is disproportionately harder, which is the mechanical basis of the constraint identified by Ng and McGregor (n.d.) and investigated experimentally by Foster et al. (n.d.).

Cracking capacity and stiffness both level off within about ten diameters. Capacity is flat beyond $L = 100$ mm for the baseline bar and stiffness beyond 125 mm, roughly eight and ten diameters. Because the stiffness is computed from the elastic solution alone, the levelling off is a property of the load transfer and not an artefact of the failure criterion. The consequences for development length are pursued in Chapter 7.

The concrete-strength exponent decomposes exactly. The measured $\tau_u \propto (f'_c)^{0.390}$ resolves into a strength contribution of 0.500 and a modulus contribution of $-0.308 \times 0.356 = -0.110$, reproducing the combined value to within 0.0001. This gives a mechanical explanation for why the empirical exponent of Zuo and Darwin (2000) lies below the square root, complementing the length-dependent explanation of Harajli (2004).

The predicted failure mode shifts with geometry. The index moves from shear-controlled to hoop-controlled as the bar diameter passes about 17 mm, and

shifts the same way as cover increases. This reproduces a pull-out to splitting transition the model was not fitted to.

The model is robust to its invented parameters but not to one assumption. An eightfold change in rib layer thickness moves the capacity by 0.15 % and a sixteenfold change in its modulus by 2.2 % (Sections 6.2.4 and 6.3.4), so the three-layer idealisation is not silently controlling the results. The exception is the assumption $f_v = f_t$: capacity changes by 133 % across a fourfold change in the assumed ratio, and the baseline failure is shear-controlled. Measuring f_v is the cheapest available improvement to the model.

Cover is understated by construction. The averaging disc has a radius of 26.4 mm, so concrete beyond that distance is never sampled directly and reaches the criterion only by stiffening the surrounding ring. The exponent of 0.281 should therefore be read as a lower bound. The same radius produced the two clipped cases of Table 6.2.3.1; repairing the response would need a criterion that recognises the free surface, which is beyond the present scope.

What follows. The capacities established here depend on four quantities a designer knows, namely bar diameter, embedment length, cover and concrete strength, and on two that must be calibrated, the critical distance l_{char} and the strength ratio f_v/f_t. Chapter 7 assembles the first four into a family of calibration curves, shows that the family collapses onto a single master curve, and inverts that curve to give a development length expression in which the bar grade appears exactly once.

Chapter 7

Calibration Curves and an Analytical Development Length Equation

Chapter 6 established which properties control the predicted bond capacity and by how much. Those results were obtained one property at a time about a single baseline, which is the right way to isolate an influence but the wrong way to produce something a designer can use. This chapter does the second thing.

The capacity is traced continuously against embedment length for each of several bar diameters, producing a family of calibration curves. The family is shown to collapse onto a single master curve whose functional form follows from shear lag rather than from curve fitting, and that master curve is then inverted to give a closed-form expression for development length in which the yield stress of the bar appears exactly once. The chapter closes by establishing the limit that the saturation of Section 6.2.1 imposes on that expression, which turns out to govern its interpretation entirely.

7.1 The calibration curve family

7.1.1 Construction

Four bar diameters between 10 and 20 mm were each run at fifteen embedments between $L/d_b = 2$ and 30, giving sixty cases. All other properties are held at the baseline of Table 6.1.1.1: 32 MPa concrete in a 150 mm cylinder, $l_{char} = 20$ mm, pull–push loading.

Figure 7.1.1.1: Calibration curves: predicted cracking load against normalised embedment length for four bar diameters ($f'_c = 32$ MPa, $d_{cyl} = 150$ mm, $l_{char} = 20$ mm, pull–push).

Every curve has the same shape: a steep, nearly linear rise, a knee, and a plateau flat to within a fraction of a per cent over the remaining two thirds of the range. Mode II governs at all sixty points; the yield load is never reached, which is the subject of Section 7.5.

7.1.2 The effective length

Section 6.2.1 established the saturation for a single bar. With four diameters available it can be quantified properly. Defining L_{95} and L_{99} as the embedments at which the load first reaches 95 and 99 % of its asymptotic value gives Table 7.1.2.1.

Table 7.1.2.1: Effective embedment length, read from the calibration curves.

d_b (mm)	L_{95} (mm)	L_{95}/d_b	L_{99} (mm)	L_{99}/d_b
10	73	7.31	98	9.77
12	81	6.73	116	9.65
16	94	5.85	158	9.85
20	111	5.55	200	9.99

The two columns behave differently. At the 95 % level neither the absolute length nor the slenderness is constant: L_{95} varies by a factor of 1.5 across the diameter range and L_{95}/d_b by 1.3. At the 99 % level, however, L_{99}/d_b takes the values 9.77, 9.65, 9.85 and 9.99 — a spread of under 4 %. The embedment at which the model is within one per cent of saturation is therefore very close to ten bar diameters, whatever the bar size.

7.2 Normalisation and the master curve

7.2.1 Single-variable normalisations fail

Plotting against L/d_b does not collapse the family: at a given slenderness the four curves differ by up to 88 %. Plotting against absolute L is no better, at 62 %. Dividing the ordinate by $d_b^{0.9}$ — the exponent established in Section 6.2.2 — reduces the spread to about 27 % but does not remove it. Optimising the exponent numerically does no better than 15 %.

No single-variable normalisation collapses the family. This is the first substantive result of the chapter: the slenderness ratio on which the design codes are built is not a similarity variable for this problem.

The reason is that two independent length scales are present. The capacity is set by the bar perimeter, which scales with d_b, while the distance over which load transfers from bar to concrete is set by the shear-lag decay of the governing equation, which has its own scaling. Any normalisation that uses one variable must conflate them.

7.2.2 The shear-lag form

The shape in Figure 7.1.1.1 is the classical shear-lag result. In an elastic bonded joint the interfacial shear decays from the loaded end over a transfer length, so the load at which the peak stress reaches a limiting value grows as the hyperbolic tangent of the bonded length divided by that transfer length. Fitting

$$F_u = F_\infty \tanh\!\left(\frac{L}{L_0}\right) \tag{7.1}$$

to each diameter separately gives Table 7.2.2.1.

Table 7.2.2.1: Parameters of (7.1) fitted to each calibration curve.

d_b (mm)	F_∞ (kN)	L_0 (mm)	L_0/d_b	RMS error (%)	max error (%)
10	10.825	44.77	4.48	3.38	9.63
12	12.766	48.94	4.08	2.53	7.32
16	16.619	60.96	3.81	3.10	10.68
20	20.269	70.47	3.52	4.37	15.25

Both parameters follow clean power laws in the diameter,

$$F_\infty = 1.343\, d_b^{0.907}\ \text{kN}, \qquad L_0 = 9.42\, d_b^{0.671}\ \text{mm}, \tag{7.2}$$

to within 0.2 and 2.1 % respectively, with d_b in millimetres. The capacity exponent 0.907 agrees with the 0.886 and 0.898 obtained independently from the two diameter sweeps of Section 6.2.2, which is a useful consistency check since those sweeps were run at single embedments and this one at fifteen.

7.2.3 The master curve

Substituting (7.2) into (7.1) and refitting all sixty points simultaneously gives

$$\boxed{F_u = 1.349\, d_b^{0.905} \tanh\!\left(\frac{L}{9.32\, d_b^{0.676}}\right)\ \text{kN}} \tag{7.3}$$

with d_b and L in millimetres. It reproduces every point in the family with a root-mean-square error of 3.4 %, a maximum error of 15.3 % at the shortest embedment of the largest bar, and $R^2 = 0.993$.

Figure 7.2.3.1: The calibration curves of Figure 7.1.1.1 replotted as F_u/F_∞ against L/L_0. The four families collapse onto $\tanh(L/L_0)$, reducing the spread from 88 to 12 %.

In the normalised variables the four curves collapse to within 12 %, against 88 % unnormalised.

7.2.4 Why the form is mechanics and not curve fitting

Two checks distinguish (7.3) from an arbitrary two-parameter fit.

The asymptotes recovered by the tanh fit agree with an independent Richardson extrapolation of the same data to within 0.1 %. The functional form is therefore capturing the true asymptote rather than merely passing near the last few points, which a form with the wrong tail behaviour would not do.

The alternatives $F_\infty(1 - \mathrm{e}^{-L/L_0})$ and $F_\infty L/(L_0 + L)$, fitted to the same data with the same number of free parameters, give root-mean-square errors of 4.7 and 8.8 % against 3.3 % for tanh. The form that shear lag predicts is also the form that fits best.

7.3 What sets the transfer length

If L_0 were simply the elastic decay length of the governing equation, the fit would carry no information the solution did not already contain. The characteristic roots of (3.17) are complex, $\lambda = \alpha \pm \mathrm{i}\beta$, and the envelope of the decay is set by the real part. Table 7.3.0.1 compares $1/\alpha$ with the fitted L_0.

Table 7.3.0.1: Fitted transfer length against the elastic decay length of the governing equation.

d_b (mm)	$\alpha = \mathrm{Re}\,\lambda$ (1/mm)	$1/\alpha$ (mm)	L_0 fitted (mm)	$L_0\alpha$
10	0.05621	17.79	44.77	2.52
12	0.05250	19.05	48.94	2.57
16	0.04717	21.20	60.96	2.88
20	0.04343	23.03	70.47	3.06

The two lengths are of the same order and both grow with diameter, but they are not proportional: $1/\alpha$ follows $d_b^{0.37}$ where L_0 follows $d_b^{0.67}$, and the ratio drifts by 20 % across the range.

The discrepancy has the same origin as the failure of the single-variable normalisation in Section 7.2. The elastic solution supplies one length scale, $1/\alpha$, which grows with the bar; the Area Method supplies a second, the averaging radius $R_{avg} = 1.32\, l_{char}$, which is a fixed 26.4 mm and does not. The length over which the *criterion* sees load transferred combines the two, and L_0 is that combined length rather than either separately.

This has a testable consequence. The exponent on d_b in (7.2) is not a constant of the material: it should move towards 0.37 as l_{char} is reduced and the averaging radius ceases to matter, and towards the opposite limit as l_{char} grows. Repeating the calibration family at two further values of l_{char} would confirm or refute it directly.

7.4 The development length equation

7.4.1 Grade independence of the calibration curve

Nothing in (7.3) depends on the grade of the bar. The Mode II capacity is a concrete cracking load determined by the geometry, the moduli and the concrete strengths; the yield stress enters the problem only through the Mode I demand

$$F_I = A_s f_{sy} = \frac{\pi d_b^2 f_{sy}}{4}. \tag{7.4}$$

The calibration curves are therefore already the grade-independent object, and one family serves Grade 250, Grade 500 and Grade 600 alike. Raising the grade does not move the curve; it raises the horizontal line that must intersect it. This is what makes a single family of curves useful as a design aid rather than a result specific to the steel it was computed with.

7.4.2 Derivation

The development length is the shortest embedment at which the concrete can deliver the force required to yield the bar. Setting $F_u = F_I$ in (7.3) and inverting the hyperbolic tangent,

$$\boxed{L_d = L_0 \operatorname{artanh}\left(\frac{F_I}{F_\infty}\right) = 9.32\, d_b^{0.676} \operatorname{artanh}\left(\frac{\pi d_b^2 f_{sy}}{4000 \times 1.349\, d_b^{0.905}}\right) \text{ mm}} \tag{7.5}$$

with d_b in millimetres and f_{sy} in megapascals. The yield stress appears exactly once, inside the inverse hyperbolic tangent; every other quantity comes from the grade-independent calibration curve.

7.4.3 Relation to the code form

Written in terms of the mean bond stress the same statement takes the form the codes use. Equilibrium of the embedded length gives $\pi d_b L_d \tau_u = \pi d_b^2 f_{sy}/4$, so

$$\frac{L_d}{d_b} = \frac{f_{sy}}{4\tau_u}, \tag{7.6}$$

which is the basic anchorage length of fib Model Code 2010 as set out by Cairns (2015), with the code's design bond stress replaced by the τ_u the present model predicts. Equation (7.5) is the explicit solution of (7.6) once the dependence of τ_u on L_d is taken from the master curve.

Recovery of the code form at low utilisation. For $F_I \ll F_\infty$ the inverse hyperbolic tangent is linear in its argument and (7.5) reduces to

$$L_d \approx \frac{\pi L_0}{4F_\infty} f_{sy} d_b^2 = 5.43 \times 10^{-3}\, f_{sy}\, d_b^{1.77} \text{ mm}, \tag{7.7}$$

linear in the yield stress exactly as Standards Australia (2018) and Cairns (2015) are. *The codes' linear dependence on grade is the low-utilisation limit of* (7.5), and the departure from it as F_I approaches F_∞ is the model's statement that a bar worked close to the capacity of its surrounding concrete requires disproportionately more length.

One difference survives this limit. The diameter exponent in (7.7) is 1.77 where Standards Australia (2018) has 1.00, so the model predicts a stronger size effect than the code — a direct consequence of F_∞ growing as $d_b^{0.91}$ while the demand grows as d_b^2. This is not merely a modelling artefact. It is the same effect that led Foster et al. (n.d.) to test bars from 10 to 37 mm rather than assume a single L_d/d_b, and the same effect that makes the anchorage of large high-grade bars the binding constraint identified by Ng and McGregor (n.d.).

7.5 The limit imposed by the saturation capacity

7.5.1 The condition

Equation (7.5) returns a real length only when $F_I < F_\infty$. The inverse hyperbolic tangent diverges as its argument approaches unity, which is the algebra expressing the fact that beyond the effective length additional embedment adds no capacity. If the force required to yield the bar exceeds the saturated capacity of the anchorage, no embedment whatever will develop the bar.

Table 7.5.1.1: Saturated capacity against the force required to yield the bar ($f_{sy} = 550\,\text{MPa}$, $f'_c = 32\,\text{MPa}$, $d_{cyl} = 150\,\text{mm}$, $l_{char} = 20\,\text{mm}$).

d_b (mm)	F_I (kN)	F_∞ (kN)	F_∞/F_I	$f_{sy,\max}$ (MPa)
10	43.2	10.84	0.250	138
12	62.2	12.78	0.205	113
16	110.6	16.58	0.150	83
20	172.8	20.29	0.117	65

The condition fails in every case and by a wide margin. The last column gives the yield stress that the saturated capacity could just develop: from 138 MPa for the 10 mm bar down to 65 MPa for the 20 mm bar, an order of magnitude below any structural reinforcing steel. The margin worsens with bar size, because the demand grows as d_b^2 while F_∞ grows only as $d_b^{0.91}$.

7.5.2 The size of the discrepancy

Two independent references fix the magnitude.

Foster et al. (n.d.) report that an anchorage length of $16.2\,d_b$ develops the nominal yield strength of a Grade 600 bar, which corresponds to a mean bond stress at failure

of

$$\tau_u = \frac{f_{sy} d_b}{4 L_d} = \frac{600}{4 \times 16.2} = 9.26\,\text{MPa}, \tag{7.8}$$

against the 2.24 MPa the model predicts for the baseline specimen — a factor of 4.1.

The code gives the same answer. The basic development length of Clause 13.1.2.2 of Standards Australia (2018) for the baseline bar, with $k_1 = k_2 = k_3 = 1$, is

$$L_{sy.tb} = \frac{0.5 k_1 k_3 f_{sy} d_b}{k_2 \sqrt{f'_c}} = \frac{0.5 \times 550 \times 12}{\sqrt{32}} = 583\,\text{mm} = 48.6\,d_b, \tag{7.9}$$

which the model cannot reach at any embedment.

7.5.3 Calibration cannot close the gap

The obvious remedy is to calibrate l_{char} upwards, since Section **??** showed it to be the only uncalibrated parameter capable of moving the capacity substantially. The capacity must rise by a factor of 4.9 for the baseline bar to be developable. Extrapolating (**??**), that requires

$$l_{char} = 20 \times 4.90^{1/0.738} \approx 172\,\text{mm}, \tag{7.10}$$

which exceeds the cover of the specimen (69 mm) by a factor of two and a half, and is an order of magnitude beyond any critical distance defensible as a material length scale.

No admissible calibration of l_{char} turns the model into an ultimate-capacity predictor. What calibration can do is fit it to measured *cracking* loads, and that is the claim Chapter [X] should make for it.

7.5.4 Why the gap exists

Three explanations contribute, and they are not mutually exclusive.

Real anchorages do not saturate, because they are not elastic. In a physical pull-out the shear near the loaded end does not grow without limit. Once the local bond capacity is reached there, the interface softens and load redistributes along the bar, bringing material further from the loaded end into play. This is precisely what the local bond–slip formulations of Tang and Cheng (2020) and Chu and Kwan (2019) represent and what the present model, being elastic to failure, cannot. It is why codes can specify development lengths of forty or fifty diameters at all, and it is the principal limitation of the formulation.

What is predicted is first cracking, not ultimate capacity. The interaction index reaches unity when the averaged stresses first satisfy the failure criterion, which corresponds to the onset of longitudinal splitting rather than to the peak of the load–slip curve. That the predicted mean bond stress sits just below the concrete tensile strength — 2.24 against $f_t = 2.83$ MPa — is consistent with a first-cracking criterion and not with an ultimate one. The factor of 4.1 is therefore largely the expected difference between cracking and ultimate.

The specimen is unconfined. The axisymmetric idealisation has no counterpart for transverse reinforcement, which in Zuo, Leon, et al. (n.d.) contributes a term scaling with $(f'_c)^{3/4}$ and in practice is what permits the shorter anchorages that codes allow. A model that cannot represent confinement should not be expected to reproduce the capacity of confined anchorages.

7.5.5 What the equation is therefore good for

Equation (7.5) should be read as a *splitting-initiation* anchorage expression, not as a replacement for (7.9). In that reading it has three uses. It predicts the embedment beyond which no further capacity against splitting is gained, which Table 7.1.2.1 puts at about ten diameters. It supplies the grade-independent shape of the capacity–embedment relationship, which the codes assume to be linear and which it shows is not. And once l_{char} is calibrated against measured cracking loads it gives a quantitative serviceability criterion, which is the limit state the elastic model is actually describing.

7.6 Comparison with code and research expressions

Setting the absolute level aside, the *form* of (7.5) can be compared with the expressions in use.

Table 7.6.0.1: Structure of the present expression against three code and research formulations.

Source	on f_{sy}	on f'_c	on c/d_b
Standards Australia (2018), Cl. 13.1.2.2	linear	$(f'_c)^{-1/2}$	through k_3
Cairns (2015) (MC2010)	linear	$(f'_c)^{-1/2}$	through α_2
Zuo, Leon, et al. (n.d.)	linear	$(f'_c)^{-1/4}$	explicit
Present model, (7.5)	artanh	$(f'_c)^{-0.39}$	$(c/d_b)^{-0.28}$

Three observations follow.

The model agrees with all three in the low-utilisation limit, as (7.7) shows, and departs from them as the demand approaches the saturated capacity, where it predicts a divergence the codes do not represent. Whether that divergence is physical or an artefact of the elastic formulation is exactly the question raised in Section 7.5; the honest position is that it is the latter, and that the codes are right to treat the relationship as linear over the range they cover.

Its exponent on concrete strength, 0.39, lies between the half power of Standards Australia (2018) and the quarter power of Zuo, Leon, et al. (n.d.), and Section 6.3.2 explains the position mechanically rather than empirically.

Its cover exponent, 0.28, is of the right sign and order but weaker than the empirical models, for the reason given in Section 6.2.3: the averaging radius makes the

criterion blind to cover beyond 26.4 mm. This is the one structural dependence the model gets qualitatively right and quantitatively wrong, and it is a limitation of the criterion rather than of the stress field.

7.7 Conclusions

The capacity–embedment relationship has a shear-lag form. A family of sixty calibration curves spanning a factor of two in bar diameter is reproduced by the single expression (7.3) with a root-mean-square error of 3.4 % and $R^2 = 0.993$. The tanh form is not chosen for convenience: it recovers the true asymptote to within 0.1 % and fits better than the exponential and hyperbolic alternatives tried against the same data.

Slenderness is not a similarity variable. No normalisation in L/d_b alone, in L alone, or in L with the ordinate scaled by any power of d_b reduces the spread of the family below 15 %; the two-parameter normalisation of (7.3) reduces it to 12 % from 88. Two length scales are present — the elastic decay length and the averaging radius — and L_0 is their combination, as Table 7.3.0.1 shows.

A grade-independent development length expression follows by inversion. Because the model's capacity contains no f_{sy}, the calibration curves serve every grade and the yield stress enters (7.5) exactly once. In the low-utilisation limit the expression reduces to the linear form used by Standards Australia (2018) and Cairns (2015), so the codes' expression is recovered as a special case, with one difference: the model carries a diameter exponent of 1.77 against the code's 1.00, predicting a stronger size effect.

The expression is bounded by the saturation capacity, and the bound binds. Development length exists only where $F_I < F_\infty$, and the saturated capacity is between 12 and 25 % of the yield force across the four diameters examined. The predicted mean bond stress is a factor of 4.1 below that implied by Foster et al. (n.d.), and no admissible value of l_{char} closes the gap: 172 mm would be required against a specimen cover of 69 mm.

The model should therefore be presented as a cracking model. The gap is largely the expected difference between first cracking and ultimate capacity, compounded by the absence of post-cracking redistribution and of confinement. Read as a splitting-initiation criterion, (7.5) is defensible and useful; read as a replacement for the code expression it is not. Chapter [X] should calibrate l_{char} against measured cracking loads rather than measured ultimate loads, and the limitations stated here should frame that calibration rather than follow it.

Chapter 8

Comparison to Other Analytical Models

8.1 Tepfers

Add calcs here to show that it is somewhere betweek Stage 1 and Stage 2 of the Tepfers model.

Chapter 9

Limitations and Future Work

9.1 Material Non-Linearity

Chapter 10

Conclusion

Appendix A

Your first appendix

A.1 The title of the first section

The appendices work exactly the same way as chapters, they are numbered with letters rather than numbers though.

NEED Coefficients A-K evaluated in the appendix and referenced in chapter 3.

A.2 Fitting of the power-law exponents

Every sensitivity in Chapter 6 is reported as the exponent n of a power law $y \propto x^{\,n}$ fitted across the swept range. An exponent is used in preference to a percentage change because it does not depend on how wide the range happens to be. This appendix sets out how each one was obtained and lists them all in one place.

Method

A power law $y = ax^{\,n}$ becomes a straight line under logarithms,

$$\ln y = \ln a + n \ln x, \tag{A.1}$$

so n is the gradient of $\ln y$ against $\ln x$ and is found by ordinary least squares. Writing $x_i = \ln(\text{input})$ and $y_i = \ln(\text{output})$ for the N points of a sweep,

$$n = \frac{N \sum x_i y_i - \sum x_i \sum y_i}{N \sum x_i^2 - \left(\sum x_i\right)^2}, \qquad \ln a = \frac{\sum y_i - n \sum x_i}{N}. \tag{A.2}$$

The quality of the fit is reported as the coefficient of determination,

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}, \tag{A.3}$$

evaluated on the fitted values $\hat{y}_i = a x_i^{\,n}$ in physical units rather than on the logarithms, so that it measures agreement with the quantity actually of interest.

Worked example: capacity against bar diameter

The six cases of Table 6.2.2.1 give the following.

Table A.2.0.1: Least-squares working for $F_{II} \propto d_b^{\,n}$, from the bar-diameter sweep at $L = 150\,\mathrm{mm}$.

d_b (mm)	F_{II} (kN)	$x = \ln d_b$	$y = \ln F_{II}$	xy	x^2
8	8.82560	2.079442	2.177657	4.528310	4.324077
10	10.76008	2.302585	2.375843	5.470581	5.301898
12	12.68776	2.484907	2.540638	6.313248	6.174761
16	16.44306	2.772589	2.799904	7.762981	7.687248
20	19.98282	2.995732	2.994873	8.971837	8.974412
24	23.25610	3.178054	3.146567	9.999961	10.100026
$\sum$		15.813308	16.035481	43.046917	42.562423

Substituting into (A.2) with $N = 6$,

$$n = \frac{6(43.046917) - (15.813308)(16.035481)}{6(42.562423) - (15.813308)^2} = \frac{4.707497}{5.313822} = 0.8859, \tag{A.4}$$

$$\ln a = \frac{16.035481 - 0.8859 \times 15.813308}{6} = 0.337754 \quad \Longrightarrow \quad a = 1.4018, \tag{A.5}$$

so that

$$F_{II} = 1.402\, d_b^{\,0.886}\ \mathrm{kN}, \qquad R^2 = 0.99975, \tag{A.6}$$

with d_b in millimetres. The residuals are given in Table A.2.0.2; none exceeds 0.7 %.

Table A.2.0.2: Residuals of the fit (A.6).

d_b (mm)	measured F_{II} (kN)	fitted (kN)	error (%)
8	8.826	8.846	+0.23
10	10.760	10.779	+0.18
12	12.688	12.669	−0.15
16	16.443	16.346	−0.59
20	19.983	19.919	−0.32
24	23.256	23.410	+0.66

Second worked example: capacity against cover, with exclusions

The cover sweep differs from the one above in that two of its eight cases are excluded before fitting. The reason is set out in Section 6.2.3; the arithmetic is as follows.

The Area Method averages over a half-disc of radius $R_{avg} = 1.32\, l_{char} = 26.4\,\mathrm{mm}$, which at $l_{item} = 2\,\mathrm{mm}$ requires $\lceil 26.4/2 \rceil = 14$ radial items. The number that actually fit between the bar surface and the outer face is $\lfloor (r_e - d_b/2)/l_{item} \rfloor$, giving Table A.2.0.3.

Table A.2.0.3: Whether the averaging disc fits inside the concrete. Fourteen radial items are required.

r_e (mm)	$c = r_e - d_b/2$ (mm)	c/d_b	items that fit	status
25	19	1.583	9	clipped
30	24	2.000	12	clipped
40	34	2.833	17	admissible
50	44	3.667	22	admissible
60	54	4.500	27	admissible
75	69	5.750	34	admissible
90	84	7.000	42	admissible
110	104	8.667	52	admissible

Applying (A.2) to the six admissible points, with $x = \ln(c/d_b)$ and $y = \ln F_{II}$:

Table A.2.0.4: Least-squares working for $F_{II} \propto (c/d_b)^{\,n}$, six admissible points.

c/d_b	F_{II} (kN)	x	y	xy	x^2
2.833	11.07883	1.041454	2.405036	2.504734	1.084626
3.667	11.28173	1.299283	2.423185	3.148403	1.688136
4.500	11.76825	1.504077	2.465405	3.708160	2.262249
5.750	12.68776	1.749200	2.540638	4.444083	3.059700
7.000	13.69603	1.945910	2.617106	5.092653	3.786566
8.667	15.09083	2.159484	2.714087	5.861029	4.663372
$\sum$		9.699409	15.165457	24.759062	16.544650

$$n = \frac{6(24.759062) - (9.699409)(15.165457)}{6(16.544650) - (9.699409)^2} = \frac{1.458410}{5.189374} = 0.2810, \tag{A.7}$$

$$\ln a = \frac{15.165457 - 0.2810 \times 9.699409}{6} = 2.073260 \quad \Longrightarrow \quad a = 7.951, \tag{A.8}$$

so that

$$F_{II} = 7.951 \left(\frac{c}{d_b} \right)^{0.281} \text{ kN}, \qquad R^2 = 0.9428. \tag{A.9}$$

Repeating the fit on all eight points, including the two clipped cases, returns $F_{II} = 9.199\,(c/d_b)^{0.199}$ with $R^2 = 0.8566$. The two inadmissible points therefore reduce the exponent by 29 % and the coefficient of determination from 0.94 to 0.86, which is why they are excluded.

Third worked example: decomposing the concrete-strength exponent

The exponent on concrete strength is not obtained from a single fit. It is built from three quantities, because f'_c reaches the capacity by two separate routes: through the strengths $f_t = f_v = 0.5\sqrt{f'_c}$ that normalise the interaction index, and through

the modulus E_c that determines how stress is distributed. Each route is measured separately and the two are then combined.

The three fits. Applying (A.2) to the three data sets gives Table A.2.0.5.

Table A.2.0.5: The three fits behind the concrete-strength exponent.

Fit	Source	N	n	R^2
τ_u against f'_c	Table 6.3.1.1, E_c tied	7	+0.3901	0.99832
τ_u against E_c	Table 6.3.2.1, f'_c fixed	6	−0.3084	0.99991
E_c against f'_c	Standards Australia (2018) Tab. 3.1.2	7	+0.3562	0.98034

Combining them. Since τ_u depends on f'_c through both f_t and E_c, the chain rule in logarithmic variables gives

$$\frac{\mathrm{d}\ln\tau_u}{\mathrm{d}\ln f'_c} = \frac{\partial\ln\tau_u}{\partial\ln f_t}\frac{\mathrm{d}\ln f_t}{\mathrm{d}\ln f'_c} + \frac{\partial\ln\tau_u}{\partial\ln E_c}\frac{\mathrm{d}\ln E_c}{\mathrm{d}\ln f'_c}. \tag{A.10}$$

The first term is known exactly rather than fitted. From $f_t = 0.5\sqrt{f'_c}$ the second factor is $\frac{1}{2}$, and the first is unity: the interaction index (6.7) is homogeneous of degree -2 in the strengths, so scaling f_t and f_v together by any factor scales the failure load by the same factor, giving $\partial\ln\tau_u/\partial\ln f_t = 1$. The strength route therefore contributes $1 \times 0.5 = 0.500$.

The second term is the product of two measured quantities from Table A.2.0.5, $(-0.3084)(0.3562) = -0.1099$. Hence

$$\frac{\mathrm{d}\ln\tau_u}{\mathrm{d}\ln f'_c} = 0.500 - 0.110 = 0.390, \tag{A.11}$$

against the directly fitted 0.3901. The two agree to four decimal places.

Read the other way, (A.10) can be solved for the one factor that was asserted rather than measured:

$$\frac{\partial\ln\tau_u}{\partial\ln f_t} = \frac{0.3901 - (-0.3084)(0.3562)}{0.5} = 1.0000, \tag{A.12}$$

which confirms the homogeneity argument numerically.

Local exponents. The exponent is not quite constant across the range, because the modulus table flattens at high strength. Table A.2.0.6 applies (A.11) to each interval in turn.

Table A.2.0.6: Local exponent on concrete strength, measured against the value predicted by (A.11) applied interval by interval.

Interval (MPa)	$n(E_c)$	predicted $n(\tau_u)$	measured $n(\tau_u)$	difference
20–25	0.478	0.353	0.348	−0.004
25–32	0.486	0.350	0.349	−0.002
32–40	0.385	0.381	0.381	−0.001
40–50	0.265	0.418	0.420	+0.002
50–65	0.275	0.415	0.417	+0.002
65–80	0.275	0.415	0.417	+0.002

The agreement holds to within 0.005 at every interval, so the decomposition does more than reproduce the average exponent: it predicts how that exponent varies across the range, from the modulus table alone.

These values use the same two-point formula as (**??**). For the interval $f'_c = 40$ to 50 MPa, over which E_c rises from 32 800 to 34 800 MPa and τ_u from 2.443 to 2.683 MPa,

$$n(E_c) = \frac{\ln(34800/32800)}{\ln(50/40)} = \frac{0.059189}{0.223144} = 0.265, \tag{A.13}$$

$$\text{predicted } n(\tau_u) = 0.500 + (-0.3084)(0.265) = 0.418, \tag{A.14}$$

$$\text{measured } n(\tau_u) = \frac{\ln(2.683/2.443)}{\ln(50/40)} = 0.420. \tag{A.15}$$

Fourth worked example: the shear-to-tensile strength ratio

The exponent quoted in Section 6.3.3 is fitted in the same way as the others, but it describes a relationship that is not in fact a power law. Both the fit and the exact expression are given here.

The fitted exponent. Applying (A.2) to the six cases of Table 6.3.3.1, with $x = \ln(f_v/f_t)$ and $y = \ln F_{II}$:

Table A.2.0.7: Least-squares working for $F_{II} \propto (f_v/f_t)^n$.

f_v/f_t	F_{II} (kN)	x	y	xy	x^2
0.50	7.480	−0.693147	2.012233	−1.394773	0.480453
0.75	10.406	−0.287682	2.342383	−0.673861	0.082761
1.00	12.688	0.000000	2.540657	0.000000	0.000000
1.25	14.415	0.223144	2.668269	0.595407	0.049793
1.50	15.709	0.405465	2.754234	1.116746	0.164402
2.00	17.418	0.693147	2.857504	1.980671	0.480453
$\sum$		0.340927	15.175279	1.624189	1.257862

$$n = \frac{6(1.624189) - (0.340927)(15.175279)}{6(1.257862) - (0.340927)^2} = \frac{4.571477}{7.430941} = 0.6152, \tag{A.16}$$

with $R^2 = 0.967$. That is the weakest fit of any reported in Chapter 6, and individual points are in error by up to 6.5 % — an order of magnitude worse than the other sweeps. The reason is that the underlying relationship is not a power law.

The exact expression. Write the averaged stresses at the governing station, evaluated at the reference load F_{ref}, as $\bar{\sigma}_\theta = a f_t$ and $\bar{\tau}_{rz} = b f_t$. Putting $f_v = k f_t$ into (6.7) gives an index of $a^2 + b^2/k^2$ at that load, and since the index is homogeneous of degree two in the load,

$$F_{II}(k) = \frac{F_{ref}}{\sqrt{a^2 + b^2/k^2}}. \tag{A.17}$$

The two constants follow from the baseline case. At $k = 1$ the normalised stresses at failure are $\hat{\sigma} = 0.612$ and $\hat{\tau} = 0.791$ with $F_{II} = 12.688\,\mathrm{kN}$, so scaling back to $F_{ref} = 50\,\mathrm{kN}$,

$$a = 0.612 \times \frac{50}{12.688} = 2.412, \qquad b = 0.791 \times \frac{50}{12.688} = 3.117. \tag{A.18}$$

Table A.2.0.8: The closed form (A.17) against the computed values and against the power-law fit.

f_v/f_t	computed (kN)	(A.17) (kN)	error (%)	power law (kN)	error (%)
0.50	7.480	7.480	0.00	7.908	+5.7
0.75	10.406	10.405	−0.01	10.148	−2.5
1.00	12.688	12.687	−0.01	12.113	−4.5
1.25	14.415	14.413	−0.02	13.895	−3.6
1.50	15.709	15.706	−0.02	15.544	−1.1
2.00	17.418	17.412	−0.03	18.554	+6.5

Equation (A.17) reproduces every case to better than 0.1 %. Its two limits are the reason the power law fails. For $k \to 0$ the shear term dominates and $F_{II} \to F_{ref} k/b$, linear in k; for $k \to \infty$ the shear term vanishes and $F_{II} \to F_{ref}/a = 20.7\,\mathrm{kN}$, a ceiling set by hoop tension alone. The local slope is therefore

$$\frac{\mathrm{d}\ln F_{II}}{\mathrm{d}\ln k} = \frac{b^2/k^2}{a^2 + b^2/k^2}, \tag{A.19}$$

which falls from 0.870 at $k = 0.5$ to 0.295 at $k = 2$. The fitted 0.615 is an average over that range and should not be extrapolated beyond it.

All fitted exponents

Table A.2.0.9 collects every exponent quoted in Chapter 6, each obtained by (A.2) applied to the corresponding sweep.

Table A.2.0.9: Power-law exponents for every factor, fitted by (A.2). A positive exponent means the output rises with the input.

Factor x	N	$n(F_{II})$	$n(\tau_u)$	$n(k_s)$
Embedment length L	8	$+0.131$	-0.869	$+0.064$
Bar diameter d_b (fixed L)	6	$+0.886$	-0.114	$+1.318$
Bar diameter d_b (fixed L/d_b)	6	$+0.898$	-1.102	$+1.318$
Cover ratio c/d_b †	6	$+0.281$	$+0.281$	-0.289
Rib thickness t_{rib}	6	-0.001	-0.001	-0.001
Concrete strength f'_c	7	$+0.390$	$+0.390$	$+0.137$
Concrete modulus E_c	6	-0.308	-0.308	$+0.384$
Strength ratio f_v/f_t §	6	$+0.615$	$+0.615$	± 0
Poisson's ratio ν_c	5	$+0.022$	$+0.022$	-0.043
Rib modulus E_r/E_c	5	-0.008	-0.008	$+0.013$
Steel modulus E_s	5	$+0.309$	$+0.309$	$+0.615$

† Fitted to the six cases in which the averaging disc is not clipped by the free surface; see Section 6.2.3. Including the two clipped cases would give $+0.199$ with $R^2 = 0.857$.

§ A local average only; the underlying relationship is not a power law. See (A.17).

Three entries in Table A.2.0.9 are not independent fits but follow from the others by identity.

Where the embedment is held fixed, $\tau_u = F_{II}/(\pi d_b L)$ differs from F_{II} only by a constant, so the two exponents are equal. This is why the $n(F_{II})$ and $n(\tau_u)$ columns coincide for every factor except the two that change a length.

For the embedment-length sweep the length itself is the variable, so

$$n(\tau_u) = n(F_{II}) - 1 = 0.131 - 1 = -0.869, \tag{A.20}$$

and for the bar-diameter sweep at constant slenderness, where $L = 10 d_b$, the bonded area grows as d_b^2 and

$$n(\tau_u) = n(F_{II}) - 2 = 0.898 - 2 = -1.102. \tag{A.21}$$

Finally the slip exponent is never fitted independently. Since $\delta_u = F_u/k_s$,

$$n(\delta_u) = n(F_{II}) - n(k_s), \tag{A.22}$$

which for the bar-diameter sweep gives $0.886 - 1.318 = -0.432$, matching the value obtained by fitting the computed slips directly to four decimal places.

Bibliography

Alavi, S. Karen, Hamid R. Valipour, and Mark A. Bradford (June 2025). "An Analytical Procedure for Failure and Stiffness Prediction of Glued-in Rod Timber Connections". In: *International Journal of Adhesion and Adhesives* 140, p. 103992. ISSN: 01437496. (Visited on 06/22/2026).

Cairns, John (Mar. 2015). "Bond and Anchorage of Embedded Steel Reinforcement in *Fib* Model Code 2010". In: *Structural Concrete* 16.1, pp. 45–55. ISSN: 1464-4177, 1751-7648. (Visited on 03/07/2026).

Chu, S.H. and A.K.H. Kwan (Nov. 2019). "A New Bond Model for Reinforcing Bars in Steel Fibre Reinforced Concrete". In: *Cement and Concrete Composites* 104, p. 103405. ISSN: 09589465. (Visited on 03/09/2026).

Foster, Stephen et al. (n.d.). "Development and Lap Lengths for 600 MPa Reinforcing Steels with an Improved Rib Pattern". In: ().

Hadi, Muhammad N.S. (Dec. 2008). "Bond of High Strength Concrete with High Strength Reinforcing Steel˜!2008-07-24˜!2008-10-28˜!2008-11-26˜!" In: *The Open Civil Engineering Journal* 2.1, pp. 143–147. ISSN: 18741495. (Visited on 04/01/2026).

Harajli, M. H. (Aug. 2004). "Comparison of Bond Strength of Steel Bars in Normal- and High-Strength Concrete". In: *Journal of Materials in Civil Engineering* 16.4, pp. 365–374. ISSN: 0899-1561, 1943-5533. (Visited on 02/27/2026).

Hassanieh, A. et al. (June 2018). "Glued-in-Rod Timber Joints: Analytical Model and Finite Element Simulation". In: *Materials and Structures* 51.3, p. 61. ISSN: 1359-5997, 1871-6873. (Visited on 10/11/2026).

Li, Hai-tao et al. (Oct. 2012). "Tensile Bond Anchorage Properties of Australian 500N Steel Bars in Concrete". In: *Journal of Central South University* 19.10, pp. 2718–2725. ISSN: 2095-2899, 2227-5223. (Visited on 03/24/2026).

Nemes, O. and F. Lachaud (Jan. 2009). "Modeling of Cylindrical Adhesively Bonded Joints". In: *Journal of Adhesion Science and Technology* 23.10-11, pp. 1383–1393. ISSN: 0169-4243, 1568-5616. (Visited on 06/22/2026).

Ng, Anthony and Graeme McGregor (n.d.). "Changes to Australian Standards to Better Utilise Higher Strength Reinforcing Steels". In: ().

Ng, Anthony, Graeme McGregor, et al. (n.d.). "Customised Rib Pattern to Reduce Development Lengths for 600 MPa Reinforcing Steels". In: ().

Orangun, C O, J O Jirsa, and John E Breen (n.d.). "The Strength of Anchored Bars: A Reevaluation of Test Data on Development Length and Splices". In: ().

Pauletta, Margherita et al. (Feb. 2020). "Bond-Slip Behavior between Stainless Steel Rebars and Concrete". In: *Materials* 13.4, p. 979. ISSN: 1996-1944. (Visited on 04/15/2026).

"Reinforced Concrete Basics" (2021). In.

Standards Australia (2018). *AS 3600:2018 Concrete Structures*. Sydney: Standards Australia.

Tang, Chao-Wei and Chiu-Kuei Cheng (Aug. 2020). "Modeling Local Bond Stress–Slip Relationships of Reinforcing Bars Embedded in Concrete with Different Strengths". In: *Materials* 13.17, p. 3701. ISSN: 1996-1944. (Visited on 03/09/2026).

Taylor, David (May 1999). "Geometrical Effects in Fatigue: A Unifying Theoretical Model". In: *International Journal of Fatigue* 21.5, pp. 413–420. ISSN: 01421123. (Visited on 10/20/2026).

– ed. (2010). *The Theory of Critical Distances: A New Perspective in Fracture Mechanics*. Amsterdam London: Elsevier. ISBN: 978-0-08-044478-9 978-0-08-055472-3.

Zuo, Jun and David Darwin (2000). "Splice Strength of Conventional and High Relative Rib Area Bars in Normal and High-Strength Concrete". In: *ACI Structural Journal* 97.4. ISSN: 0889-3241. (Visited on 04/15/2026).

Zuo, Jun, Roberto T. Leon, et al. (n.d.). "408R-03 Bond and Development of Straight Reinforcing Bars in Tension". In: ().